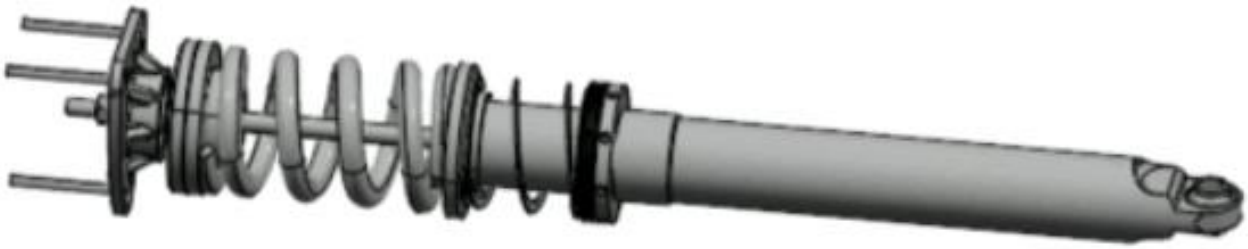
ADVANCED AUTOMOTIVE ENGINEERING

MOTORVEHICLE UNIVERSITY OF EMILIA-ROMAGNA



Magnetorheological Suspensions for Pikes Peak Hill Climb

Group 3:

Edoardo Barera

Francesco Pio Cotugno

Martina Donzelli

Pierluigi Maria Teotini

AUTOMOTIVE COMPUTER AIDED DESIGN

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1. Executive Summary

Pikes Peak Hill Climb and, more generally, all the off-road competitions are known to be extremely tough for both the driver and the car, that must withstand strong excitations and extreme road and environmental conditions.
The main aim of this project is to study and replace the usual suspensions with a more high-performing option, namely the magnetorheological suspension system. The goal is to achieve a more comfortable drive, allowing the driver to push the car to the limit and, consequently, to gain as much performance as possible.

2. Application and Product Overview

At first, to fully understand the real **application** of this project, the considered competition should be better described.
The "Pikes Peak International Hill Climb" is a timed hill climb to the summit of the Pikes Peak Mountain in the USA. The race is 19.99 km long and highlights the presence of more than 155 twists and turns with great altitude difference. The path was once composed by both gravel and paved sections, but nowadays it is fully covered by asphalt.
However, conditions are not ideal as it is a mountain public road, hence it lacks the maintenance required to be a racetrack. Moreover, pressure and atmospheric conditions are very different too, changing usually from sunny to snowy weather with icy roads.
In such a competition, one of the most excited elements is the suspension system, especially because of the irregular roadbed and the sudden and continuous swerves.



Figure 1 - Pikes Peak Hill Climb Competition

That's why the proposal of this project is the application of new active suspensions called magnetorheological suspensions (from now on briefly called "MR" suspensions).
More generally, active suspensions can keep the tires in correct position with respect to the road also during cornering and can adjust their own rigidity too. Everything is aimed at better traction and control. Therefore, they allow to reach a better-quality drive, improving comfort, performance and safety, that is crucial in races where guard-rails and barriers are not present, as the Pikes Peak competition.

MR suspensions can be a real advantage during a Pikes Peak Hill Climb because the race is extremely tough for both car and driver and, consequently, for the suspension system: steep climbs, tight turns and strong elevation changes must be withstood.
Hence, this particular type of suspension can ensure stability during cornering, adaptability to quick surface changes, vibrations reduction and grip maximisation.
That's why the main goal of this project is to design an MR suspension that is able to improve safety and handling, together with a very quick and predictable response to the external inputs.

Giving a brief **overview of the product**, the specificity of this suspension system is that MR suspensions have magnetorheological dampers.

These dampers are containing the magneto-rheological fluid that changes its viscosity when subjected to a magnetic field that is controlled by electro-magnets.
The damping fluid contains metallic particles, so the electro-magnet is able to control the optimal stiffness of the shock absorbers. This optimum is calculated through an onboard computer and wheel sensors that measure suspension extension, steering and acceleration and send all the data to the car control unit that, in its turn, send the correct input to the electro-magnet.
As a result, a quicker reaction of the system allows, for example, a smoother passage over bumps.
Essentially, with higher current flow (and the magnetic field strength) the particles align increasing the viscosity of the fluid and, consequently, its resistance to flow; this rises the compression/rebound rate making the damper stiffer.
Vice-versa, decreasing the flow, particles are less aligned letting the fluid pass more easily and softening the effect of the dampers with less resistance to compression/rebound.

That's why this kind of suspension could be very useful on irregular road races, ensuring a great improvement in comfort and safety, so enhancing the driving style and the performance, which is always the main focus in such a fiercely competitive field.

3. Competitor Analysis

In the racing world nothing is more important than performance and results.
MR suspensions are nowadays still a novelty in the Pikes Peak competition: that's why the analysed competitors before starting the conceptual design phase are the actual, more traditional, suspensions that widely appear during the Pikes Peak climb.
The attention is therefore focalized not on other factories that are able to design, produce and sell MR suspensions, but on the factories that have their manual suspensions mounted on the Pikes Peak cars, to find out all the benefits that MRs could bring with respect to the current ones.

More, listening to the **customers' voice** thanks to a brief interview to Vittorio Roberti, engineer for Lancia Martini Racing engaged in Rally and Safari races, it was possible to define some key points to be able to compare the different possible suspensions solutions.
Specifically, he emphasized the first parameters to be taken always into account: track condition and the subsequent grip level. When conditions are not comfortable, active suspensions could be a real solution. However, he warned about reliability as one of the most important features in every innovative project.
As a consequence, the two main areas to be focused on are performance and reliability, together with the price and the economic aspect.

3.1. Performance

First, it is important to analyse the strengths of the MR suspensions compared to the passive/traditional systems, finding out the main points letting MR suspensions to be better than the competitors.
Since MR suspensions are considered as "active", they provide an almost real-time adaptation of the system (and therefore of the car) to the irregularities, asperities, difficulties of the road, including bumps and kerbs. Stiffness of the damper is continuously changing, allowing to always retrieve the optimal value for each part of the track under each condition (also altitude change) and for each driving style. This makes MR suspensions versatile, with no need of tuning and being able to deliver the maximum possible performance improving the handling of the car and the tires' correct usage.

MR suspensions are avoiding the risk of a wrong setup choice, that usually bring to poor driving performance with drivers complaining and not being able to push the car to its real limit.
On the contrary, traditional suspensions have fixed stiffness settings that are decided before each run and are kept the same throughout the entire session. Hence, the best setup is a compromise between all the possible conditions encountered during a lap.
Also compared to other possible active solutions (as purely hydraulic or air-based systems), MR are usually more compact, lightweight and energy efficient with simpler control mechanism. Those features can be all considered as crucial when dealing with high performance and competition cars.
On the contrary, MR can just provide adjustable damping, without touching geometry or ride heights as other types of active suspensions.

3.2 Reliability

When a new product is used for the first times in a new application field, it is important to detect in advance all the possible weaknesses that can make competitors' product better.
Talking about MR, temperature change has a big impact on the fluid, and consequently on performance, because the fluid is strongly excited so there is the risk of overheating. The issue is addressed by meticulously controlling tolerances during the manufacturing process to minimize unwanted friction while ensuring the highest standards of sealing quality for all components.
Leaks and contamination from the outside (particles and air "in primis") can be both critical points altering the correct functioning of the suspension and leading to possible failure and contamination of the environment. A solution is given by putting great attention on chosen materials (analysing deeply their properties). There are two possible and widely used MR fluid: Lord MRF-132DG and AMT DAMPRO+. In this project the chosen is the first since, together with other important properties, it is unsensitive to contamination and, consequently, solving this major problem. This topic will be better highlighted in Paragraph 7.3.3 during the Conceptual Design phase.
Traditional suspension systems, vice-versa, are usually intrinsically more reliable since adopting simpler components and being an already well-defined and developed technology, widely tested and adopted on a variety of vehicles.

3.3 Price

MR suspensions are much more complex than traditional systems, that's why they are more expensive and with higher need of maintenance.
In addition, MR can have shorter life because magnetic particles may degrade quicker: in the racing environment this can be considered as a minor problem since, to always keep the highest possible level of performance, no component is exploited until its real end of life. To gain achievements, maintenance is continuous and components replacement can be more frequent too.
That's why it is possible to talk about "**cost efficiency**", meaning that a slightly higher price can be justified to gain a much higher performance, bringing to better results.
Traditional suspensions are simpler so cheaper and requiring less maintenance.

To conclude, the cons of the MR suspensions can be overcome with an accurate design and, as written above, price is always an important driver, but it is not the crucial factor in such a competitive environment. Teams are usually more prone to pay a bit more if the return in terms of performance is adequate.

4. Requirements: a preliminary list

To better understand all the requirements needed to reach the desired functionality, MR suspensions can be subdivided into several subsystems.
Finally, the creation of a preliminary list is fundamental to put the basis of the project and future development defining at first all the obvious requisites.
To define the correct fundamental requirements a distinction between demands and wishes has to be made: the first are requirements that must be fulfilled under all the circumstances setting minimum and maximum performance, while the second should be considered whenever possible.

SUBSYSTEM	PROPERTIES & FUNCTIONS
Piston	- Allow the liquid to flow - Allow variable damping → changeable electric resistance - Heat dissipation - Sealing → fluids and electricity - Sliding without friction →attention to roughness
Cylinder	- Lightweight - Load resistance → toughness - Sealing and piston guidance - Sliding without friction → attention to roughness
Spring, Bumpstop, Top mount	- Store and release energy to absorb impacts - Provide suspension stiffness and ride comfort - Maintain vehicle height and load distribution - Limit excessive suspension travel - Prevent damage to suspension components
Bottom seal	- Piston guidance - Sealing → fluids
Sealing piston	- Sealing → gas and oil chambers
Rod	- Electric control connection - Load resistance toughness - Piston guidance - Piston stroke definition

Table 1 - Preliminary Requirement List

Moreover, all these subsystems must fulfil another common and primary requirement: strict tolerances must be necessarily satisfied to have a correct functioning of the entire assembly.

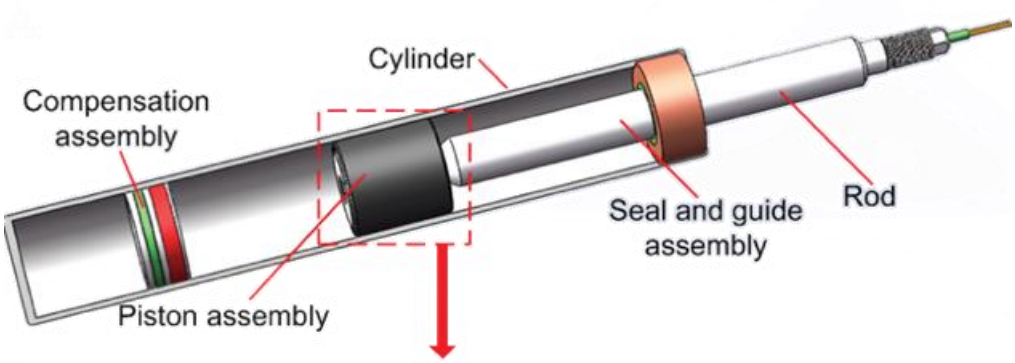


Figure 2 - MR Damper Overview

5. Quality Function Deployment

Quality Function Deployment (briefly QFD) is a fundamental tool to compile and deepen the requirement list in the above paragraph. The **house of quality** summarizes and correlates all the requirements, giving them a weight too.
Here below, the entire house of quality for the MR suspensions design is shown: it answers to the three fundamental questions which are "What?" (customer requirements), "How?" (target requirements) and "How much?" (weighted metrics). More, benchmark is included comparing the developed product with the competitors' products, both by the technical and the costumers' point of view.
Relations between technical requirements are represented by the roof (with positive and negative correlation); the relations between technical and customers' need are in the so-called matrix of inter-relations setting strong/moderate/weak correlation.

The evaluation of "what" and "how" has been carefully performed through a brainstorming process, since setting the priorities between demands is a crucial element of the systematic approach. This is a key point of the entire process because it has an impact on the final design and all the future steps.

5.1. QFD results

5.5.1. Technical Requirements and Interrelations

Here, each part of the house of quality is explained with more details.
During the House of Quality creation, the first focus has been on the technical requirements definition, together with their interrelations, hence highlighting the trade-offs.
Positive and negative correlations have been decided with the following reasoning: if increasing the value of a technical requirement means that reaching another one in easier, then the correlation is considered as positive (+), while if it would be harder to reach another one then the correlation is considered as negative (-).
This can be easily shown with a simple example: increasing the temperature range, the damper will be able to withstand higher temperatures too. However, higher temperatures mean greater expansion and, consequently, more difficulty to get dimensional stability and a leak proof device.

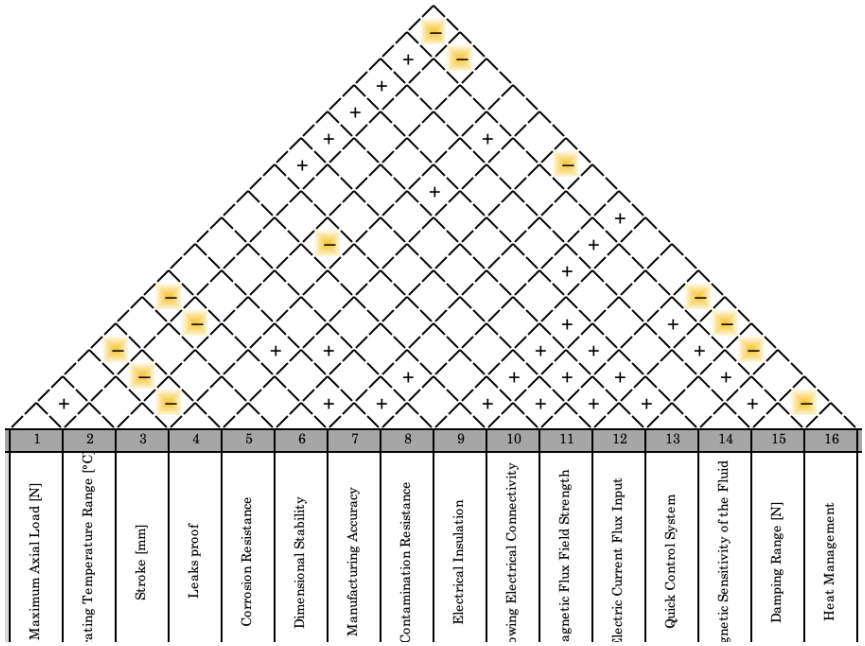


Figure 3 - Technical Requirements and Relations

Some general trends can be detected:

- Heat Management has a negative correlation with all the others related requirements taken into account. This is due to the fact that increasing loads, temperatures, electric/magnetic quantities, the generated heat would be much higher with a much more complex dissipation.
- Damping Range has positive correlation with all the others related requirements taken into account. Damping can achieve wider values if the suspension is working properly and quick with a highly sensitive fluid without contaminations and leaks.
  A lot of similarities can be noticed when dealing with the "Quick Control System" technical requirement.
- "Mixed correlations" are related to the Maximum Axial Load requirement. As an example, higher the load, higher the electric and magnetic quantities and the temperatures, bringing to more difficult heat management and more probability of having leaks.

5.1.2. Customers' Requirements and Matrix of Interrelations

The study continued with the customers' requirement definition, finally filling the matrix of inter-relations.

Customer Importance	Customer Requirements (Explicit and Implicit) \ Functional Requirements	Maximum Axial Load [N]	Operating Temperature Range [°C]	Stroke [mm]	Leaks proof	Corrosion Resistance	Dimensional Stability	Manufacturing Accuracy	Contamination Resistance	Electrical Insulation	Allowing Electrical Connectivity	Magnetic Flux Field Strength	Electric Current Flux Input	Quick Control System	Magnetic Sensitivity of the Fluid	Damping Range [N]	Heat Management
4	Ease of Maintenance				●	○		○	○								○
4	Driver Tunability	●	▽					○			○	●	●	●	●	●	●
1	Cost Efficient	○	○	▽	●	●		●	●	○		▽	▽	○	▽	●	●
4	Fail Safe	○	○		●	○		▽								○	○
5	Handling	●	○	●	○	▽	○	○	○			○	○	●	○	●	●
3	Comfort			○	▽			▽				○	○	●	○	●	▽
3	HMI				○	○		○	○								○
3	Lightweight	●		○												●	▽
5	Safety	●	●		●	○	▽	○	○	○							▽
4	Reliability	●	●		●	○	▽	○	○	○	○			○		○	▽
3	Wear Resistance & Durability [service hours]	●	○	○	○	▽	▽	○	●	▽					▽	○	○

Figure 4 - Matrix of Interrelations

As before, some meaningful examples can be extracted:

- Handling has the highest number of relations: the reason is due to the fact that this project is entirely oriented to the racing world. Handling is crucial during races and has a strong link with performance, that's why every technical requirement is devoted to help to reach this major goal.
- Reliability is crucial too and has big importance for the customers (with consequent big impact on their choice). It has strongest links with maximum load, temperatures and leaks since increasing these quantities, the product must withstand more extreme conditions, and it can become more fragile and prone to failures.
- Driver tunability has the highest number of "Strong Correlations" because it is a key point of MR suspensions. This is related to the capability of the damper to be adjusted directly by the driver while driving and to provide a quick response and adaptation to the new settings. It depends mainly by how the control system is built and how electrical/magnetic quantities interacts with the fluid.

5.1.3. Benchmarks

Going on with the House of Quality analysis, the proposed product has been compared with the competitors chosen and explained in the Competitor Analysis (Chapter 3).
The first to be considered is the so-called competitive benchmark.

	Customer Competitive Assesment				
Customer Requirements (Explicit and Implicit)	Our Product: MR Suspensions	Competitor #1: *Passive*	Competitor #2: Active Air	Competitor #3: Active Hydraulic	0 1 2 3 4 5
Ease of Maintenance	3	5	3	3	
Driver Tunability	4	0	5	5	
Cost Efficient	4	3	2	2	
Fail Safe	4	3	0	4	
Handling	4	3	5	4	
Comfort	5	3	4	4	
HMI	3	5	3	3	
Lightweight	4	5	3	3	
Safety	3	4	0	3	
Reliability	3	4	2	3	
Wear Resistance & Durability [service hours]	4	4	3	4	

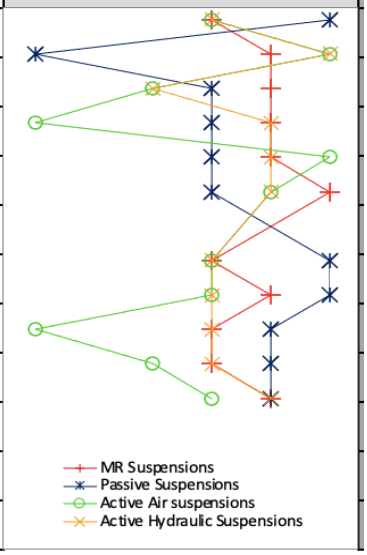


Figure 5 - Competitive Benchmark

Thanks to the coloured lines it is easy to spot weaknesses and strengths of all the considered solutions: as written in the competitor analysis too, traditional passive suspensions have their key factors in reliability and safety (since they are an already well-known technology), together with lightweight and an easy human-machine interaction that brings straightforward to an easy maintenance.
MR suspensions (that are only slightly inferior considering these drivers) can reach higher comfort thanks to the "live" tunability. They are following the fail-safe principle too, since in case of failure they can act as traditional suspensions.
Hydraulic active suspensions can be comparable with the MR solution, but they have a 5 in the important "Driver Tunability" requirement. The reason is that with MR damper only the damping can be changed during running, while active hydraulic suspensions have the possibility to adjust ride heights too.
Finally, active pneumatic suspensions have maximum value when related to "Driver Tunability" and "Handling" because they allow real-time adjustments in damping and stiffness, enabling the driver to optimize ride comfort and performance based on road conditions and driving preferences. Additionally, they enhance vehicle stability and control by dynamically adapting to load changes and cornering forces, improving overall handling characteristics, but they are not competitive at all by the safety/damages/reliability point of view since when a damage occurs the entire suspension system has no remaining capability of fulfilling its function, hence it is completely useless and could also be dangerous for the driver if this happens suddenly during a race.

The other benchmark is related to the comparison of the set technical requirements with respect to the other competitors. As before an easy overview is given by the coloured lines.

Functional Requirements	Maximum Axial Load [N]	Operating Temperature Range [°C]	Stroke [mm]	Leaks proof	Corrosion Resistance	Dimensional Stability	Manufacturing Accuracy	Contamination Resistance	Electrical Insulation	Allowing Electrical Connectivity	Magnetic Flux Field Strength	Electric Current Flux Input	Quick Control System	Magnetic Sensitivity of the Fluid	Damping Range [N]	Heat Management
Target	4200 N	-40; 150 °C	85 mm	no leaks @ max load	no corrosion in harsh environment	respect given tolerences in the temperature range	respect manufacturing tolerances	0.092 ±0.015 [Pa*s]	1*10e6 - 1*10e8 Ω	good connectivity	0,13 T	2 A	5 ms	14,3 H/m	-2500; 4200 N	781 W
Max Relationship	231	124	73	198	77	27	100	99	33	24	61	61	123	64	177	147
Our Product: MR Suspensions	5	5	5	4	3	2	3	5	4	4	5	5	5	5	5	4
Competitor #1: Traditional Passive	5	3	5	4	3	3	4	3	5	5	0	0	0	0	2	5
Competitor #2: Active Air	5	2	5	1	4	4	2	3	3	3	0	3	3	0	4	3
Competitor #3: Active Hydraulic	5	2	5	4	3	4	2	3	3	3	0	1	2	0	4	3

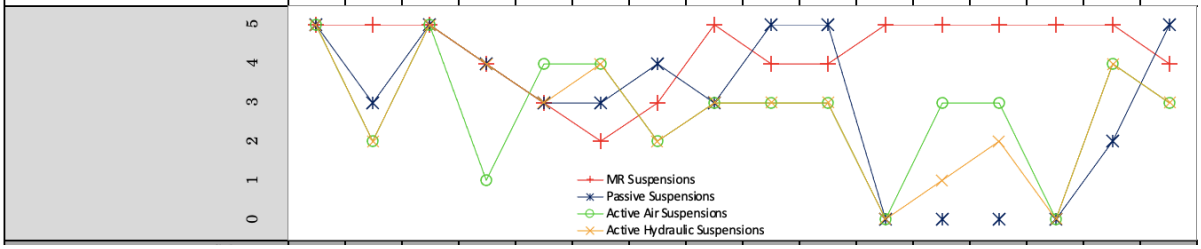


Figure 6 - Benchmark and Final Values

It is easy to notice that all the alternatives start with maximum value 5 when dealing with the "Maximum Axial Load": to be a real option that can be adopted in the considered race this requirement must be fulfilled, otherwise the solution is not acceptable.

All the competitors have very low values for all the columns concerning and considering electrical/magnetic requirements: this can be explained because it is difficult to compare solutions that are exploiting very different working concepts by the technical point of view. In this case all the competitors do not have or do not require high performance characteristics linked with electrical/magnetic quantities, that are on the contrary key points for the correct functioning of MR suspensions (5 value everywhere). More, MR dampers can allow a wider operating range of temperatures since the fluid is able to change its properties.

"Damping Range" is crucial and better when dealing with MR suspensions: the fluid viscosity changes can allow a wider range, letting the suspension to be more adaptable to every road/weather/driving condition, hence permitting to reach better drivability and comfort.

5.1.4. Final Values and HOQ Final Shape

Finally, all the calculations are done giving a number to the relationships between technical and customers' requirements weighted by the given (and previously explained) level.

This is shown in Figure6 too, and the following values have been used to get the final result:

- Weak relationship: 1 (Triangle in the matrix of interrelations)
- Moderate relationship: 3 (White circles in the matrix of interrelations)
- Strong relationship: 9 (Black circles in the matrix of interrelations)

The results obtained are totally consistent with the expectations, since it is easy to notice that the most connected technical parameter is still the "Maximum Axial Load", so that the suspension must be able to withstand the possible maximum load under every condition. It is followed by the capability to be leaks proof (to keep good functionality and prevent damages) and the damping range (that is a strength that can lead to the choice of MR suspensions).

Here below, the full House of Quality is reported.

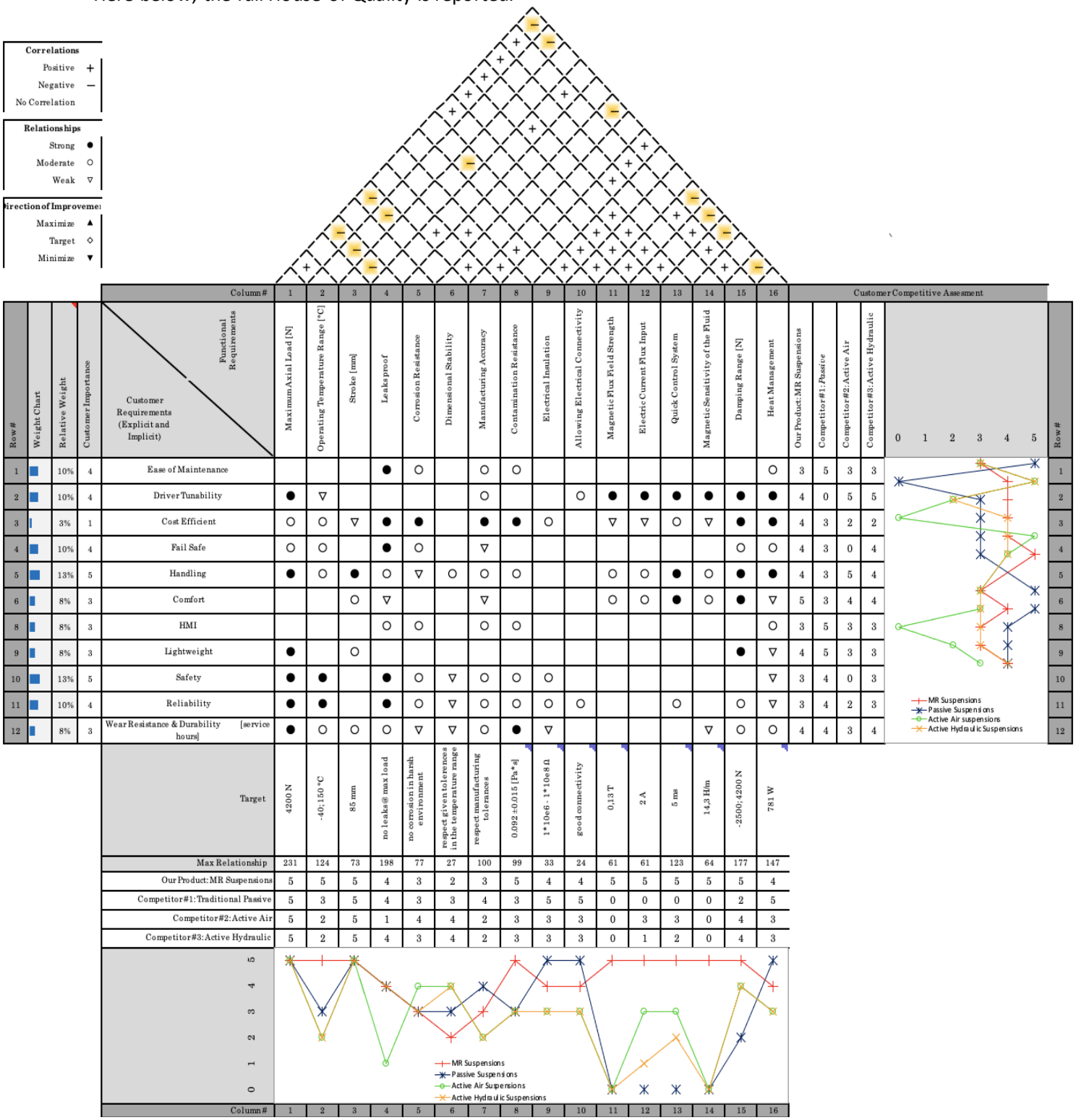


Figure 7 - Final House of Quality

6. Costs: a preliminary estimation

The cost analysis for the MR suspension design project involves estimating the production costs by considering both non-standard in-house produced parts and standard purchased components. On this purpose, the MR suspension system is split into several subsystems, considering the same division of the preliminary requirement list.
To correctly proceed in the cost estimation, few assumptions were used in the cost calculation, so, for clarity, some remarks are necessary:

- The non-standard parts, which require custom manufacturing processes, are expected to contribute significantly to the overall production cost. However, due to the absence of preliminary geometric constraints, and the flexibility afforded by a high number of design parameters, providing an accurate cost estimation for these parts is challenging.
  As a result, the cost for these non-standard parts was derived by referencing the dimensions and specifications of competitor products, specifically the Nitron racing suspensions and Porsche's GT suspension systems, which served as the benchmark for the design.
- The parameters used for selecting and producing these parts are sourced from the Granta database, which provides detailed data on materials, manufacturing processes, and associated costs.
- For standard parts, such as the piston assembly, which requires complete redesigning, costs were derived from existing market prices of similar parts to establish a competitive reference.

This approach allows to the estimate the cost for both non-standard and standard components, with particular attention to the custom non-standard parts due to their complexity.
In this way a rough but insightful estimate of the total production expenses can be yielded, helping to identify potential cost drivers and areas for optimization in the MR suspension development.

To reach the final estimated price for each subsystem, the cost for each non-standard component is computed according to the following formula.

$$C = \left[\frac{mC_m}{1-f}\right] + \left[\frac{\sum C_t}{n}\right] + \frac{1}{\dot{n}} \bullet \left[\sum\left(\frac{C_c}{L \bullet t_{wo}}\right) + \dot{C}_{oh}\right]$$

Equation 1 - Cost Equation used for non-standard components

Where the parameters provided by the Granta software to correctly fill this formula for each process and material are shown in the table below.

	C_{oh}	**60**	**€/h**
Machining	C_c	100000	€
	$1/n_{dot}$	1,50	h/unit
	$L*t_{wo}$	21900	h
Hard anodizing	C_c	10000	€
	$1/n_{dot}$	0,5	h/unit
	$L*t_{wo}$	21900	h
Honing	C_c	10000	€
	$1/n_{dot}$	0,1547	h/unit
	$L*t_{wo}$	21900	h
Induction hardening	C_c	10000	€
	$1/n_{dot}$	0,1	h/unit
	$L*t_{wo}$	21900	h

Table 2 - Granta Values

It should be remarked that the tooling cost is not taken into account since no specifically designed instruments are supposed to be needed, like in a casting process.

Here below, results are displayed for each considered subassembly and put together to get total price.

	COMPONENTS	PROCESS	MATERIAL	PRICE
Damper body	**Cylinder**	machining + honing + anodizing	anodized Al 7075	**189,47 €**
	Piston + Shims	standard part		**92,72 €**
	Piston rod	Machining + induction hardening	induction hardened high strength SS	**21,16 €**
	Free piston			**7,00 €**
Sealings and Lubrication	**Seals set + Grease**	standard part		**13,31 €**
Springs and Accessories	**Lock ring x2**	machining + hard anodizing	ergal	**14,47 €**
	Main spring	standard part	54SiCr6	**120,00 €**
	Helper spring + Supports	standard part	steel	**127,79 €**
	Bumpstop	standard part	polyurethane	**53,00 €**
Pick-up points	**Lower-end eye**	machined	high strenght steel	**30,89 €**
	Upper uniball	standard part		**235 €**
	Top mount + Uniball	standard part	aluminum/steel	
	Lower uniball	standard part		**26,08 €**
Oil	**MR fluid (0,15L)**	standard part	Lord MRF-132DG	**147,75 €**
Magnetic spools	**Set of 5 spool wire**	standard part		**13 €**
C_{TOT}				**1092 €**

Table 3 - Final Cost Estimation

It is important to point out that the tuning of the damping coefficient is supposed to be based on the frequencies of the stresses coming from the ground. This means that in the cost estimation list the cost of a sensor able to provide feedback to the control unit which will tune the intensity of the magnetic field has to be taken into account. However, since this aspect will be addressed in the conceptual design Paragraph 7.3, for the purposes of his preliminary investigation a *maximum budget of 1000€ is accounted* in the final cost. Thus, to keep the product cost efficient, while keeping the performance.

7. Conceptual design

This fundamental phase of the design process is devoted to the elaboration of a solution principle starting from basic solutions.
First, a decision has to be made, and it must take into account the requirements considered during the previous "Task Clarification" phase checking and answering the following questions:

- Is the task sufficiently clarified to allow a design solution?
- Is the conceptual phase really needed or is it possible to directly skip to the embodiment?
- How to develop the concept systematically? So, which is the clear path to follow?

The key word of the design phase is "**Abstraction**", but there are several more detailed steps to be followed during this phase.

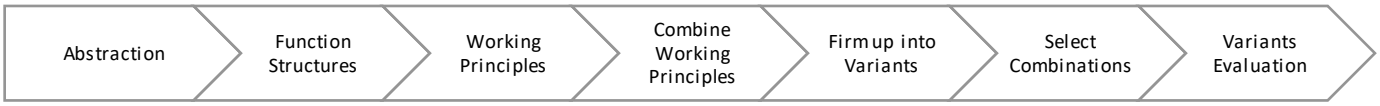


Figure 8 - Conceptual Design Flowchart

1. Abstraction: deleting personal preferences and identifying essential problems that must be shown in solution-neutral terms for an easy turning into functions (which are technical processes based on a flow of energy, materials and signals).
2. Function Structures: facilitating solutions discovery. Technical systems are split into subsystems to better identify function structures which belong to specific technical functions that can be in their turn divided into subfunctions separated by boundaries.
3. Working Principles: searching working principles reflecting physical effects, geometric and material characteristics of the considered subfunction.
4. Combine Working Principles: combining them into a working structure to get the overall system synthesis, checking compatibility of principles too.
5. Firm up into Variants: understanding the key characteristics of the working principle, embodiment and task-specific constraints.
6. Select Combinations: using a specific schematic chart.
7. Variants Evaluation: estimating both technical and economic criteria in order to reduce the number of solutions to be finally evaluated.

7.1. Abstraction

The first step to get into the conceptual design is the abstraction, to better identify the basic solution principle that is behind the concept and have easier possibility to detect problems.
The basic solution principle consists of the **simplest representation of the damper**. In this case it can be described as the motion of a piston inside a cylinder filled with a fluid. The piston has one orifice to allow the fluid motion.
From the requirement list, the essential constraints are:

- provide damping
- allow the fluid motion
- keep the fluid inside the main chamber

This kind of representation allows to focus on the damping working principle.
In particular, it stresses the role of the fluid viscosity and the force developed by its flow through the orifice. This leads to the volumetric flow rate Q:

$$Q = C_d * A_v * \sqrt{\frac{2P_{1-2}}{\rho}}$$

Equation 2 - Volumetric Flow Rate

The abstraction allows the designer to focus primarily on the fluid–piston interaction, so that there is not any pre-defined solution principle. In this stage of the development the magnetic interaction with the fluid has to be considered as just a wat to tune the fluid viscosity, not as a design constraint.

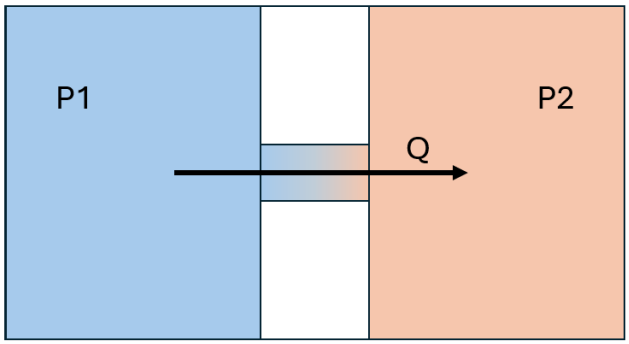


Figure 9 - Fluid-Piston Abstract Interaction

This basic solution principle includes just the piston without the connecting rod. Clearly this is not how the real damper can work because the rod is essential to transfer the damping force. However, the rod is not essential for the basic working principle and its motion inside the chamber adds the concept of volume compensation.
This last parameter has to be assessed further on the development as it can bias the initial design towards a specific solution (monotube vs twin tube design specified in Paragraph 7.3.4).

7.2. Function Structures

It is now possible to summarize the overall function of the product in a general schematic. The main function structure derives from the abstraction process, and it associates the main task of the product with the main **flows of material, energy and signal**:
First of all, to correctly create the structure, both inputs and output must be clearly identified.

- Inputs:
  - External force: it comes from the suspension motion. It is the force that has to be damped.
  - Electric power: it is composed by the current and voltage that are supplied to the coils, their magnitude gives the magnetic flux that modifies the oil properties.
  - Electric signal: it the electric power modulation that operates the coils and it comes from the damper control unit.
- Outputs:
  - Heat: it is the result of the energy dissipation that comes from the oil flow through valves and coils.
  - Output signal: it is the suspension motion. It can be measured by a linear potentiometer, positioned at the damper extrema. This signal can be fed back to the damper control unit.

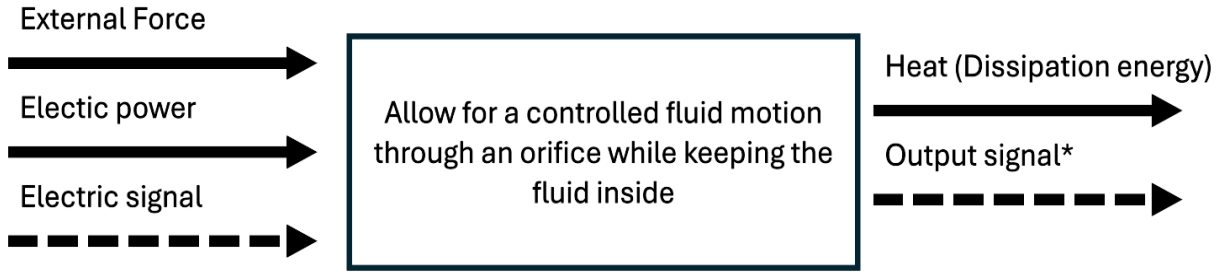


Figure 10 - Main Flows (Inputs and Outputs)

Finally, the main task has been subdivided into subfunctions in order to find the best solution principles for each subfunction. The figure below should be considered as a "zoom" of what's inside the main function shown in the previously explained Figure10, indeed it is easy to notice that inputs and outputs are exactly the same.

Thanks to this more precise and specific representation more fundamental flows can be highlighted:

- Internal material flow inside the valve subfunction: the oil can pass through the valves located at the piston in both directions depending on bump and rebound movements (that's why the arrow has both orientations); its passage is affected by the coil magnetic field.
- Magnetic field "flow" inside the valve subfunction: the electric current flowing in the coils generates magnetic field, that consequently is able to change the fluid viscosity changing the alignment of the particles present inside the fluid itself. The flow rate Q is, in its turn, affected by this magnetic field flow since the fluid viscosity appears in its direct calculation formula (see Equation2 in Paragraph 7.1) and a different quantity of fluid can pass through the valves in a considered time.
- The piston starts moving, excited by the external force, reaching a certain displacement. Coming from the piston displacement we have a further "energy flow" leading to a second displacement that is fundamental for the so-called volume compensation that should be considered to avoid damages, pressure peaks and risk of leaks.

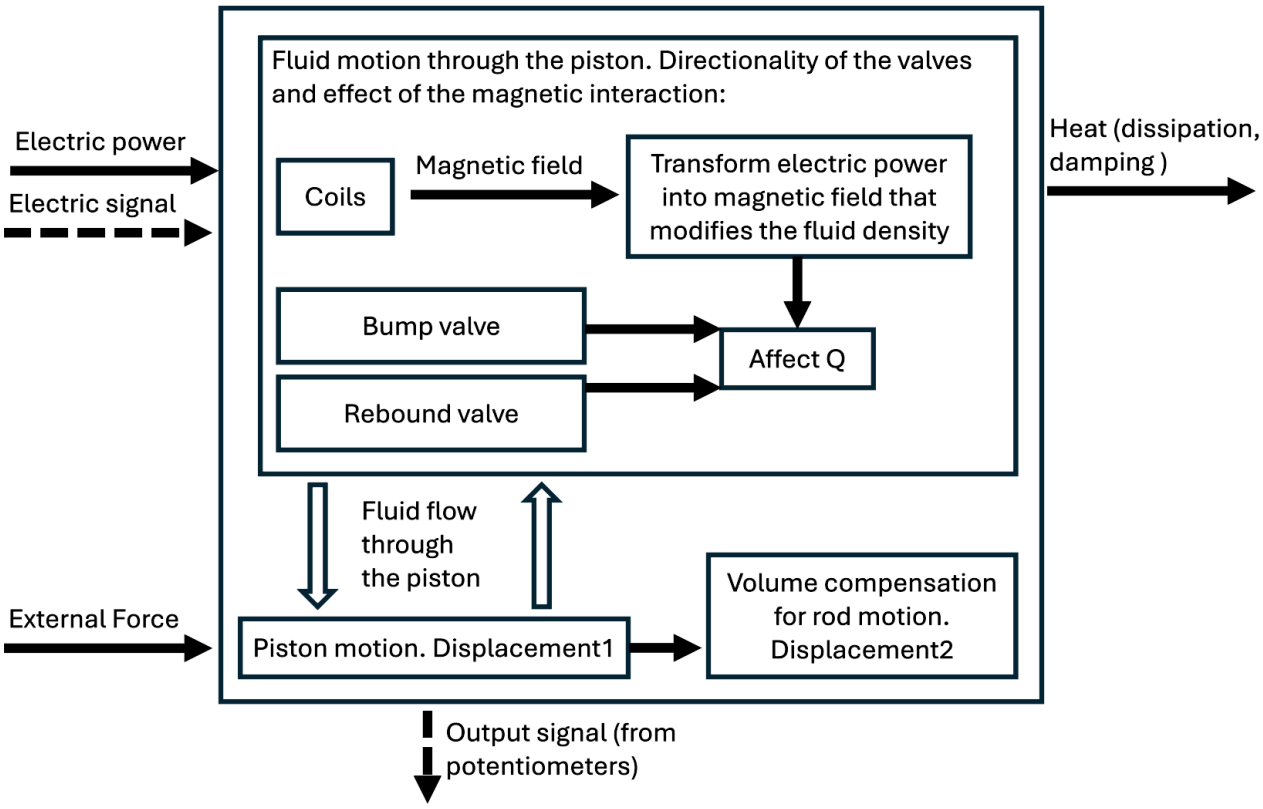


Figure 11 - Main Flows in the Valve Subfunction

7.3. Working Principles

In this section several components are considered with their different alternatives. The main goal is to different concepts that will be then evaluated during the next conceptual design steps to finally choose the most appropriate one to fulfill all the set requirements.

7.3.1. Rebound and Bump Valves

The first fundamental component that should be taken into account are the valves. Valves can have very different layouts and functioning. Here below two alternatives are considered analyzing their main properties together with their key pros/cons (check valves and spool valves):

- **Check Valves**, also known as one-way valves, are a simple type of directional control valve and a simple schematic is shown in Figure12. They allow fluid to flow in one direction while preventing reverse flow, so they are commonly used in hydraulic systems to maintain pressure and prevent the backflow of hydraulic fluid.
  The usual way in which check valves are applied in shock absorbers is using shim valves; by tuning the shim stiffness, it is possible to achieve the desired damping curve.
  At least two valves are needed, one for bump and one for rebound motion, since check valves allow the flow in one direction only as displayed in Figure13.
  Therefore, it is possible to have different shim stiffnesses, thus damping curves, for the two motions. Shims are arranged in packs to have better modularity of the curve. Shim shape can vary much as well as the number of orifices in order to get the desired damping curve; some possibilities are shown in Figure14.

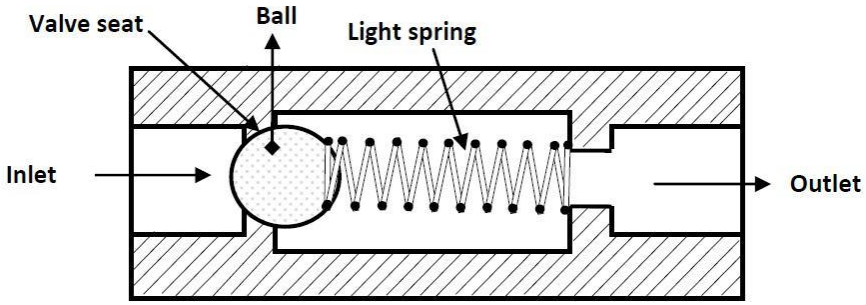


Figure 12 - Check Valve Schematic

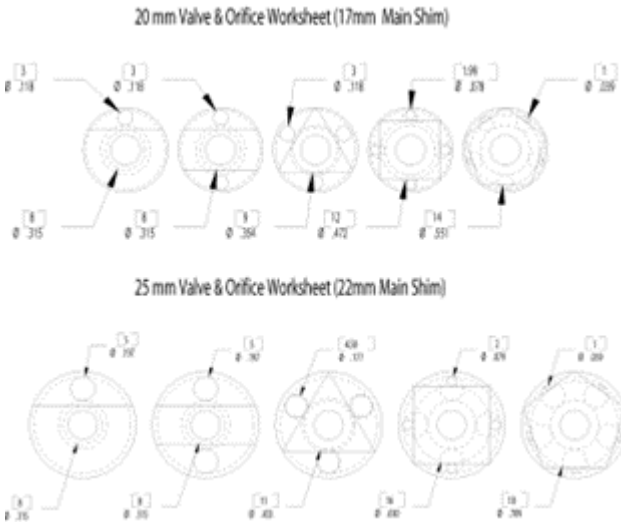


Figure 14 - Shim Design Arrangement

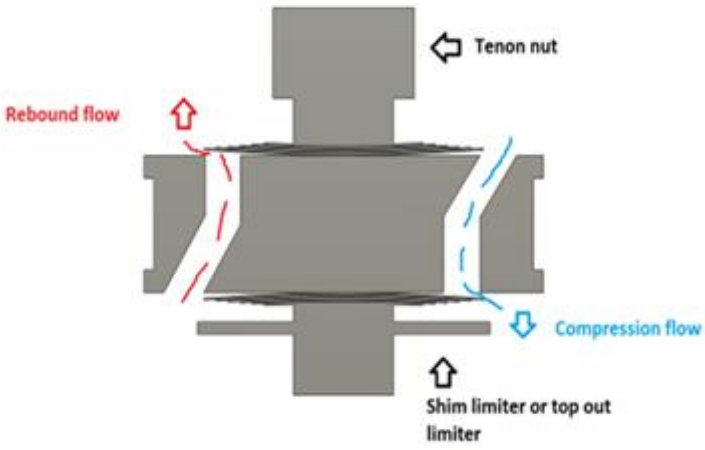


Figure 13 - Shim Packs for bump and rebound valves

The following picture shows a damping graph in the most general speed-force domain.

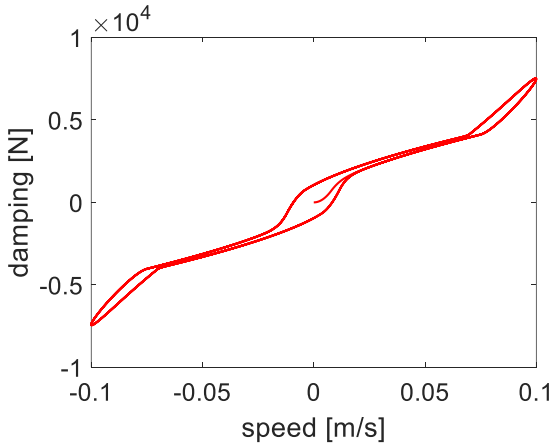


Figure 15 - Speed-Force Curve

The Figure15 plot comes from a custom-made Simulink model for a damper with check valves and orifices with variable area opening as the displayed in the following Figure16.

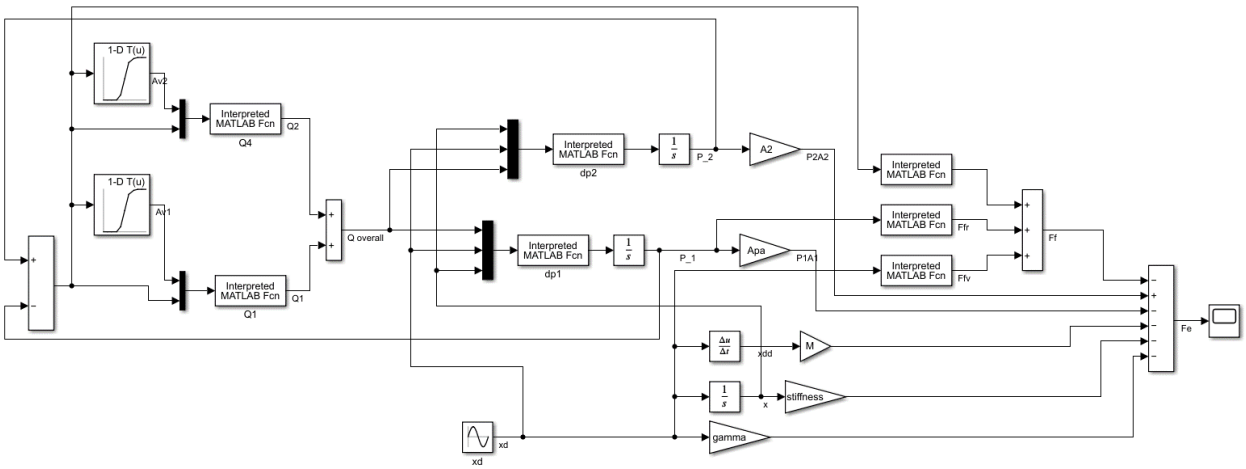


Figure 16 - Simulink Model for a damper equipped with check valves

- **Spool valves** are versatile and widely used directional control valves in hydraulic systems. They consist of a cylindrical spool that moves within a housing: by shifting the spool, the valve controls the path of hydraulic fluid, directing it to the desired actuator. Spool valves can be further classified into various configurations, such as 2-way, 3-way, and 4-way, depending on the number of ports they have.
  Spool valves offer a much higher degree of repeatability, predictability, and accuracy when tuning compared to traditional valve shim stacks. Since spool valve damping curves are predictable, they can easily be calculated using the math that governs hydraulic equations. This means that the desired damping curves can be calculated and executed without the extensive trial and error of typical shock tuning. Also, due to the flow characteristics of spool valves the shock's oil viscosity has far less of an effect on performance.

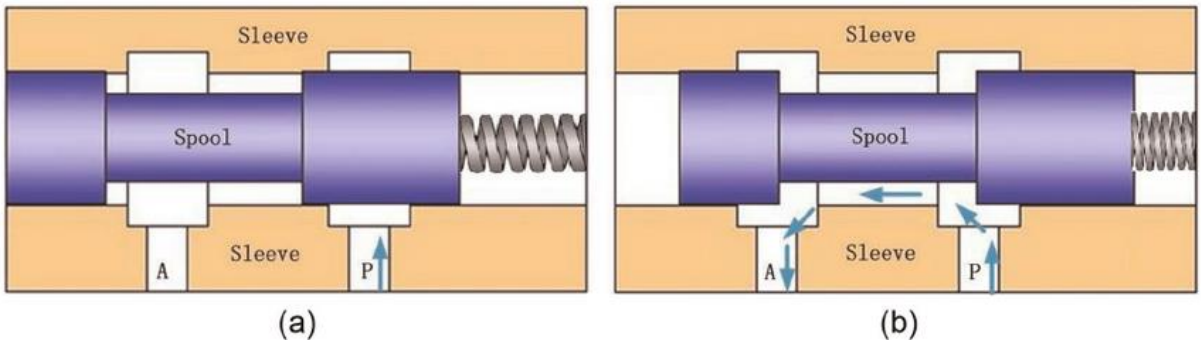


Figure 17 - Spool Valve Schematics

SUMMARIZING

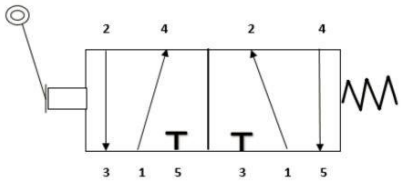
Check valve	Spool valve
	
Pros: - Easy to manufacture - Good flow control - Many design options Cons: - High damping sensitivity with heat for traditional non-MR fluids - Poor repeatability	Pros: - Very accurate - Fine tuning of damping characteristic - Faster design process - Less sensitive to oil viscosity changes (when dealing with non-MR fluids) Cons: - More expensive to manufacture due to high tolerances required - Shorter life of the internal components

Table 4 - Valves Summary

7.3.2. Coils and Positioning

The principle behind MR dampers is the magnetic flux produced by the coils inside the piston. By the electromagnetic laws, coils are defined as a solenoid and the magnetic flux is given by the following formula:

$$B = \mu_0 \mu_r \frac{N}{L} I$$

Equation 3 - Magnetic Field

$$\mu_{eff} = \frac{\mu_r}{1 + k(\mu_r - 1)}$$

Equation 4 - Permeability of the Core

Where μ_0 is the magnetic permeability of the void, μ_{eff} the magnetic permeability of the core, μ_r the relative permeability, N is the number of turns, L is the length, and I is the current intensity. It is clear from that formula that the key design parameters for the solenoid are the number of turns, the solenoid length and the current intensity. On the other hand, the length is constrained by the piston height. The last condition does not apply in the case of external coils, which are embedded in the damper cylinder thus they are not constrained by the piston height. The electric permeability is not enhanced by the aluminum core of the piston itself, being the material with relative permeability equal to one.

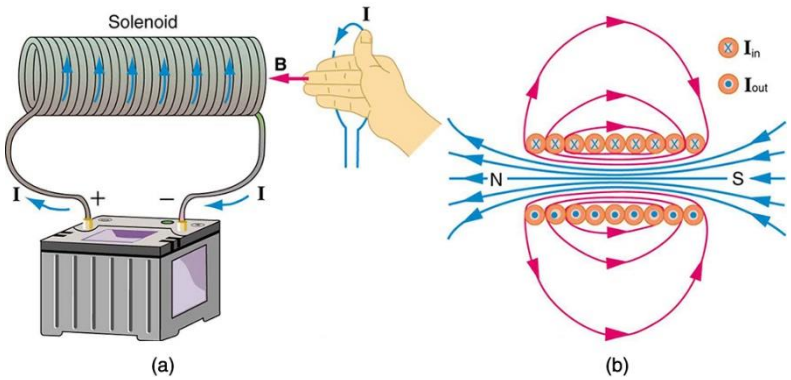


Figure 18 - Physics behind Magnetic Induction

The following Table5 is showing the effect of coil turns and current intensity variation. It is clear that when both factors are increasing the magnetic flux increases as well. However, to have a more deep insight of the problem the following plots are needed (Figure 19 in the next page).

Current Intensity	100 coil turns	500 coil turns	% Increased
0.5 A	0.02 T	0.12 T	83.3%
1 A	0.05 T	0.24 T	79.16%
1.5 A	0.07 T	0.36 T	80.5%
2 A	0.09 T	0.47 T	80%

Table 5 - Magnetic Flux Variation with current intensity and number of turns

The right graph shows the relationship between N, I, number of turns and the fluid yield stress. An MR fluid works by opposing to the damper motion thanks to its shear stress, therefore the higher the better for maximum shear stress. In this graph there is a clear bump for the 2 A curve.

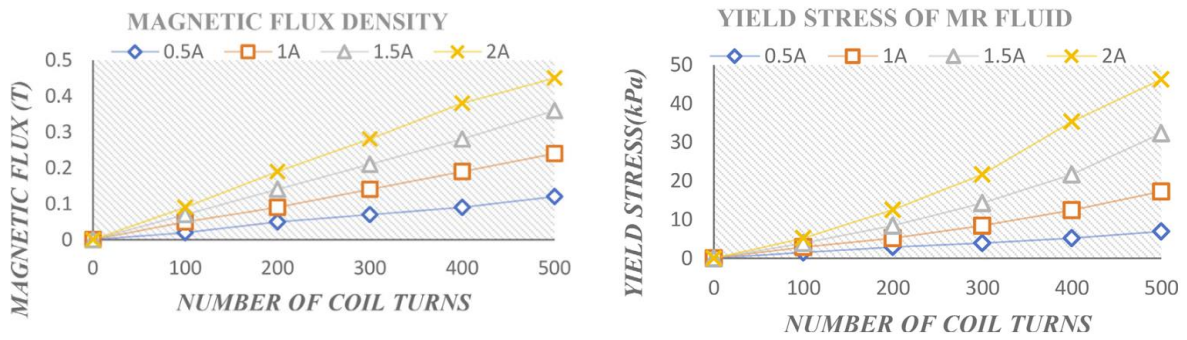


Figure 19 - Effect of Current Intensity Variation

A design variation for the coils can be internal or external positioning. Internal is thought to be within the piston, whereas external is embedded in the cylinder.

The internal design brings the following advantages (**Pros**):

- lightness, since the external coils should be as long as the piston stroke, whereas in the internal variant the solenoid moves with the valves.
- external dimensions, since an external coil would require thicker cylinder walls.

On the other hand (**Cons**), the external coils provide a longer solenoid with more room for a higher number of coil turns so its behaviour is closer to the one of an ideal solenoid.

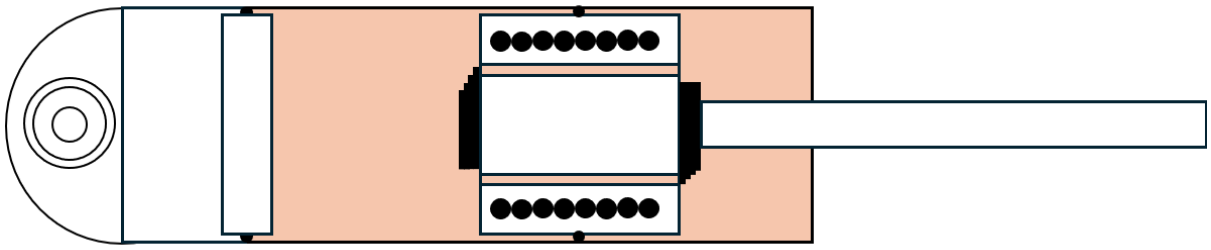


Figure 21 - Internal Coils Schematic

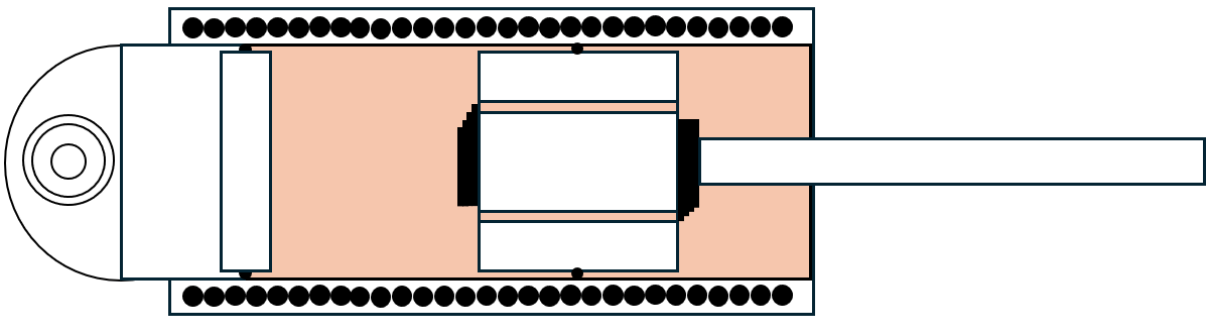


Figure 20 - External Coils Schematic

7.3.3 Magnetic Fluid

The choice of the magnetic fluid is one of the key parts of this "Working Principles" paragraph, since it is thanks to the fluid that the MR damper can change its damping value so it can be considered the core of the entire MR suspension system. MR fluids can have different characteristics and properties, hence here below two alternatives are considered: **Lord MRF-132DG** and **AMT DAMPRO+**.
Generally, all the MR fluids are composed metallic solid particles in a liquid carried; it is exactly thanks to these magneto-sensitive metallic part that the fluid can change its viscosity thanks to the different alignment of the particles. As explained in the introductive Chapter 2, when particles are aligned, the viscosity is higher and the damper is stiffer; vice-versa, with lower current and less alignment the fluid can pass more easily leading to softer damper.

The picture below is showing the behaviour of the magnetic fluid when subjected to different magnetic field B with governing formula:

$$\tau = \tau_0(B) + \mu d\gamma/dt$$

Equation 5 - Magnetic Fluid Governing Formula

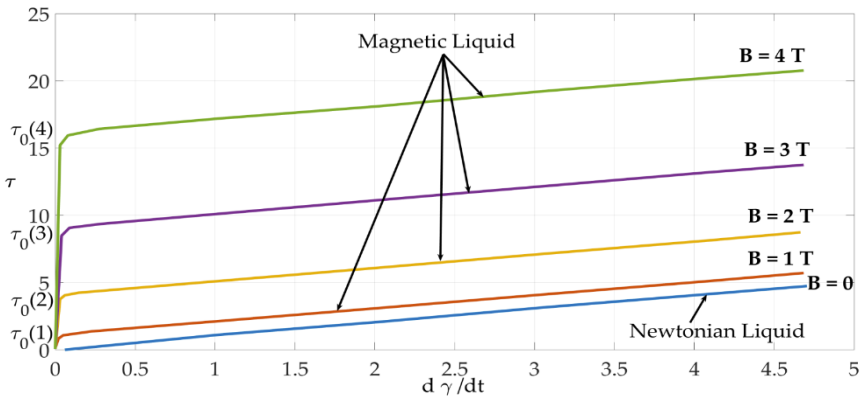


Figure 22 - Magnetic Fluid subjected to different B

Providing more details about the graph and the calculation terms:

- The x-axis represents the shear rate ($d\gamma/dt$), which is how fast the fluid layers slide past each other.
- The y-axis represents the shear stress (τ), which is the force per unit area to make the fluid flow.
- Each curve corresponds to a different magnetic field strength (B). As B increases, the fluid becomes more resistant to flow, meaning higher shear stress at the same shear rate.
- Without a magnetic field (B = 0, blue curve), the fluid behaves like a Newtonian fluid, where shear stress increases linearly with shear rate.
- As the magnetic field increases, the fluid exhibits a yield stress ($\tau_0(B)$), meaning that it needs a minimum force before it starts flowing.
- Higher B values shift the curves upward, showing that stronger magnetic fields create a more solid-like behaviour in the fluid.

This behaviour is what makes MR fluids useful in adaptive dampers and suspension systems, where changing the magnetic field allows real-time control over damping characteristics.

Now, the main comparison between the two chosen fluids is done highlighting all the technical properties of both fluids, together with performance analysis and suitability for the competition conditions.
This is done because the choice of magnetorheological (MR) fluid is critical for optimizing the dynamic performance of an MR damper designed for the Pikes Peak International Hill Climb competition, hence for an MR damper with should withstand very high and quick solicitations.

Characteristic	Lord MRF-132DG	AMT DAMPRO+
Appearance	Dark grey liquid	Dark grey liquid
Density g/cm³	2.98 - 3.18	2.50 – 2.65
Operating temperature range	-40°C to +130°C	-50°C to +150°C
Base viscosity (@40°C)	~0.28 Pa·s	~0.20 Pa·s
MR particle concentration	~40%	~30-35%
Dynamic damping force	Higher force due to a greater concentration of particles.	Slightly lower, but sufficient for dynamic applications.
Stability under thermal stress	Excellent resistance to sedimentation during prolonged operation.	Good stability but may experience rheological changes under extreme loads.
Performance at low temperatures	Slightly more viscous but still suitable for cold conditions.	More fluid, offering better initial response at low temperatures.
Track record	Widely used in automotive and motorsport applications.	Relatively new, with less documentation in high-performance scenarios.
Adaptability and customization	Standardized formula, ready for use.	Customizable for specific applications.
Suitability for Pikes Peak	Proven performance in required conditions (dynamic precision and stability).	Potentially suitable but may require optimization to maximize stability.

Table 6 - Magnetic Fluids Comparison

The **performance analysis** is here reported, going through the main drivers that must be taken into account to have a reliable and high-performance product that matches all the needs of the chosen Pikes Peak competition:

- **Thermal resistance**: both fluids operate well within the expected temperature range for the Pikes Peak competition (-10°C to +25°C, with peaks of -5°C or lower at higher altitudes). However, *Lord MRF-132DG* offers superior rheological stability under prolonged thermal stress.
- **Dynamic damping performance**: the higher particle concentration in *MRF-132DG* enables greater damping force and more precise response compared to *DAMPRO+*. This is particularly useful in rapid load changes typical of the track conditions, which include technical corners, hard braking, and sudden acceleration.
- **Low-temperature efficiency**: *DAMPRO+*, with its lower base viscosity, may provide a smoother response in the colder sections of the track. However, *MRF-132DG* still maintains an acceptable viscosity level that does not compromise performance.
- **Reliability and stability:** *MRF-132DG* is a well-established fluid widely used in high-performance automotive applications, ensuring predictable and reliable results. *DAMPRO+*, though innovative, may require additional attention to prevent sedimentation or rheological changes during extended use.

While both fluids are technically viable for magnetorheological applications, *Lord MRF-132DG* stands out as the optimal choice for an MR damper intended for the Pikes Peak competition due to its higher dynamic damping force, excellent stability under stress, and proven performance in competitive conditions. That's why, in the following paragraph, no concept will be developed using the *AMT DAMPRO+* magnetic fluid.

7.3.4. Volume Compensation

As named in paragraph 7.1., in the market there are two strategies for volume compensation management: monotube and twin tube. Volume compensation is crucial when dealing with dampers because of the rod movement inside the oil chamber. When the damper is in compression stroke, the rod enters the chamber and an equal volume of oil needs to be displaced, since liquids are incompressible. The opposite happens in the extension stroke, where the oil needs to be restored inside the chamber. For the two available solutions, the first exploits the compressibility of the air, while the second uses a second set of valves which communicate with an external cylinder.

SUMMARIZING

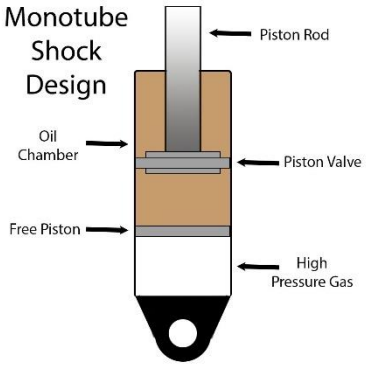
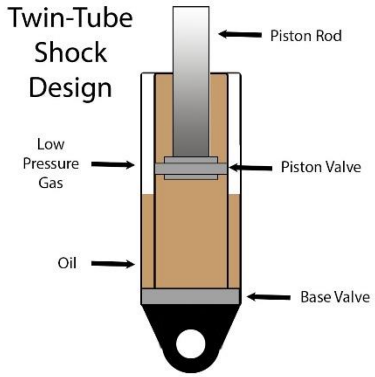
Monotube	**Twin tube**
Presence of a second coaxial piston that separates oil chamber and gas chamber	Presence of an external cylinder with no gas and oil separation
	
Pros: - More consistent through extensive usage, less oil cavitation issues and contaminations - Possibility to have the gas chamber in an external tank for better oil cooling - Allows a bigger primary piston diameter for equal external size compared to a twin tube Cons: - More expensive due to the second piston and higher manufacturing tolerances	Pros: - Cheaper - Base valves can offer higher tuning possibilities Cons: - Oil contaminations and cavitation issues due to the direct contact between oil and gas - Smaller piston diameter

Table 7 - Volume Compensation Summary

7.3.5. Output signal

The output signal is fundamental for the active control system of the damper. Such closed loop system can receive the signal coming out of the shock absorber through a feedback.

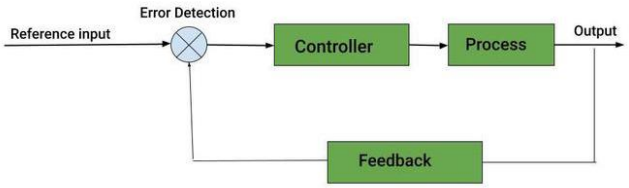


Figure 23 - Feedback General Schematic

The control system is bought from an external supplier, so it is not our task to define its working principles. On the other hand, our product should be able to provide the right output signal for the controller. The feedback can be delivered as:

- displacement by means of linear or rotary potentiometers
- force from loadcells
- flux measurements of the oil though the valves
- pressure difference inside the oil chambers

All the following sensors are already available on the market, the choice will be upon the best integration within our shock absorber.

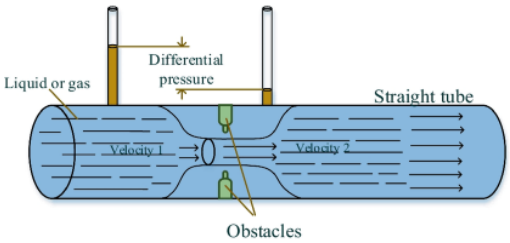
Displacement 	Pros: - Easy to measure - Easy to derive speed and accelerations - Widely available on the market Cons: - Increased external dimensions of the assembly
Force 	Pros: - Direct force measurement Cons: - Expensive - Short life in case of extensive use
Flux Measurement 	Pros: - Easy comparison with the governing parameters of the flux as the volumetric flow rate Q Cons: - Cumbersome to be fit inside the already busy piston assembly
Pressure Difference 	Pros: - Easy comparison with the governing parameters of the flux as the volumetric flow Q. Cons: - Difficult to tune - At least one more sensor to be installed

Table 8 - Sensors providing Feedback

7.3.6 End eye connection

The double wishbone architecture of the described car's suspension requires 3 rotational degrees of freedom to be free at both ends of the shock absorber. Therefore, the best solution is to use a spherical joint. The most common piece of hardware available on the market is the Unibal, like the one shown in the picture below.



Figure 24 - Unibal

7.4. Combine Working Principles

Every solution to each subfunction is then summarized and carefully analysed in the following table: the so-called "Classification Table" (Table9). A first selection is then made to select the most promising solutions or at least to discard the worst ones.

- **Valve**: the sleeve valve is a better option if a traditional oil is used. Thanks to the MR technology, the drawbacks of the check valves disappear.
  Moreover, the sleeve valve makes it more difficult to fit the coils inside the piston, therefore this option is discarded.
- **Solenoid positioning**: both options are kept since they can be both valid for their own characteristics. The internal design is good because the magnetic field moves with the valves, while the external design is good because it can be closer to an ideal solenoid thanks to its length.
- **Magnetic fluid**: as described in Paragraph 7.3.3, the best solution is to exploit the Lord MRF-132DG because of its higher dynamic damping force, excellent stability under stress, and proven performance in competitive conditions.
  On the contrary, the AMT DAMPRO+ magnetic fluid, can be still a good alternative with better reaction at low temperatures, but a less precise response together with a limited knowledge of its reliability because of its recent introduction led to the final discard of this possibility.
- **Volume compensation**: Both options are valid since they successfully solve the volume variation issue. Monotube design have its strength in the volume control precision. Vice-versa, twin tube design is worth to be explored as an option for the increased degree of freedom of the second set of valves.
- **Output signal**: flux and pressure difference signals are discarded from the beginning, since their sensors are cumbersome to fit in the relatively narrow space of the cylinder. Another cause is the lower availability of purposely built sensors, making them custom made and therefore way more expensive.
  On the other hand, displacement sensors are cheap and well proven, while force transducers are more precise but more expensive, but these two possibilities can still be considered.
- **End eye connection**: there is just one widely spread solution on the market that is the spherical joint, namely the Unibal.
  It has very good mechanical properties in term of friction and precision of motion, but it can wear fast bringing some unwanted clearance and it can be relatively expensive.

Subfunction	Solution	Pros	Cons	Keep
Valve	*Check Valve*	Easy to manufacture. Good flow control. Many design options.	High damping sensitivity with heat for traditional non-MR fluids. Poor repeatability.	Yes
	Sleeve Valve	Very accurate. Fine tuning of damping characteristic. Faster design process. Less sensitive to oil viscosity changes (non-MR fluids).	More expensive to manufacture due to high tolerances required. Shorter life of the internal components.	No
Solenoid positioning	*Internal*	Light and compact.	The solenoid is very short in axial direction, bringing some undesired edge effects.	Yes
	External	The solenoid is closer to the ideal case of infinite length.	Heavier and less accurate. Bigger external dimensions.	Yes
Magnetic Fluid	*Lord MRF-132DG*	Higher stability under stress. Greater damping force and precise response.	Less reactive under cold conditions.	Yes
	AMT DAMPRO+	Smoother response at low temperature	Possible sedimentation.	No
Volume compensation	*Monotube*	More consistent through extensive usage (less oil cavitation issues and contaminations). Possibility to have the gas chamber in an external tank for better oil cooling. Can have bigger primary piston diameter for equal external size.	More expensive due to the second piston and higher manufacturing tolerances.	Yes
	Twin tube	Cheaper. Higher tuning possibilities thanks to the second set of valves.	Oil contaminations and cavitation issues (direct contact oil-gas). Smaller piston diameter.	Yes
Output signal	*Displacement*	Easy to measure. Easy to derive speed and accelerations. Available on the market.	Increased external dimensions of the assembly.	Yes
	Force	Direct force measurement.	Expensive. Shorter life in case of extensive use	Yes
	Flow	Easy comparison with the governing parameters of the flux.	Difficult fit inside the piston assembly.	No

	Pressure difference	Easy comparison with the governing parameters of the flux.	Difficult to tune. One more sensor needed.	No
End eye connection	*Unibal*	Three rotational degrees of freedom. Low friction. Constraints all translatory motions.	Wear. Costs.	Yes

Table 9 - Classification Table

After a first selection, a **morphological matrix** is used to obtain proper solution variants. In this table, the solutions are connected to the subfunctions which are then combined to arrive at an overall solution. The possible combinations are selected through a systematic method.
Finally, it is possible to select the most remarkable variants and to pick the best one.

Function	Solution		Concept variant			
	A	B	1	2	3	4
Valve	Check valve	X	A	A	A	A
Solenoid positioning	Internal	External	A	A	A	B
Magnetic fluid			A	A	A	A
Volume compensation	Monotube	Twin tube	A	A	B	A
Output signal	Displacement	Force	A	B	A	B
End eye connection	Unibal	X	A	A	A	A

Table 10 - Morphological Matrix

In the following page, “free hand” **sketches** are used to firm up the solution variants The following sketches represent the four concepts deriving from the morphological matrix. All these sketches will be explained with more details in Paragraph 7.5, that is directly related to the “Firm up into variants” phase, highlighting all their principal differences.

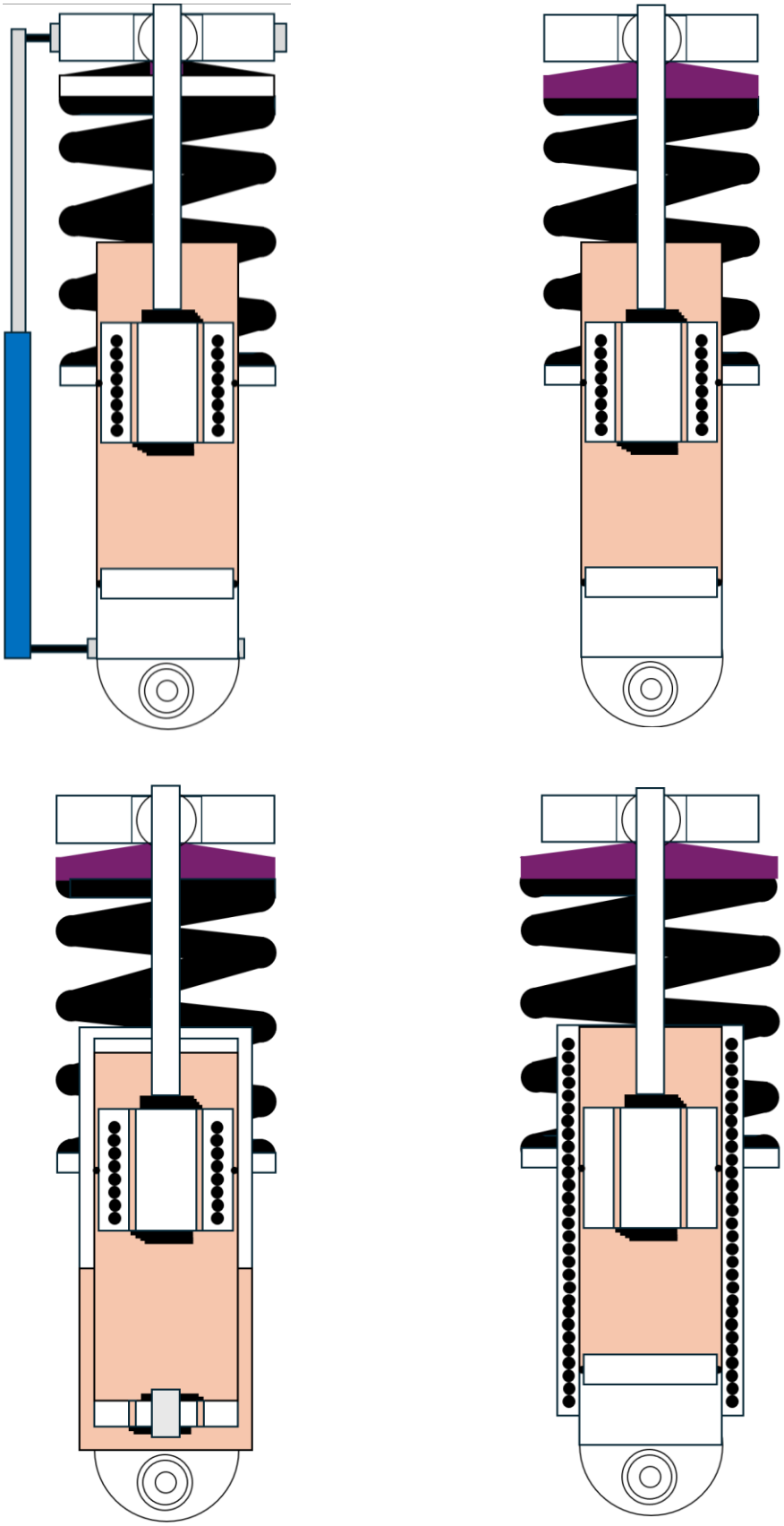


Figure 25 - Resulting variants - From Top Left and Clockwise: Concept 1,2,3,4

7.5. Firm up into Variants

In this paragraph all the previously found variants are considered one by one putting together sketches, preliminary calculations and crucial drivers to better evaluate all the differences that will be used in the following Paragraph 7.6 in the definition of some of the "evaluation criteria" to carry out the VDI 2225 Analysis.

7.5.1. Concept 1

Concept 1 is composed by check valves, internal coils, magnetic fluid, monotube design, potentiometer sensors and Unibals, hence it is differing from the other alternatives because of the presence of the potentiometers.

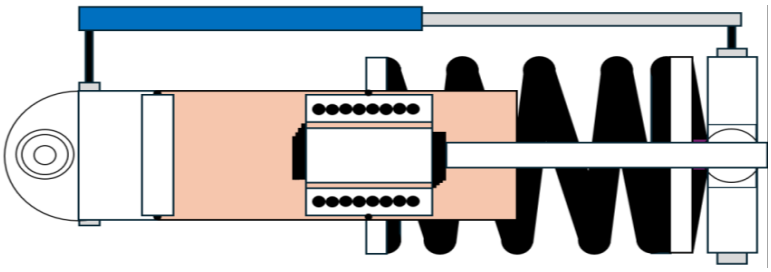


Figure 26 - Concept 1

The magnetic field can be easily calculated depending on the coil design using a basic mathematical equation together with the correct coil-related values that can be found in the following Table11:

$$B = \mu_0 \mu_r \frac{N}{L} I = 2.51x10 - 3\ T$$

$$\mu_0 = 4\pi \times 10 - 7 \frac{N}{A^2}, \qquad \mu_{r,Al} = 1, \qquad N = 50, \qquad L = 0.05\ m, \qquad I = 2A$$

Equation 6 - Magnetic Field Calculation for Concepts 1,2,3

Ease of maintenance	Average, potentiometer brackets take some time to be removed
Driver Tunability	Infinite options given by the MR design
Cost Efficiency	Estimated to cost around 1300 € with average performance
Performance	Average performance due to the lower accuracy of the feedback signal given by the potentiometers compared to loadcells
Coil turns N	50
Solenoid length L	0.05 m
Current intensity I	2 A
Relative permeability μr	1
Magnetic field strength	B = 2.51x10-3 T
Lightweight	Estimated to be around 5.3 kg
External dimensions	Envelope cylinder of ø 85 mm
Reliability	Life around 8000 km

Table 11 - Concept 1 Characteristics

7.5.2. Concept 2

Concept 2 is composed by check valves, internal coils, magnetic fluid, monotube design, load cell sensors and Unibals, hence it is differing from the other alternatives because of the presence of load cells.

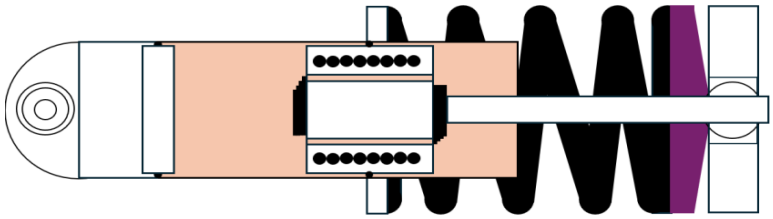


Figure 27 - Concept 2

The magnetic field calculation is equivalent to Concept 1 (Equation6), since the coil design is not varied.

Ease of maintenance	Very good, loadcells are easy to be removed with no additional effort
Driver Tunability	Infinite options given by the MR design
Cost Efficiency	Estimated to cost around 1900 € with very good performance
Performance	Very high due to the fast and precise feedback of loadcells
Coil turns N	50
Solenoid length L	0.05 m
Current intensity I	2 A
Relative permeability μr	1
Magnetic field strength	B = 2.51x10-3 T
Lightweight	Estimated to be around 4.9 kg
External dimensions	Envelope cylinder of ø 48 mm
Reliability	Life around 15000 km

Table 12 - Concept 2 Characteristics

7.5.3. Concept 3

Concept 3 is composed by check valves, internal coils, magnetic fluid, twin tube design, potentiometer sensors and Unibals, hence it is differing from the other alternatives mainly because of the presence of the twin tube design.

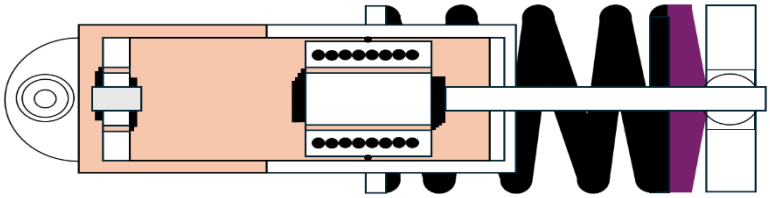


Figure 28 - Concept 3

As before, the magnetic field calculation though Equation6 is equivalent to Concept 1 and 2, since the coil design is not modified.

Ease of maintenance	Average, potentiometer brackets take some time to be removed
Driver Tunability	Infinite options given by the MR design
Cost Efficiency	Estimated to cost around 1100 € with very good performance
Performance	Poor due to the twin tube design
Coil turns N	50
Solenoid length L	0.05 m
Current intensity I	2 A
Relative permeability μr	1
Magnetic field strength	B = 2.51x10-3 T
Lightweight	Estimated to be around 5.8 kg
External dimensions	Envelope cylinder of ø 75 mm
Reliability	Life around 2500 km

Table 13 - Concept 3 Characteristics

7.5.4. Concept 4

Concept 4 is composed by check valves, external coils, magnetic fluid, monotube design, load cell sensors and Unibals, hence it is differing from the other alternatives mainly because of the presence of the external coils design.

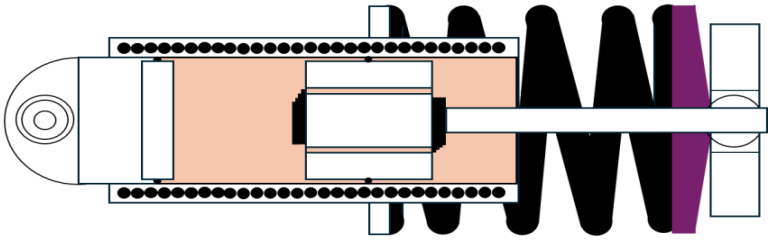


Figure 29 - Concept 4

That's why in this case the magnetic field B should be recalculated using the new coil design parameters.

$$B = \mu_0 \mu_r \frac{N}{L} I = 2.36x10 - 3\,T$$

$$\mu_0 = 4\pi \times 10 - 7 \frac{N}{A^2}, \qquad \mu_{r,oil} = 1, \qquad \mathrm{N} = 800, \qquad L = 0.85m, \qquad I = 2A$$

Equation 7 - Magnetic Field Calculation for Concept 4

Ease of maintenance	Very good, loadcells are easy to be removed with no additional effort
Driver Tunability	Infinite options given by the MR design
Cost Efficiency	Estimated to cost around 2080 € with average performance
Performance	Average due to moving valves with respect to the solenoid
Coil turns N	800
Solenoid length L	0.85 m
Current intensity I	2 A
Relative permeability μr	1
Magnetic field strength	B = 2.36x10-3 T
Lightweight	Estimated to be around 6.1 kg
External dimensions	Envelope cylinder of ø 68 mm
Reliability	Life around 15000 km

Table 14 - Concept 4 Characteristics

7.6. Select Combinations

Generally, the evaluation to select the best option between the four found variants can be finalized following two different paths: the cost-benefit analysis (setting hierarchical order between objectives) and the VDI 2225 (combining technical and economical evaluations without a hierarchy).
In this project the VDI 2225 analysis is employed.

Thanks to what is anticipated in Paragraph 7.5, this analysis is started with the identification of the evaluation criteria and is then completed in the following Paragraph 7.7 with the calculation of technical and economic ratings.
Here, several key parameters are considered and rated with the VDI 2225 points. They go accordingly to a scale from 0 to 4, depending on the parameter magnitude itself. It is thanks to these points that the correlation chart below can be determined.
More, it is easy to notice that this VDI 2225 points can be both qualitative and quantitative depending on the needs carried by the considered parameter.

Parameter	**VDI 2225 Points**				
	0	1	2	3	4
Ease of maintenance	Extremely poor	Poor	Average	Good	Extremely good
Driver Tunability	0 options	2 options	5 options	10 options	∞ options
Cost Efficiency	Extremely poor	Poor	Average	Good	Extremely good
Fail Safe	Very dangerous	Dangerous	Average	Safe	Very safe
Performance	Very low	Low	Average	High	Very High
Magnetic field strength	1.5x10-3 T	1.7x10-3 T	2x10-3 T	2.3x10-3 T	2.5x10-3 T
Ease of Manufacture	Extremely poor	Poor	Average	Good	Extremely good
Lightweight	> 2.25 kg	2÷2.25 kg	1.75÷2 kg	1.5÷1.75 kg	< 1.5 kg
External dimensions	> ø 100 mm	ø 80÷100 mm	ø 60 ÷80mm	ø 50÷60 mm	ø 48÷50 mm
Reliability	< 1000 km	1000÷5000 km	5000÷7500 km	7500÷10000 km	>10000 km
Wear resistance and durability	Extremely poor	Poor	Average	Good	Extremely good

Table 15 - Correlation Chart

Going on with the VDI 2225 analysis, a weight is given to the most important parameters and, starting from Table15, the final **VDI 2225 matrix** is retrieved.
All the parameters are one-by-one considered for all the concept variants and the corresponding VDI 2225 points are given. Starting from this table it will be possible to calculate and evaluate the technical and economic ratings and the final overall rating of each concept.

No.	Parameter	W	Concept variants			
			1	2	3	4
1	Ease of maintenance	3	2	4	0	4
2	Driver Tunability	3	4	4	4	4
3	Cost Efficiency	1	3	2	4	2
4	Fail Safe	2	4	4	3	3
5	Performance	4	2	4	1	2
6	Magnetic field strength	1	4	4	4	3
7	Ease of manufacture	1	3	3	4	1
8	Lightweight	3	3	4	1	0
9	External dimensions	3	1	4	2	2
10	Reliability	3	3	4	1	4
11	Wear resistance and durability	2	2	4	2	3

Table 16 - VDI 2225 Matrix

Some examples about how to read this table can be given:

- Driver Tunability: equal shared among all the concepts because of the "infinite" tunability of the MR technology.
- Performance: concept n.2 is the best due to the monotube design, coils inside the piston and very precise feedback given by the loadcells. Concept n.1 is similar, but it has less precise feedback due to the potentiometer.
  On the contrary, concept n.3 has the lowest value because of the twin tube design, while concept n.4 is negatively influenced by the external coils.
- Cost Efficiency: concept n.4 has load cells and external solenoid (that turns into a generally bigger component with more material needed and more difficult assembly phase). This brings to a highly expensive solution and VDI Points equal to 2.
  Vice-versa concept n.3 is using the twin tube, that is widely used in the automotive industry also thanks to its low price (hence the given VDI Points are 4).
- Reliability: Concepts n.2 and 4 have the highest reliability because of the monotube design and load cells. Concept n.1 has slightly lower reliability for the potentiometers that can wear, damage or misalign more easily.
  On the other hand, concept n.3 can encounter problems with contamination and cavitation, bringing to lower predicted mileage and resulting in the lowest VDI Score of 1.

7.7. Variants Evaluation

To correctly evaluate all the found variants, it is important to determine the overall value of each possible solution. This can be calculated starting from the weights given to each factor and previously explained in VDI Matrix (Table16).

7.7.1 VDI 2225 Procedure

As anticipated in Paragraph 7.5, the VDI 2225 procedure is the one chosen to evaluate both technical and economic criteria, and it considers several ratings.
The so-called absolute rating (R_j) is defined as the summation of all the sub-values when evaluation criteria are at least approximately independent; more, it can be evaluated both considering and not considering the weights of the factors.

Several values play a key role in the final formula shown below, specifically:

- v_{ij} = assigned value to a certain parameter (without considering its weight), hence VDI Points
- w_i = weights to be assigned to each factor
- V_{max} = maximum possible achievable value 44
- n = number of considered parameters

$$\mathrm{URj} = \frac{\sum_{i=1}^{n} (v_{ij})}{v_{max} * n}$$

Equation 8 - Unweighted Rating

$$\mathrm{WRj} = \frac{\sum_{i=1}^{n} (w_i * v_{ij})}{v_{max} * \sum_{i=1}^{n} (w_i)}$$

Equation 9 - Weighted Rating

Considering the VDI 2225 procedure for a complete evaluation of the alternatives, two other fundamental ratings must be considered.
They are both again computed exploiting the previously explained Table16 and are:

- R_t = **technical rating**. It is calculated using the previous formula (both as weighted and unweighted) but keeping into account only the evaluation criteria related to technical characteristics, such as:
  - *Driver tunability*: coming directly from the active suspensions' characteristics
  - *Fail safe*: preserving a bit of functionality even after failure
  - *Performance*: handling and comfort needed during the race
  - *Magnetic field strength*: managing the correct damping value
  - *Lightweight*: reaching high strength and performance with the minimum possible weight
  - *External dimensions*: managing the correct fit into the car
  - *Reliability*: ensuring mileage and safety
  - *Wear resistance and durability*: to reduce dangers and costs
- R_e = **economic rating**. It is calculated using its specific formula, where C_{min} is the lowest price between all the considered variants.

  $$R_e = \frac{C_0}{C_{variant}} = 0.7 * \frac{C_{min}}{C_{variant}}$$

  Equation 10 - Economic Rating

  As for the technical rating, the Re rating considers only the evaluation criteria related to economic characteristics, such as:
  - *Ease of maintenance*: corresponding to quicker and cheaper maintenance operations
  - *Cost efficiency*: performance can justify a slightly higher price
  - *Ease of manufacture*: bringing to quicker and easier process

Finally, the overall rating (R) can be calculated following two different strategies:

- The **straight-line method**:

  $$\mathrm{R} = \frac{R_t + R_e}{2}$$

  Equation 11 - Straight-Line Overall Rating

- The **hyperbolic method**, that is preferrable when a balance between large rating differences must be done:

  $$\mathrm{R} = \sqrt{R_t * R_e}$$

  Equation 12 - Hyperbolic Overall Rating

	TECHNICAL RATING		ECONOMIC RATING	OVERALL RATING	
	Unweighted UR_t	Weighted WR_t	R_e	Straight R	Hyperbolic R
Concept 1	0.70	0.66	0.59	0.63	0.62
Concept 2	0.93	0.97	0.40	0.69	0.63
Concept 3	0.59	0.48	0.70	0.59	0.58
Concept 4	0.64	0.65	0.37	0.51	0.49

Table 17 - Final Ratings

The results can be summarized in the table below, where it is easy to notice that, according to the Overall Ratings, the preferrable variant is concept n.2, even if some weak points can still be present and will be detected in Paragraph 7.8.

7.7.2. Strength Diagram

Another visualization that can be useful in the variants' choice is given by the so-called strength diagram. Its aim is to put into relation the two fundamental ratings Rt and Re, both in the weighted and unweighted case.

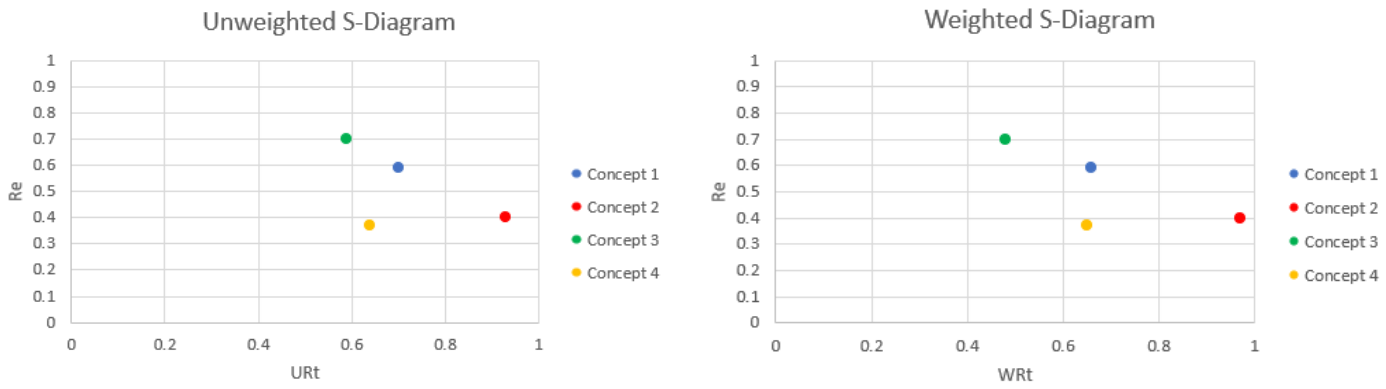


Figure 30 - S-Diagrams

From these two graphs some easy results can be immediately seen:

- Concept 3 is better from the economic point of view, with the highest economic rating and lowest estimated price. This option, as said above, is the one concerning the twin tube solution that is adopted for a variety of everyday cars, resulting in a more competitive price at the expense of performance.
- Concept 2 has the highest technical rating both considering the unweighted and weighted case. This, as previously explained, is due to the adoption of a monotube with internal coils and feedback provided by a precise loadcell leading to higher performance.

The same visualization can be achieved for the hyperbolic calculation and the same conclusions are straightforwardly reached.
The further "upper-right" a hyperbole is, the better solutions lying on that curve will be; this is due to the fact that hyperboles are iso-lines where the product of the ratings is constant, so at a greater k value is corresponding a better curve.
Consequently, it can be noticed that Concept 2 (red dot) is the one lying on a hyperbole with a bigger k with respect to all the other alternatives, so with a $k \approx 0.40$ instead of $k<0.35$ as in all the other cases.

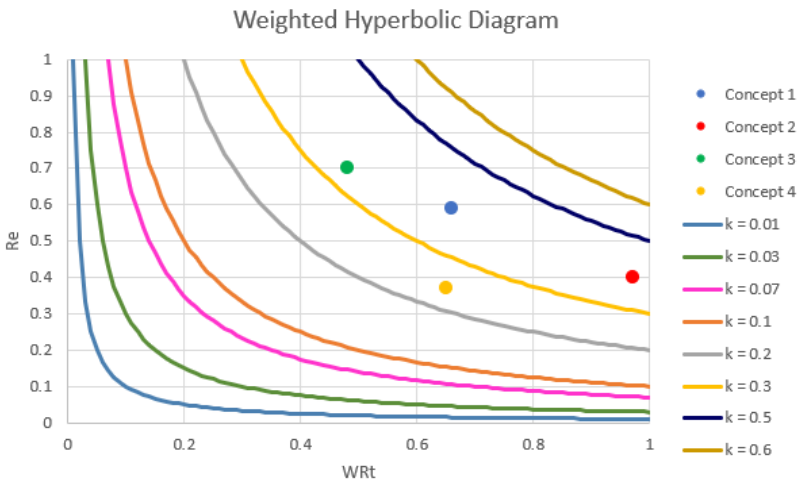


Figure 31 - Hyperbolic Diagram

7.8. Search for Weak Spots

In all the variants considered, also the most promising one, some weak spots can be detected.
It is fundamental to detect weaknesses of the chosen concept (and more generally of the solutions with higher overall rating) before going on with the following steps because big problems can arise when defining the embodiment design if weaknesses are well-definite and established.
Here, the search for weak spots is done considering the two variants with the highest overall rating from Table17 and it is fundamental to find out if the chosen solution is really the one not presenting weak spots, otherwise other solutions must be reconsidered.
In particular, in order to give a proper representation of the balance of the evaluation criteria profile of the 2 most evaluated concepts, the weights of the overall weighted rating $(W R_j)$ are expressed in terms of percentage of importance. In this way it is possible to compare each value assigned to the evaluation parameters with the global rating of the concepts.
The results, shown in Figure32, is the profile of the evaluation criteria related to each concept under investigation, that allows to find out possible weak spots.

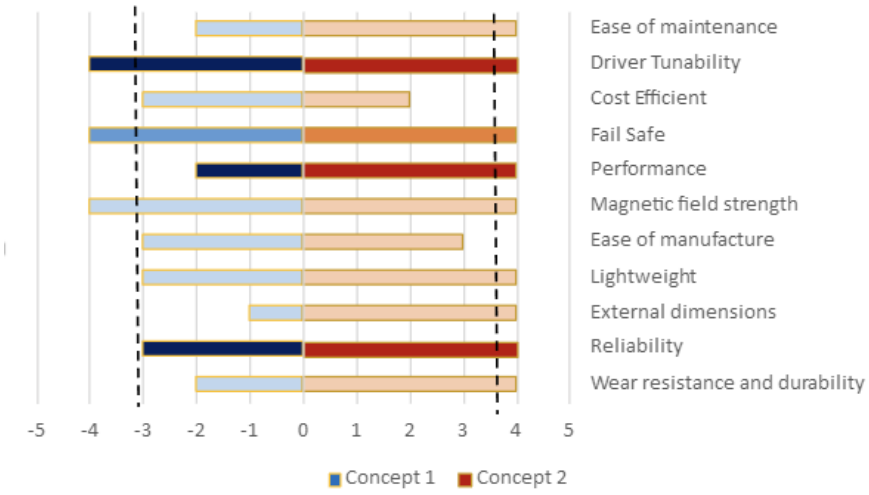


Figure 32 - Weak Spots Evaluation

Few general considerations can be made to better understand the figure:

- The higher the weight, the higher the color intensity of each block.
- The lower the width of the block, the lower its rating.
- The most balanced concept is the one getting nearest to the dashed line, and the one having a smaller variance between the different evaluation criteria.

As Figure32 shows, even though the two concepts differ for one component only, the evaluation criteria profile is strongly different. This is due to the different working principle used to provide the damper feedback: the load cell indeed is a more compact and consistent solution with respect to that provided by the potentiometer, since, perfectly fitting on the cylinder, overcome the reliability issues while keeping the encumbrance of the standard damper.

From the plot, it is easy to see that the most rated solution, resulting from the analysis performed so far, is also the one characterized by a more uniform and less discontinuous profile, hence the previously chosen Concept 2. Moreover, for this solution the most weighted criteria are more balanced with respect to the target line, thus suggesting being less prone to problems during the following development.

All these observations can be furtherly supported by an analytical investigation of the variance. Indeed, this statistical index, provide a measure of how much a set of values move away from a mean value, hence it works as measure of dispersions of the data. This means that the lower the variance, the lower the dispersion.

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n}}$$

Equation 13 - Variance

Relating these concepts to our weak spot analysis, the Concept 2 show a much smaller variance with respect to the one of Concept 1, as listed here below:

- $\sigma_1 = 0.93$
- $\sigma_2 = 0.41$

This supports the qualitative analysis performed so far.

8. Embodiment Design

After all the steps related to the conceptual design phase (entire Chapter7), the proper "Embodiment Design" can be executed. During the embodiment only the winning concept is considered and carried on determining precise dimensions, materials and processes, hence defining at first a preliminary and then its definitive layout. This is completed together with a final check for errors and faults, with safety and warning advice and with the analysis of eventual auxiliary functions.

During the embodiment phase, design is developed in accordance with the found technical and economic criteria to underline all the details that starting from the concept can bring to the final production. These steps are therefore devoted to optimize and complete form designs for shapes, materials, processes and to prevent predicted possible faults and disturbing factors.

8.1. Embodiment Flowchart

As for the conceptual design, some steps are followed during the embodiment too.
From the chosen concept, the embodiment-determining requirements together with the main function carriers regarding key points as safety, assembly and recycling should be identified and related to size, arrangement, material. That's why simple scale drawings are created to clarify all the special constraints that must be respected (as clearances and installation requirements). In this way a first **preliminary layout** is created. When all the major and key requirements are fulfilled, the search for eventual auxiliary functions can start leading to a new integrated version of preliminary design.
Once reached and defined the preliminary layout, form designs should be optimized as much as possible, with particular attention to possible faults and problems, hence exploiting tools as the fault-tree analysis to better understand the impact of all the possible arising misfunctions. This leads to the **definitive layout** definition and to the final draw up of parts list and processes.

8.2. Layout and Sizing

To determine the overall layout of the MR damper, the three main guidelines have been adopted:

- **Clarity** removing all the ambiguous relations between subfunctions and subsystems design, hence between fulfilled functions and their relative inputs and outputs, creating a specific cause-effect link. This brings to an easy performance estimation since for example loads can be explicitly calculated under all the operating conditions.
- **Simplicity** adopting the minimum number of needed components and giving them a shape that is as simple and as regular as possible. This brings to a time-saving production and, more interesting for the topic of this project, maintenance since during races parts substitutions should be easy and quick.
- **Safety** relating this key driver to both humans and environment. This is carried on exploiting different principles related to both direct and indirect safety, including fail-safe that has a crucial role on the design choices of this project. Safety principles come usually together with all the specific risk evaluations, where the most spread is the so-called fault-tree analysis.

In this paragraph, starting from the winning concept, all its main dimensions and constraints are defined, and they must be respected by the final assembly.
Here below, as anticipated in Paragraph 8.1, a refined but still simple schematic of the chosen variant is shown in Figure33.

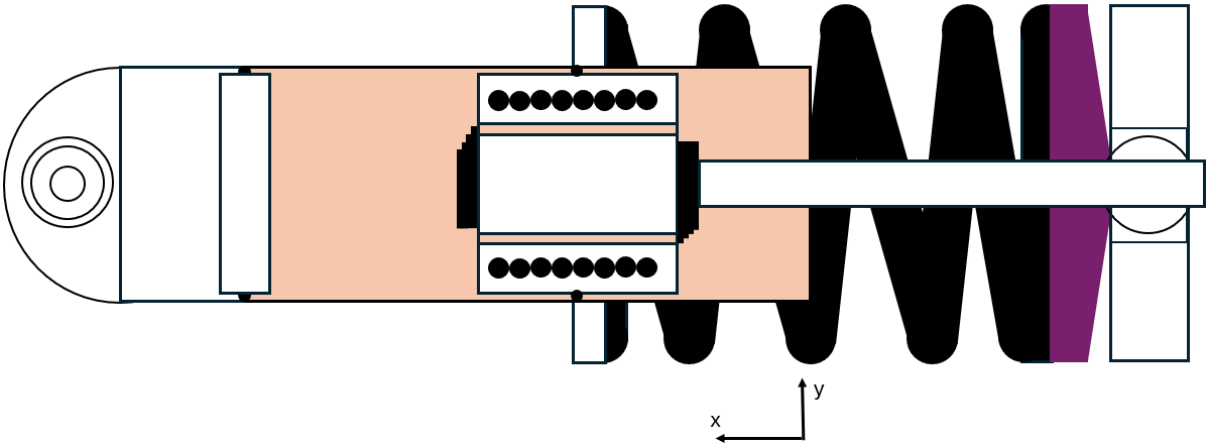


Figure 33 - Shock Absorber Final Version Schematic (and reference system)

8.2.1. Case Sizing

Hardpoint locations and suspension kinematics give the external dimensions of the shock absorber.

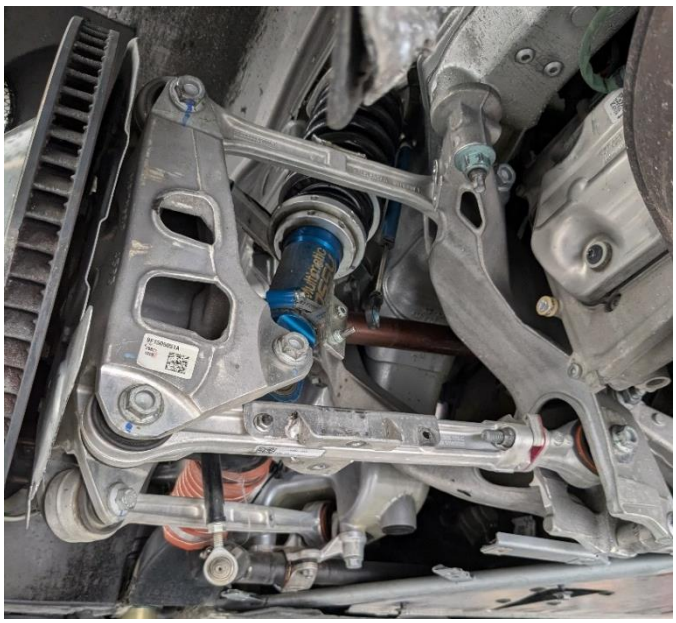


Figure 34 - Actual suspension Schematic (reference car)

The fully extended length accounts for the maximum extension during rebound travel. In this position, both helper and main springs are fully extended, and the piston is in its top position inside the chamber.

Vice-Versa, the fully compressed length happens during maximum bump motion. In this position both springs are fully compressed.

For the embodiment design, it is very important that the shock absorber is able to successfully link both hardpoints on the upright and on the chassis. On the other hand, the external dimensions should not interfere with the piston travel during bump motion. After such assessment, it is possible to estimate the maximum and minimum dimensions of the assembly and the case. Moreover, from the dimensions of the two springs, it is possible to determine the location and the length of the case's thread. It is worth noting that the length of the threaded part is a very important feature of the shock absorber because it the only tool for ride height adjustments in a GT car.

Furthermore, there are some constraints in the y direction too, always within the frame of reference depicted in Figure33. The maximum cylindrical envelope of the assembly is given by the free space left in between both the hole inside the upper wishbone and the chassis fixing point of the shock absorber. Moreover, it is very important to consider all the dimensions in the surroundings of the damper for mechanic's accessibility and ease of work, as described in more detail in the Paragraph 8.6.2 related to warnings and user safety.

The maximum dimension of the damper case is given by the springs internal diameter. Therefore the dimensions of the ring gears should be in between the dimensions of the case and the assembly, or in other words, as the thickness of the two springs plus some clearance for the cylinder.
The cylinder minimum diameter is given by its internal components. It should be able to fit the piston that in turn has to locate the coil. The following table shows the aforementioned dimensions.

Damper body free length [mm]	731
Damper body fully compressed dimensions [mm]	646
Case length [mm]	480
Main spring length [mm]	151
Main spring outer diameter [mm]	93
Main spring coil diameter [mm]	15
Helper spring length [mm]	73
Helper spring outer diameter [mm]	82
Helper spring thickness [mm]	8
Thread positioning from x0 [mm]	129
Thread length [mm]	55
Cylindrical max envelope [mm]	100
Ring gear diameter [mm]	86
Case external diameter [mm]	52
Case thickness [mm]	3

Table 18 - Case sizing

8.2.2. Hydraulic components: "X-Axis" Sizing

The hydraulic components of the damper in the x-direction are listed in the following table. The frame of reference is the one depicted in Figure33.

Main piston height [mm]	53
Oil chamber minimum length [mm]	150
Second piston height [mm]	15
Rod length [mm]	352

Table 19 - Hydraulic components, x-direction sizing

8.2.3. Hydraulic components: "Y-Axis" Sizing

The hydraulic components of the damper in the y-direction are listed in the following table. The frame of reference is the same depicted in Figure33. The rod internal diameter allows for the two supply wires channelling (this aspect will be described more precisely in Paragraph 8.4).

Main piston diameter [mm]	45
Rod external diameter [mm]	12
Rod internal diameter [mm]	2.5
Second piston diameter [mm]	45

Table 20 - Hydraulic components, y-direction sizing

8.2.4. Gear Rings, Plates and Auxiliar Elements

The auxiliar elements are dimensioned in the Table21 below. Those elements are all the plates required in the shock absorber assembly, from top to bottom:

- Top mount: it connects the shock absorber to the chassis.
- Upper plate: it is the housing for the bumpstop, it is in between the load cell and the upper plate. The top part has to be conical to allow for the three rotational degrees of freedom given by the spherical joint connection.
- Helper plate: it separates the main spring from the helper spring. It provides y-direction constraint for both springs.
- Upper and Lower rings: they are located below the helper spring. They are both internally threaded and their movement allows for car's ride height regulation. Two gears are required to provide locking in x-direction.

Top mount thickness [mm]	9
Top mount bolts positioning	As prescribed from the chassis hardpoint of the suspension
Top mount central height	35 mm to accommodate the Unibal NMB ABYT12
Upper plate base thickness [mm]	10
Bumpstop housing depth [mm]	13
Upper plate external diameter [mm]	95
Upper plate internal diameter [mm]	11
Helper plate thickness [mm]	3
Helper plate external diameter [mm]	94
Helper plate internal diameter [mm]	58
Upper ring external diameter [mm]	86
Upper ring internal diameter [mm]	54
Upper ring thickness [mm]	10
Lower ring external diameter [mm]	70
Lower ring internal diameter [mm]	54
Lower ring thickness [mm]	10

Table 21 - Auxiliar elements dimensioning

8.2.5. Unibal Sizing

When dealing with end eye Unibals, the most important parameters to consider are the loads and the dimensions. Being the shock absorber fit in between a pair of Unibals, it behaves as a rod which is subjected just to axial loads. From the conceptual design phase, the target load is 12600 N. As for the dimensions, the most important one is the internal bore diameter of the Unibal that should accommodate with clearance fit its respective pin.

The lower one is the bolt that connects the upright to the damper, while the upper one is the damper shaft itself.

Furthermore, the external diameter is constrained with interference fit by the hole in the case and the top mount for the lower and upper Unibals respectively.

The last dimension is the height, that is given by case and top mount thicknesses. Being the liner of the bearing made from polymeric material, the maximum allowed temperature has to be within the specified target of 150 °C. The very next step is to navigate through the catalogues in order to find the best match in between the offered products. Both of the following solutions match perfectly the dimensional constraints and overcome by far the load requirements. The upper one has an axial static strength of 38698 N, while the lower one has a radial strength of 231200 N.

This is a make or buy decision, better explained in Paragraph 8.11.1, where all the elements inserted in the final BOM, hence all the elements composing the MR suspension are analyzed to correclty choose what to internally produce, so considering resources and time to develop new specific production lines, and what to find already available on the market to match the wanted performance. Here below, the needed dimensions together with the target load are better shown for clarity.

Upper Unibal		Lower Unibal	
Max axial load*SF	12600 N	Max radial load*SF	12600 N
Internal bore diameter	19 mm, clearance fit	Internal bore diameter	16 mm, clearance fit
Outer diameter	39 mm, interference fit	Outer diameter	35 mm, interference fit
Height	33 mm	Height	31 mm

Table 22 - Unibals Sizing

As previously written in this paragraph, there are unibals perfectly fitting the MR suspension needs that are already on the market. The choice, after a selection looking at several catalogues, has fallen upon the NMB ABYT12 as upper unibal and upon the HAB-M16TG-1 as lower unibal.
Their technical representation, together with their fundamental technical data can be summarized as in the below Table23.

	NMB ABYT12 (Upper)	HAB-M16TG-1 (Lower)
Bore	0.750" (19,05 mm)	16 mm
Outside Diameter	1.5625" (39,69 mm)	35 mm
Ball Width	1.280" (32,51 mm)	30.5 mm
Race Width	0.620" (15,75 mm)	14 mm
Ball Diameter	1.375" (34,93 mm)	31.75 mm
Shoulder Diameter	0.970" (24,64 mm)	21.5 mm
Misalignment α	18°	21°
Fit/no load Breakaway Torque	1-5 In-Lbs	Preloaded Fit
Radial Static Limit Load	54800 lbs (243750 N)	231200 N
Axial Static Limit Load	8700 lbs (38698 N)	
Dynamic Oscillating Radial Load	24200 lbs (107642 N)	
Approximate Weight	0.32 lbs (143 g)	109 g
Max Operating Temperature	151°C (250°F)	163°C (325°F)
Alternative Part-number	ABYT12(R)	
Manufacturer	NMB Minebea	Aurora Bearing Company
MATERIALS		
Ball	CRES 440C stainless steel	Alloy steel, heat treated, hard chrome plated
Race	CRES 410 stainless steel, heat treated	17-4 PH stainless steel, heat treated
PTFE Liner	PTFE/Dacron (or glass fibre)	AT3200 (PTFE/Nomex laminate)

Table 23 - Unibals Technical Sheet

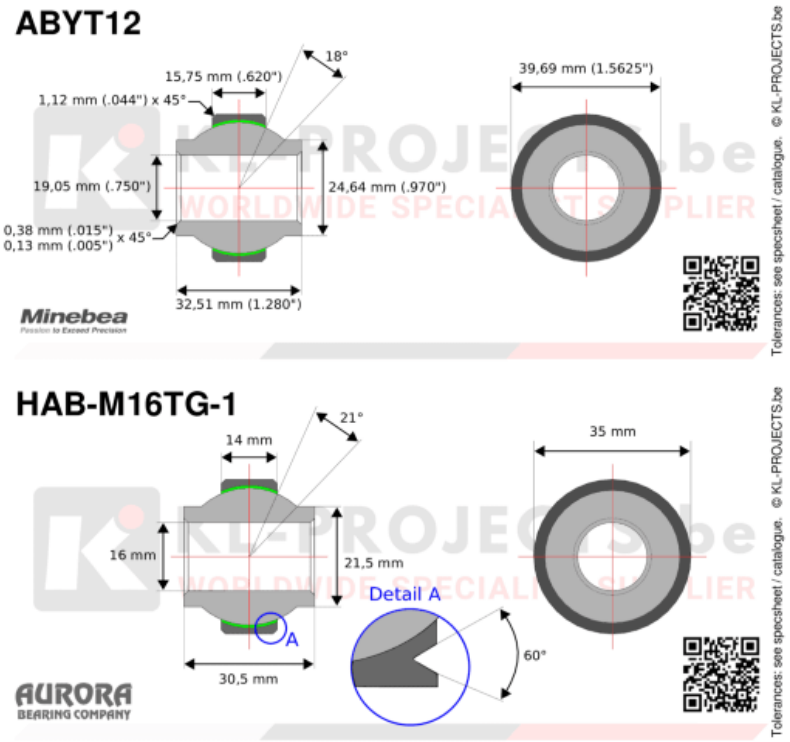


Figure 35 - Upper (above) and Lower (below) Unibal

8.2.6. Bump stop Choice

The bump stop is a fundamental component in the shock absorber because it avoids the end run of the suspension when fully compressed that can lead damage to piston, shims, oil chamber, electrical components and the spring. Secondly because its tuning can strongly affect the handling of the car. The bump stop is made by elastomeric material and its stiffness is not linear like a spring, but it usually increases parabolically with displacement. The right bump stop for our suspension has the following characteristics and it comes from the Porsche Catalogue.

Exactly as in the unibals case, a make or buy analysis better explained in Paragraph 8.11.1 resulted in a buy decision where the reference from Porsche Catalogue is BUMP STOP 9F1412303B with the following technical data.

Height [mm]	23
External diameter [mm]	40
Internal diameter [mm]	13
Material	Microcellular polyurethane elastomer MPU

Table 24 - Bump Stop Sizing

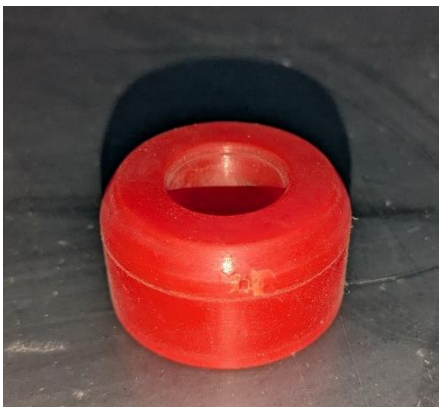


Figure 36 - Bump Stop (Reference Car)

8.3. Shims and orifices

The designed product of this project does not rely only on MR technology, but it exploits also the more traditional shims to produce damping even if the electric system fails, as described in the Fail-Safe principle in Paragraph 8.6.
The device is made by four orifices, two in each flow direction. Having just one orifice per flow direction would make an imbalance around the piston axis, therefore creating a bending moment and more stresses in the structure (more detailed stress analysis in Paragraph 8.5). This issue is avoided by employing a total of four holes, symmetrically distributed.
The orifice diameter is a very important parameter to define the damping properties. The goal now is to define the hole diameter and the shim stiffness. The damping obtained by those parameters has to be not to as strong as the overall maximum damping target, otherwise the MR technology would not be of any help. Therefore, this force has to be set as the minimum in the damping range in order to let the electric system tune the force as desired.

As described in the following chart (Figure38), blue lines are the wanted damping curves coming from the orifice and shims, namely Reduced_Bump and Reduced_Rebound. The grey dashed lines represent the overall damping range as set on the targets. The red curves are the reference setup for handling performance when the MR fluid is activated.

8.3.1. Hole Area Design

The design process for the orifice area is very complex, therefore the Simulink model presented in Paragraph 7.3.1 (namely Figure16) has been used to get damping curves as the ones in the described above Figure38.

Hole area Bump motion [m²]	2e-6
Hole area Rebound motion [m²]	1e-5

Table 25 - Hole Area Sizing (bump and rebound)

8.3.2. Shim Stiffness Design

The physical principle behind the shims is the cantilever beam, whose deflection is:

$$x = \frac{FL^3}{3EI}, \qquad I = \frac{bh^3}{12}$$

Equation 14 - Beam Deflection

From the blue curves of the Figure38 chart it is possible to retrieve the maximum forces that the shims will experience, both in bump and in rebound:

F_{bump} [N]	2661
$F_{rebound}$ [N]	1593

Table 26 - Max Force

The deflection of the shims at this stage of damping are expected to be fully opened, therefore as the hole diameter:

x_{bump} [mm]	1.6
$x_{rebound}$ [mm]	3.6

Table 27 - Displacement

The shims are expected to cover the whole orifice, therefore their half-length is as the distance between the furthest point of the hole and the piston axis:

L_bump [mm]	12.75
L_rebonud [mm]	11.75

Table 28 - Shim Half Length

The base length of the shim is set to let free the other two holes as shown in Figure37. In other words, this is made to let the system work as a proper check valve, therefore the bump valves will just cover the bump holes and let free the rebound holes and vice versa.

b_bump [mm]	20
b_rebound [mm]	18.5

Table 29 - Shim Width

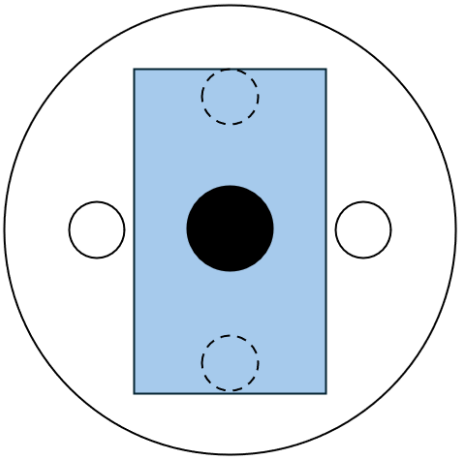


Figure 37 - Top View of the Bump Shim

The final formula for the shim height calculation shows like:

$$h = \sqrt[3]{12\frac{(F/2)L^3}{3Ex}}{b}$$

Equation 15 - Shim Height

By employing steel (E = 210 GPa), the two shim heights are:

h_{bump} [mm]	2.8
$h_{rebound}$ [mm]	1.4

Table 30 - Shim Height Results

SUMMARIZING

The following table (Table31) summarizes the designed parameters of the shims that are previously independently shown from Table28 to Table30.

Parameter	Bump Shim	Rebound Shim
Length [mm]	25.5	23.5
Width [mm]	20	18.5
Height [mm]	2.8	1.4
Material	Steel	Steel

Table 31 - Design Parameters for the Shims

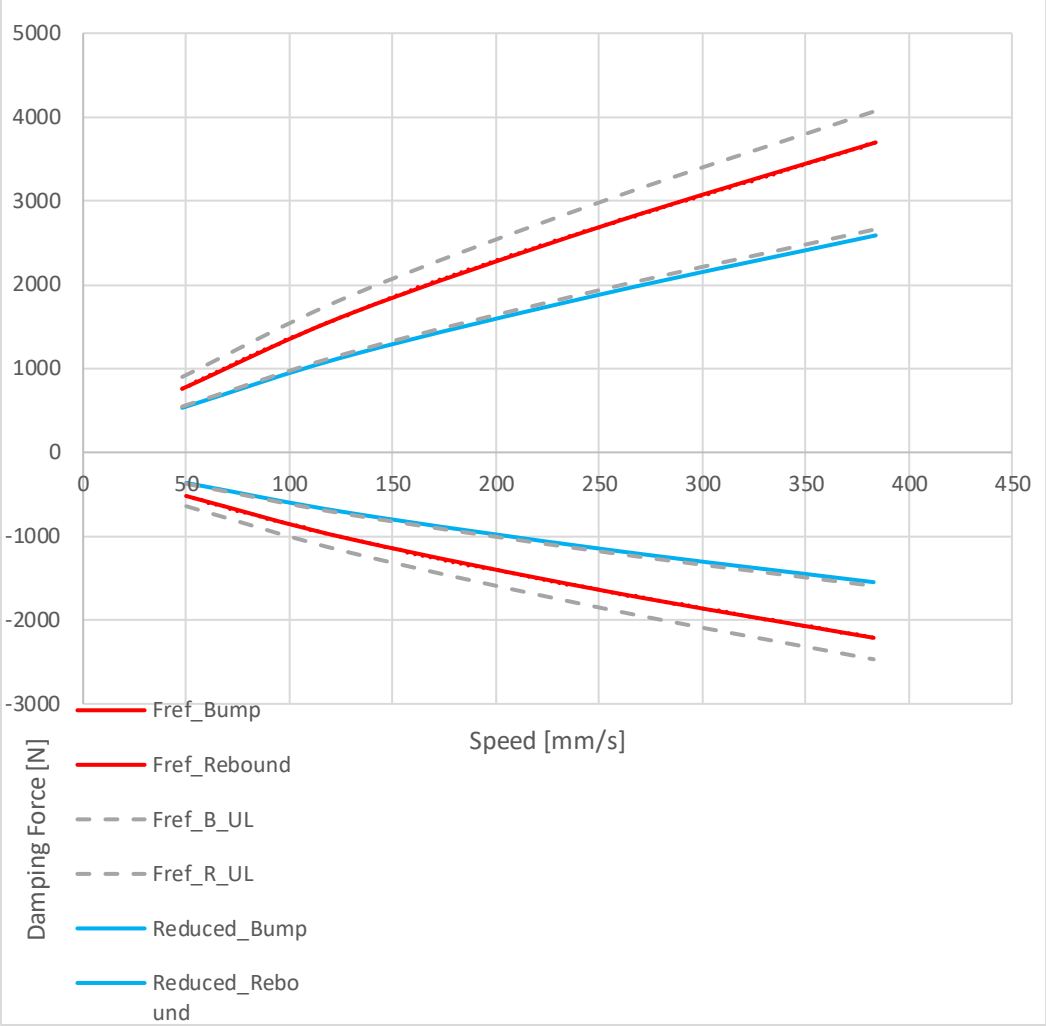


Figure 38 - Target and Reached Damping Curves

8.4. Coil and Wiring Design

The wire has to be big enough to carry 2 A of current intensity without melting or damaging itself. For this reason, a table is used as reference for the choice of the right wire diameter. The chosen reference is the "American Wire Gauge (AWG) Conductor Size Table" where the higher the AWG number, the smaller the physical size of the wire. Specifically, the used table can be found at: https://www.solaris-shop.com/content/American%20Wire%20Gauge%20Conductor%20Size%20Table.pdf?srsltid=AfmBOopuOI5es7eZvgbwXHbK-D2fjxGGVqkNav2zsaC6DQ5BXs9Cx1Wy.

The chosen one is the size 18 of the AWG system and here below all the needed dimensions taken from the AWG table are displayed.

AWG	18
Diameter [mm]	1.024
Max Current Supply [A]	2.3
Max Supported Frequency [kHz]	17

Table 32 - Wires Parameters

The coil turns are arranged one on top of the other, with no overlapping in order to be as close as possible to an ideal solenoid. Therefore, it is possible to check if the coil length exceeds piston's height.

$$50\ turns \cdot 1.024mm = 51.2\ mm$$

Equation 16 - Coil Length

The coil will be housed inside the piston, inside a groove, just outside the orifices. The solenoid is surrounded by a polymeric resin for two main reasons: to provide electrical insulation and to successfully lock the coils in its position.
The wires have two functions: connect the control unit to the coils and provide the electric supply. Those wires go all the way inside the hollowed rod, see Paragraph 8.2.3. Once the piston is reached, the wire goes through two channels of diameter ø1.5mm and it is then wrapped inside the groove.

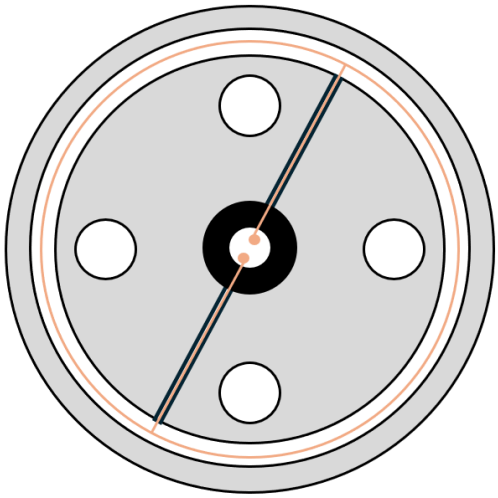


Figure 39 - Top View of Piston Grooves and Wire Channels

8.5. Structural Load and Material Thickness

As anticipated in Paragraph 8.3, the rod resistance must be verified only in in case of *pure axial force* since the chosen layout of the orifices is related to put symmetrical valves on the piston so that, consequently, there are not bending forces.
In this paragraph both the designed rod and case will be verified, splitting the structural calculations in the two fundamental cases: bump and rebound.

8.5.1. Rod verification

The bump and rebound forces, as outlined in Table26 of Paragraph 8.3.2., are defined as follows:

- Bump Load: F_{bump} =2661 N
- Rebound Load: $F_{rebound}$=1593 N

The rod has a diameter of **14 mm**, so now the objective is to verify whether this dimension is sufficient to withstand the applied loads.

The first step is to go through **material selection** and two candidates are considered for the rod:

- Ergal, with σ_{amm} = 450 N/mm^2
- High Strength Stainless Steel (HSSS), with σ_{amm} = 900 N/mm^2

Then, it is possible to finalize the **stress calculation** for both bump and rebound scenarios. The axial stress on the rod is calculated using the standard formula:

$$\sigma = \frac{4F}{\pi d^2}$$

Equation 17 - Axial Stress

As a consequence, substituting the given values, the following stresses are retrieved:

- Bump Scenario: σ_{bump} = 17.29 N/mm^2
- Rebound Scenario: $\sigma_{rebound}$ = 10.35 N/mm^2

Finally, the calculated stresses are compared with the allowable stress of each material, hence performing the **material verification**.

- Ergal: σ_{amm} (Ergal) = 450 N/mm^2 >> 17.29 N/mm^2 > 10.35 N/mm^2 **SATISFIED**
- HSSS: σ_{amm} (HSSS) = 900 N/mm^2 >> σ_{amm} Ergal **SATISFIED**

Therefore, it is possible to conclude that both materials are more than capable of withstanding the applied stresses.

To further validate the rod design, the **minimum diameter required** to withstand the worst-case load (using Ergal) is calculated using the inverse stress formula

$$d_{min} = \sqrt{\frac{4F}{\pi \sigma_{amm}}}$$

Equation 18 - Minimum Required Diameter

The worst-case scenario is the one where the force applied is the bump force and the chosen material is Ergal. After the d_{min} calculation, it is easy to find out that the minimum allowed diameter is:

- d_{min} = 2.74 mm

The current rod diameter of 14 mm far exceeds this minimum requirement, ensuring a significant safety margin.

8.5.2. Case verification

The cylinder case is verified only considering the bump scenario, as it represents the most demanding condition.

First, the **internal pressure on the piston** generated by the bump force is calculated as:

$$p = \frac{F_{bump}}{A_{piston}}$$

Equation 19 - Internal Pressure on Piston

Where the piston area is simply calculated as:

$$A_{piston} = \frac{\pi * {d_{piston}}^2}{4}$$

Equation 20 - Piston Area

And, substituting all the values, the found pressure is:

- p = 1.67 N/mm^2

Going on, the **hoop stress on the cylinder walls** is calculated using the formula:

$$\sigma_{hoop} = \frac{p \cdot r}{e}$$

Equation 21 - Hoop Stress on Cylinder Walls

Where:

- e = wall thickness
- r = mean radius $= (\frac{Dext}{2} - \frac{e}{2}) = 23.25\ mm$

Substituting the values:

- σ_{hoop} = 25.87 N/mm^2

As in the previous case, **material verification** must be done.
The case material, Ergal, has the previously mentioned allowable stress of $\sigma_{amm} = 450\ N/mm^2$, so:

- σ_{amm} (Ergal) = 450 N/mm^2 >> σ_{hoop} = 25.87 N/mm^2 **SATISFIED**

The verification is fully satisfied, ensuring that the damper walls can safely withstand the internal pressure.

To **conclude**, the analyses conducted for both the rod and the cylinder case demonstrate that the current design exceeds the required safety margins under the most demanding load conditions. This robust design ensures reliability and structural integrity, making the damper suitable for high-performance applications such as the Pikes Peak race.

8.6. Safety and Fail-Safe Principle

As anticipated in Paragraph 8.2, safety is always crucial and must be carefully evaluated especially when dealing with a new product and a new design. Safety must be both direct, so developing solutions precluding danger, and indirect, hence building special protective systems, and must be devoted to the protection of operators, customers and environment.
A lot of principles are related to safety, as the "safe-life principle" that is used when an arising fault can cause very serious and dangerous problems or the "redundancy principle" that provides multiple or reserve elements in order to have repetitions in case of failure.
For this project's purpose, the most useful principle is the so-called "**fail-safe principle**" that consists in having a small functionality preserved to prevent dangers even when a malfunction occurs. This means that no big consequences are expected after a failure during the service life and, consequently, unexpected faults are strongly reduced.
Talking about MR suspensions, the fail-safe principle is linked to the presence of shims. As described in Paragraph 7.3.1, the used check valves allowing the fluid passage are equipped with shims, whose stiffness tuning determines the damping curve. In case of failure of the electronic control system on which the MR damper relies on, the shims are still present and are still able to provide a bit of damping. This turns the MR suspension into a "traditional" passive suspension: performance will not be anymore as high as the one of the fully functioning system, but the functionality related to damping is so far preserved. Handling "in primis" and then comfort are consequently a bit conserved. This is crucial during races because a sudden malfunction can be really dangerous leading the driver to loose the car, hence it is fundamental specifically during an exhausting climb with bad road surface and hairpins, as the Pikes Peak is.

8.6.1. Fault-Tree Analysis

A fundamental part when dealing with safety principles is the evaluation of the possible faults and all the possible disturbing factors. The main goal is to minimize the number of potential faults. This analysis is only carried on for the most important and critical areas that is this case are all related to the MR fluid.
As widely previously stated, MR suspensions are exploiting fluids whose viscosity can be rapidly changed by applying a magnetic field. However, while MR dampers offer significant advantages in terms of controllable viscosity, there are several potential reliability problems that can arise.

Here below, a list of critical faults related to the MR fluid is provided together with their possible impact on the whole MR system.

- **Magnetic field sensitivity**: MR suspensions rely on the presence of a magnetic field to alter their viscosity. Variations in the intensity or direction of the applied magnetic field can affect the performance and consistency of the suspension.
    - **Impact:** Inconsistent or fluctuating magnetic fields can result in unpredictable behaviour of the suspension, reducing the reliability of systems.
- **Sedimentation**: MR fluids consist of fine magnetic particles suspended in a carrier liquid. Over time, especially under low or no magnetic field conditions, these particles can settle or clump together, reducing the effectiveness of the suspension.
    - **Impact:** The settling of particles can lead to poor responsiveness to changes in the magnetic field and reduced overall performance, particularly in long-term use.
- **Wear**: magnetic particles can cause wear and abrasive damage to the components they come into contact with, especially under high shear rates or in poorly maintained systems.
    - **Impact:** continuous wear can lead to the degradation of system components (e.g., seals, valves), reducing performance, sealing, lifespan and reliability of the system.
- **Viscosity drift over time**: viscosity of MR fluids can degrade over time due to factors such as particle degradation, exposure to temperature variations, and chemical reactions between the particles and the carrier fluid.
    - **Impact:** gradual decrease in the fluid's responsiveness and a loss of performance, making long-term reliability a concern.
- **Temperature sensitivity**: MR fluids are temperature-sensitive, and elevated or fluctuating temperatures can alter their performance. High temperatures can cause the carrier fluid to thin, while low temperatures can cause it to become too viscous.
    - **Impact:** temperature changes can cause inconsistency in the suspension's behavior, reducing the efficiency of systems that rely on precise control of the MR fluid.
- **Leakages and contamination**: MR fluids are susceptible to contamination from external particles, moisture, or air, especially if the system is poorly sealed or maintained.
    - **Impact:** contaminants can affect the fluid's rheological properties, leading to poor performance or even failure of the system. Leakage of MR fluids can also lead to loss of functionality and contamination of the environment.
- **Aging**: over time, the chemical stability of MR fluids may degrade, particularly if they are exposed to aggressive environmental conditions such as UV light or chemicals.
    - **Impact:** aging can lead to changes in the composition of the fluid, such as particle aggregation or the breakdown of the carrier fluid, leading to lower performance and reliability.
- **Electromagnetic interference**: systems using MR fluids can be prone to electromagnetic interference, which can affect the control electronics or other nearby devices.
    - **Impact:** electromagnetic interference can cause malfunctions in the systems, leading to performance instability, poor and slow control and potential failure.
- **System calibration and control**: performance of MR suspensions is highly dependent on precise control of the magnetic field.
    - **Impact:** inaccurate calibration or control of the parameters can lead to suboptimal or unreliable performance. System malfunctions, as previously described failures, can also increase these issues.

- **Incompatibility with other materials**: MR fluids may not be compatible with all materials, particularly in the context of seals, containers, and other system components.
    - **Impact:** chemical incompatibilities can lead to quick degradation or leakage, reducing the overall system's reliability and lifetime.
- **Cost and supply chain**: production of MR fluids can be expensive due to the specialized materials as the magnetic particles and to the manufacturing processes required.
    - **Impact:** high costs and potential supply chain disruptions could lead to reliability problems for manufacturers who rely on MR fluids, particularly in large-scale or critical applications as the racing application field.

In summary, while MR suspensions offer significant advantages in adaptive systems, these reliability issues need to be carefully managed through proper design, maintenance, and operational procedures to ensure their long-term effectiveness and predictable performance.

8.6.2 Warnings and User Safety

Warnings are fundamental to point out dangers to support both the direct and the indirect safety. User safety is directly linked with the adopted safety measures and in this Paragraph is related to the driver (as in Paragraph 8.6 with the "fail-safe" principle), but especially to operators that have to quickly fix, mount and dismount the entire MR system.
This is directly related to other fundamental guidelines that must be taken into account during the embodiment design phase: the "**Design for Ergonomics and Design for Maintenance**".
MR suspensions must be maintained to avoid sudden breakdowns that would be extremely unsafe for the driver during the race: inspections must be done after the runs and not only when approaching the predicted end-of-life of the critical components.
In addition, frequent checks to the correct functioning of the MR damper are important for the environment too, avoiding dispersion of the toxic fluid because of leakages.
Moreover, the key word for both those design principle is in this case accessibility. By the point of view of the operators, the entire system should be easily reachable, paying also attention to the proximity to dangerous sharp edges belonging to near components. Therefore, the position of the assembly inside the car gains great importance and is taken into account during the dimensioning of the suspensions as explained in Paragraph 8.2.1. Enough space to work for mechanics is indeed considered, together with the attention to an easy connectivity of all the cables required by the control system, as stated in Chapter 5 when dealing with the House of Quality and in particular with the Human-Machine Interface (HMI) requirement.

8.7. Materials

The main characters of the embodiment to be designed are represented by the components which constitute the body of the damper: the cylinder, the piston rod and the piston which has to accommodate for the magnetic spool. On this purpose, considering the main qualitative and technical requirements defined so far, as well as the problems highlighted in the fault-tree analysis in Paragraph 8.6.1, a screening of the eligible material has been carried out with the aid of Ashby diagrams.

Before proceeding further, it is important to point out that even if they are components which carry out different functions, they share the same general requirements. For this reason, the first material screening will be performed irrespective of the component under scrutiny.
Therefore, as to the mechanical properties, the specific strength, the stiffness and cost per unit mass was used as main parameter for the first material discretization, whereas to consider the overheating problem a "good thermal conductivity" constraint was applied.

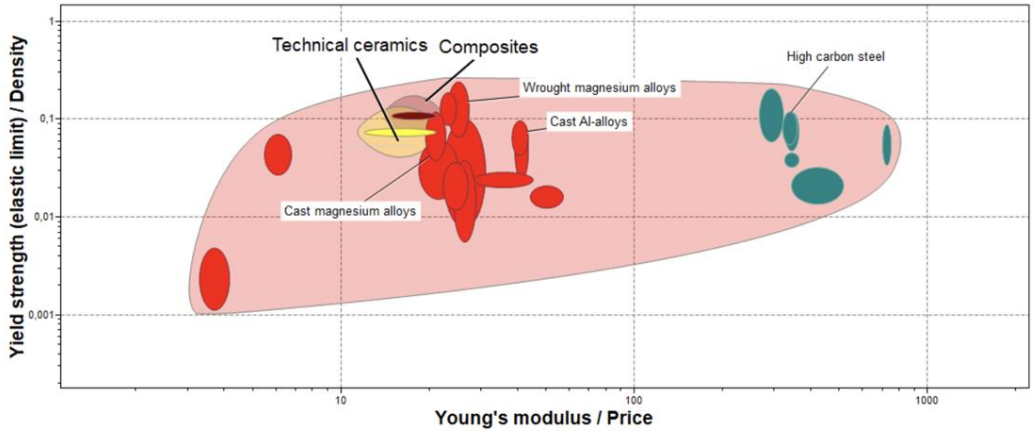


Figure 40 - ASHBY Diagram

The results of the analysis provided different material families:

- Steel
- Aluminum
- Metal matrix composites
- Technical ceramics
- Cast magnesium alloys

We can immediately discard the less promising candidates that are ceramics and magnesium alloys, which does not consider some main requirements that damper must have.
The high sensitivity to corrosion of magnesium would result in reliability issues of the assembly, other than requiring additional surface finishing to provide the required protective layers, which imply additional costs and a more intense maintenance.
For what concerns the ceramic family instead, we can easily eliminate them because of their brittle behavior which does not enjoy of intrinsic fail-safe properties of the plastic deformation that limit the consequences of misuse loads or unexpected loads.
As a consequence, a comparison between the most suitable alternatives can be done to correctly choose the right material.

PROPERTY	STEEL	ALUMINUM
Strength to Weight Ratio	Generally Higher (**stronger but heavy**)	Generally Lower (**weaker but 1 /3 of the weight**) Exception: Al7075 (ergal) is higher
Corrosion	Stainless steel (SS) resists to corrosion	High corrosion resistance (natural passivation layer or artificial anodizing process) Not suitable in acid environments
Cost	SS more expensive Others are cheaper	Cheaper than SS
Service Temperature	High operating temperature (constant mechanical properties with T) Provided corrosion resistance	90-180°C (decreasing mechanical properties)
Thermal Conductivity	Not so conductive	Higher conductivity
Thermal Stability (CTE)	Higher (CTE = 10^{-5} K^{-1})	Lower (CTE = 2 CTE_{steel})
Magnetic Property	Ferromagnetic	Non-magnetic (Paramagnetic)
Workability	Very resistant to wear and abrasion Stiffer	Easy to cut and form
Castability	Lower (high T_m) limited to few casting technologies	higher due to lower melting temperature Suitable for all the casting technologies
Electrical Conductivity	Good conductor	Very good conductor
Wear Resistance	Higher	Anodizing provides corrosion and wear resistance with good adhesion of the substrate and low price

Table 33 - Steel and Aluminum properties comparison

Looking at the comparison table between steel and aluminum it is possible to select the best material for each component according to their requirements and criticalities.

Generally speaking, having to do with racing components, the weight is an important parameter to take into account. Therefore, the higher specific strength of steel would allow to achieve a more compact geometry with respect to that of aluminum, thus saving mass. However, due to the absence of severe restriction in radial direction we can achieve the same performance with a weaker but lighter material like the Al, even though the ergal can provide a solution as strong as the steel but one third lighter.

Moreover, aluminum is cheaper and easier to work with, but it is affected by a worst thermal stability both in terms of mechanical properties, which tend to decrease faster than steel with temperature, both in terms of dimensional stability, because of the twice larger coefficient of thermal expansion (CTE). This latter criticality could not only compromise the effect of the sealings, leading to some unexpected leakages if not carefully managed, but also problems to maintain the roundness and straightness tolerances necessary to guide the piston during its stroke. Such a problem, that would lead to prefer a more stable material like steel, can be easily solved considering the higher thermal conductivity of the aluminum, whose improved heat management allow to keep under control the working temperature, and hence limit the thermal expansion.

Finally, the hard anodizing of the aluminum allows to simultaneously deal with its higher corrosion sensitivity and lower wear resistance, thus discarding the use of metal matrix composites materials which raise the cost and the complexity of the design and production process.
Therefore, aluminum results to be a very good candidate for the design of the **cylinder**, being also very versatile to many different manufacturing technologies (whose choice will be addressed in the following chapter 8.8) and easy to work.

The only problem, when dealing with Al, is the paramagnetic property that, due to its low magnetic permeability, in presence of a magnetic field, does not intensify its intensity as the steel would do. This aspect could drive the choice of material of the piston toward a more ferromagnetic material like the steel. Indeed, since the main function of the piston is to tune the damping coefficient through the oil flux, a magnification of the magnetic field acting on the MR fluid will afford for a reduced electric power (lower intensity) or a smaller coil (lower coil turns, hence weight) required to generate the desired magnetic intensity. However, in order to keep the tight tolerances of the piston-cylinder interface as constant as possible, irrespective of the wear, aluminum results to be again the best choice. Indeed, this kind of component is the one that is exposed to the higher number of heat sources (friction, oil heating, electric resistivity), Therefore, building it with the same material family of the cylinder will allow to better control the tolerance variability caused by temperature. And las but not least, aluminum would help to keep the mass but also the inertia of the component low, that is an important property for an alternatively moving component like the piston.
Talking about the **piston rod** instead, it is a critical component since, being a relatively slender bar subjected to compression, other than tension, the buckling stability can be one of the main sources of failure or even deformations that could compromise the correct behaviour of the part. Indeed, the bending of the bar would compromise the guiding function of the piston such that, increasing the wear, the resistance of the cylinder could be affected or even worse cause the stuck of the system in case of a too high deformation. For this reason, due to its high strength and, in particular, stiffness a steel alloy is more suitable to produce the piston rod. This choice is furtherly defended if we consider the fatigue behaviour: the improved fatigue resistance of steel allows to withstand the number of cycle provided by the high frequency oscillations caused by uneven gravel roads typical of this kind of races.

Last but not least, it is important to take into account that the piston is an external moving component inside the cylinder. Therefore, particular care has to be taken not only about the corrosion resistance but also to the impact resistance, because of possible gravel stones impacts which can either notch the surface, reducing the fatigue life, either deform the bar because of the unexpected load application. Finally, to guarantee an as smooth as possible displacement a good surface finishing is required to reduce as much as possible the friction with the sealing cup and hence its wear.

SUMMARIZING

COMPONENT	MATERIAL	REASONS	IMPROVEMENTS
Cylinder	Aluminum 7075	Weight saving Heat management	T6 + Hard Anodizing
Piston Rod	SS (AISI 304)	Stiffness Strength	Induction Hardening
Piston	Aluminum 6082	Thermal Stability Matchup	T6

Table 34 - Material Choice

8.8. Production Processes

Once the material is selected, it is possible to proceed with the choice of the most suitable manufacturing process. Of course, when doing that, what it is important to take into account are not only the geometric properties that can be achieved with a specific process with respect to the component's targets, but also the cost that will affect the final price of the product and hence its competitiveness on the market.

8.8.1. Process comparison: CNC, sand casting, extrusion

An overview of the most suitable process for all the different components and materials is carried out: machining from a raw billet material, sand casting (that is the only suitable manufacturing process for steel) and extrusion, since having to do with all axisymmetric components.

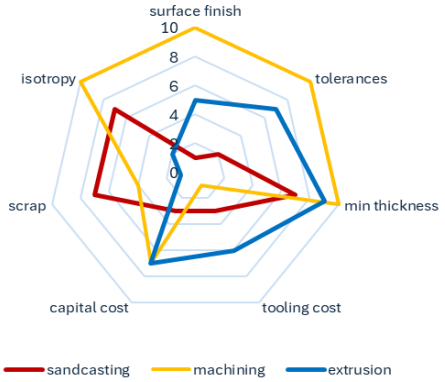


Figure 41 - Casting vs Machining vs Extrusion

As can be observed by the spider chart, compared to sand casting, the extrusion and machining processes are the most promising ones in terms of precision and quality of the final product (with machining being the best one), even though the initial investment cost required is quite high because of the cost of the press and the mold, for extrusion, and of the CNC machine for machining. Moreover, the extrusion process allows to minimize the waste of material because of the almost null scrap fraction produced by forcing all the raw material to pass through the designed die. This, however, implies a certain anisotropy of the product which either maximizes the mechanical performance of the component in the axial direction and can be detrimental for the loading conditions of the part.

Sand casting, instead, is the most economic one but it is characterized by the highest waste of material and lower tolerances and surface finish, an aspect that can be improved with more performant casting techniques (as die casting, pressure die casting), only in case of aluminum, or through an additional finishing process.

So, since in general the complexity of the part to be produced is not so high, we should choose the most efficient component in terms of precision to cost ratio.

8.8.2. Piston Rod Manufacturing

The piston rod is a very small, long hollow bar made of steel which require very tight tolerances of straightness and roundness. Because of its very simple shape, and the not so large production volume expected for this kind of applications, the most cost-efficient technology is selected for its production: machining.

Despite the lower production rate and the higher material utilization, extrusion is much more expensive with respect to machining due to additional cost of the mold and less precise. Therefore, it is possible to achieve the same or even better results at a lower cost, and almost same time with machining. The high geometric simplicity of the part and the very poor quality instead does not justify casting as a proper choice.

Weight	Property	Sand Casting	Extrusion	Machining
0,1	scrap	↓↓	↑↑	↓
0,5	tolerances	↓	↑	↑↑
0,1	production rate	↓↓	↑↑	↑
0,3	cost	↑↑	↓	↑
Score		2.4	2.9	3.4

Table 35 - Piston Manufacturing Techniques Comparison

This can also be demonstrated by a more systematic approach. Indeed, by consider 4 level of quality for the most important properties of the manufacturing process, according to their importance, machining results to be the best one.

Finally, to provide the correct level of straightness, wear resistance and hardness, a grinding and induction hardening processes are respectively considered in the manufacturing cycle of the piston that is shown below, while a threaded interface has to be taken into account for the ends components connections.



Figure 42 - Piston Rod Manufacturing Cycle

8.8.3. Cylinder Manufacturing

The damper cylinder is the real body of the shock absorber. It is subjected to a pressurized ambient, provided by the oil, as axial load. Therefore, due to the multiaxial stress state, isotropic mechanical properties are preferred in order to keep the part safer.

In this case, due to symmetry of the component and very small thickness to diameter ratio ($t/\emptyset$) a hot extrusion process (to reduce the grain deformation) would be more suitable to reduce the waste of material compared to the other technologies.

However, to reduce the higher complexity of the assembly and the source of problems caused by an higher number of component and interfaces, one of the two end-eye housing of the unibal is directly obtained from the cylinder body. For that reason, machining is again the best choice since despite the higher complexity of the different geometric features of the part which would lead to choose casting, the higher material scrap and the successive refinements through specific machining operations would not justify casting as a good choice.

The cylinder indeed requires sufficiently high geometric tolerances of the internal hole for the guiding of the piston which can be achieved by subsequent honing process, achieving at the same time a good surface finish to reduce the inner wear too. Moreover, a thread on the external surface must be provided to accommodate the lock ring of the springs. Thus, it is possible to summarize the manufacturing cycle for the cylinder as follows.



Figure 43 - Cylinder Manufacturing Cycle

8.8.4. Piston Manufacturing

The piston has the function to provide the housing for the magnetic coil and let the oil flow through it, other than to be able to travel along the cylinder stroke. For that reason, a simple machining from a raw billet is enough to reach the required properties and geometric features.

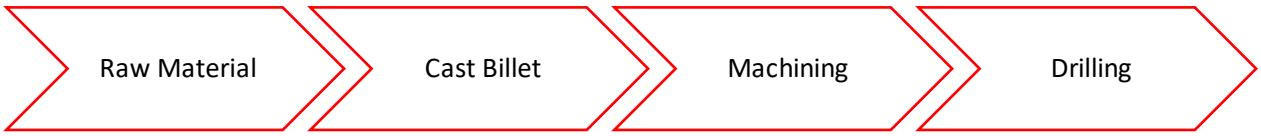


Figure 44 - Piston Manufacturing Cycle

8.9. Environmental Effort and Recycling

A great effort must be put in the environmental protection, hence there are guidelines to be followed that are all under the more general name of "**Design for Environment and Design for Recycling**".
It is fundamental to not leave toxic waste in the environment, hence the critical point is concerning MR fluid, since all the other components can be recycled if properly disposed as they are made by "simple" common metals as steel, aluminium or Ergal. On the other hand, the toxicity of these types of fluid is depending on their specific composition but, as written in Paragraph 7.3.3, they are all made by a liquid carrier and by metallic particles to have the sensitivity to the magnetic field changes.
Before going more deeply in the recycling issues, it is important to underline that the precise formulation of the used Lord MRF-132DG fluid is a secret, however some generic information can be still retrieved. The carrier is a mineral petrol-based oil, while the magnetic particles are made by high-purity iron or carbonyl iron and are around the 70% of the entire weight.
Talking about dispersion in the environment and recycling, mineral oils can pose environmental risks since they are derived from petroleum and can harm especially aquatic life if incorrectly released. On the other hand, particles are made by materials that are not theoretically toxic, but the fine particles can pose a risk to soil and water if improperly disposed. Moreover, additives as anti-wear agents can be added, and they can be harmful since related to non-biodegradable chemicals.
However, it is difficult to write a "universal" recycling and disposal strategy, since the actual disposal depends on the legislation of the considered country. This means that MR fluid disposal should strictly follow the local hazardous waste disposal regulations.
In general, iron particles can be separated, and the liquid carrier can be usually treated as it is done for motor oil, anti-freeze and similar products. In addition, to minimize environmental risks, a Material Safety Data Sheet (MSDS) is provided by the manufacturer of the specific MR fluid and should be always consulted and followed.

Another key point is the **reconditioning** of the product to reuse the entire system or a part of it. In the case of this project this can be done thanks to the presence of the shims. When the MR suspensions cannot be used anymore for racing and high-performance purposes it can be still exploited for "normal" car and usage since they can behave as traditional passive suspensions because of the shims that can work exactly as already described in Paragraph 8.6 when dealing with fail-safe principle.
After the disassemble of the assembly and its inspection in order to be sure that all the suspension components are still in a good condition even with a previous highly stressful racing use, they can be reassembled keeping the original shape and mounted on road cars, hence reused.

8.10. Auxiliary Functions

After the definition of the main embodiment functions and layout, the presence of auxiliary functions must be taken into account. Usually, already known solutions should be preferred, while keeping special custom solution only when really needed. When all the auxiliary functions and their carriers are defined, the overall layout of the product can be determined.
After the definition of the fundamental damping function that must be carried on by the developed MR suspension system, the search for eventual desired auxiliary functions started. However, no really wished auxiliary function has been found. This can be due to the fact that a damper is exploiting simple mechanism and movements, and all its related function carriers are crucial for its functioning.
Also, apparently secondary needs as the easy accessibility must be considered as primary for reaching the goal of this project, as underlined in Paragraph 8.6.2. The reason is that in racing the easy mounting/dismounting/inspection is crucial because mechanics have limited time between the sessions and limited space where to operate.

8.11. Final Bill of Materials (BOM)

Here below the final BOM representing all the parts needed to build the entire MR assembly. As last column it is highlighted the make or buy decision. A more detailed analysis about how this choice has been made is provided in the following Paragraph 8.11.1.

N.	Qty	Title	Description	M/B
1	1	Piston cylinder	Al 7075 T6 hard anodized	Make
2	1	Lower unibal	STD: uniball HAB-M16TG-l	Buy
3	1	Lower uniball ring	STD: uniball HAB-M16TG-l	Buy
4	1	Second piston	Al 6782	Make
5	2	Pistons O-rings	STD: OR 40.0X3.O-N70	Buy
6	1	Locking gear	CNC Al7075 anodized	Make
7	1	Locking gear counternut	CNC Al7075 anodized	Make
8	1	Closing cap		Make
9	1	Cap inner O-ring	STD: OR 14.0X2.O-N70	Buy
10	1	Cap outer O-ring	STO OR 45.0X2.0-N70. Compression ratio 25%	Buy
11	1	Cylinder seager	PART-Wuerth_043948 ON 472, spring steel, phosphated and lubricated, type J LOD Medium I-DESC	Buy
12	1	Helper spring support	CNC Al7075 anodize	Buy
13	1	Helper spring	OEM Coil spring	Buy
14	1	Main spring	OEM Coil spring	Buy
15	1	Piston rod	AISI 304	Make
16	1	Piston seager	PART-Wuerth 043814 OIN 471 spring steel, phosphated standard vers. LOD Medium I-DESC	Buy
17	1	Piston nut	M14x15 DIN 934, galvanized steel 181, fine pitch LOD High I-DESC	Buy
18	1	Upper uniball plate nut	M10x1 DIN 934, galvanized steel 181, fine pitch LOD High 1 DESC	Buy
19	1	Bumpstop	STD: BUMP STOP 961412303B	Buy
20	1	Upper plate	Al7075 hard anodized	Make

21	1	Load cell		Buy
22	1	Main piston	Al CNC machined	Make
23	1	Bump shim 1	L=25,5 mm	Make
24	1	Bump shim 2	L=24 mm	Make
25	1	Bump shim 3	L=22,5 mm	Make
26	1	Bump shim	L=21 mm	Make
27	1	Rebound shim 1	L=23,5 mm	Make
28	1	Rebound shim	L=22,5 mm	Make
29	1	Magnetic coil	copper wire ϕ1,024 x 50 turns	Make
30	1	Top mount		Make
31	1	Upper uniball	STD; NMB ABYT12	Buy
32	1	Upper uniball ring	STD NME ABYT12	Buy
33	1	Plate closing cap		Make
34	2	Bushing	Steel	Make
35	3	Top mount screws	DIN 961, galvanozed steel 8.8 , fine pitch VITE-TE-DIN961-8.8-(A2K)-M8x1 x60	Buy

Table 36 - BOM

8.11.1. Make or Buy Strategy

The make or buy decision strategy involves evaluating whether to produce the component internally (make) or outsource it to an external supplier (buy). This decision is mainly affected by costs, but for a more refined analysis, quality, timelines and strategic alignment factors should be accounted for as well.

Talking about **costs**, it's important to the different source of costs. From the "make" point of view, this means to consider the "investment costs", like machinery and tooling, and "variable costs" such as materials, energy a labour. From the "buying" point of view, it's the supplier's price and shipping and potential hidden costs (e.g.: quality inspections).
Moreover, there's also long-term effect, since making in-house could be cheaper if the **production volume** is high enough. This is due to economies of scale which is strongly related to the expertise (technical skill, know-how, specialized equipment) that make the cost per part decrease in a non-linear way. Therefore, in order to decide whether to make or buy some components, comparing total costs for both options, basing on the production volume expected, is important to identify the break-even point which will drive this choice.
Because of the very small popularity of the Pikes Peak races and the limited number of competitors, the production volume has to be very small. By considering 4 dampers for each car and at least one spare part for each wheel then no more than 1000 products have to be produced per year.

Starting from the cylinder, which constitute the main part of the body of the damper and the most expensive one, the comparison between an in-house CNC machined one and a outsourced part to buy is based on the average minimum cost of a common high quality hydraulic cylinder damper of automotive application of 500€. Basing on this reference, after a rough economic production analysis of the component designed so far, it is possible to verify from the make or buy decision plot below (Figure45) that the break-even point is far below the 500 units, thus suggesting a cheaper in-house production (make) of this component.

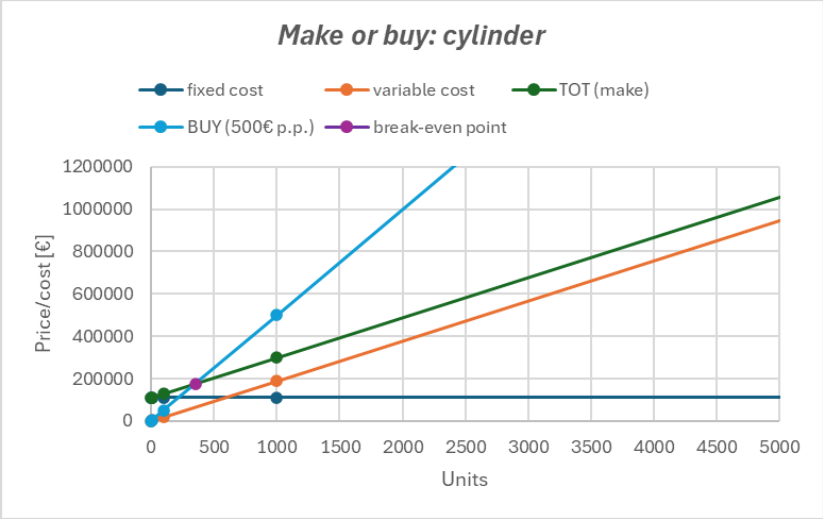


Figure 45 - Cylinder Make or Buy Analysis

The same occurs also for the piston rod and locking gears of the spring, whose plot are being omitted for the sake of simplicity. For what regard the piston instead, no analysis is required being that a specifically designed component for his kind of product that no one produce, thus requiring necessary an in-house production.

As to the more common components (from now on termed "standard" components) like uniballs, springs, screws, nuts and so on, the other factor affecting the make or buy decisions strategy become predominant over the income that would be obtained from a possible in-house production. Indeed, this kind of components are produced by companies that are so specialized in producing such products that the **level of quality and knowledge** acquired in producing them for years will be never reached from a first in-house production. Moreover, this companies produce such components for so many applications that their cost per part would be much less than one obtained from an own made one. In this way we can justify the make or buy attribute present in the bill of material (BOM), mentioned before.

9. Detailed Desing

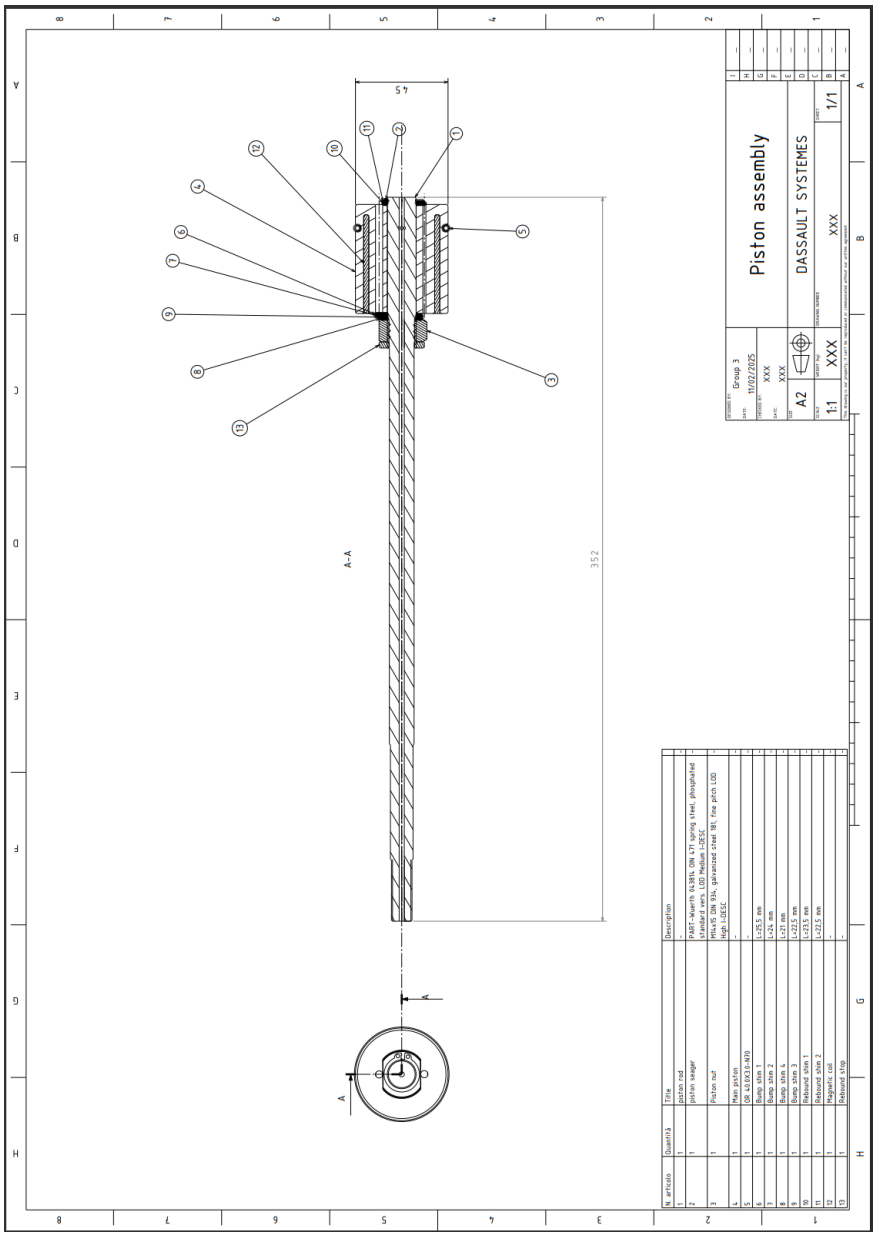
During this final phase, production and operating details are finalized; the way to exchange and show information changes allowing technical drawings, together with reports, to be their main carrier. That's why the detailed design steps are generating the so-called "Technical Product Documentation" related to both components and assembly design.
This is a key point of each design "procedure", since technical drawings are considered a universal language to communicate and represent shapes/geometries. This brings to easier transferability of information with simplicity and uniqueness of interpretation. The graphical part is therefore including not only shapes, but dimensions, references and tolerances too.
Technical drawings are not needed for standard components, so they are only added to the assembly as 3D part in order to have a clear and finished idea about the final structure and shape of the mechanism; drafts are not created since these parts are chosen after technical analysis of properties and geometries of the bought part itself.

Here below, the entire suspension system is divided into subassemblies to have more visually clear drawings to reach an easier and more immediate reading of all the data. Only at the end of the report, the final definitive MR assembly is composed.
In each paragraph, the MR subassembly taken into account is shown both with its exploded view to have a precise overview of how the part is actually mounted and its final technical draft.

9.1. Piston and Rod Assembly

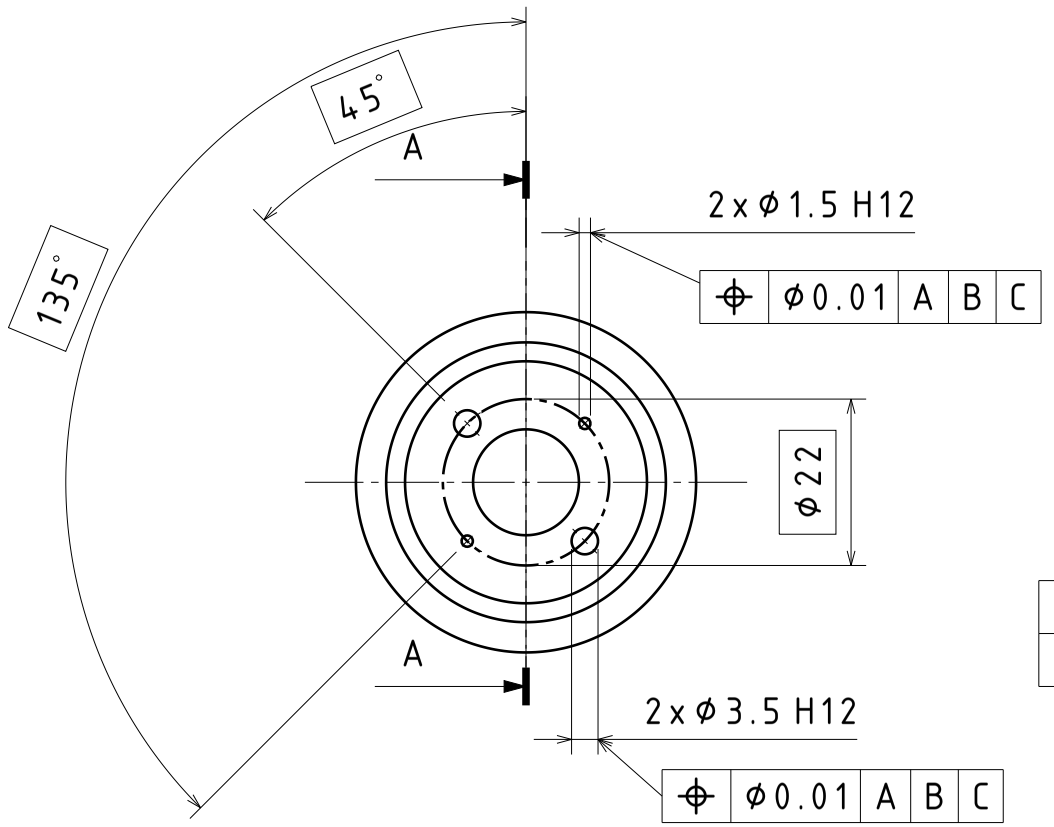
Until Paragraph 9.4, the structure of each section will be the same: after a brief description of the considered sub-assembly, the technical drawings of all the customized and in-made components are shown. Lastly, a final layout of the entire sub-assembly is provided together with the list of all the parts composing it.

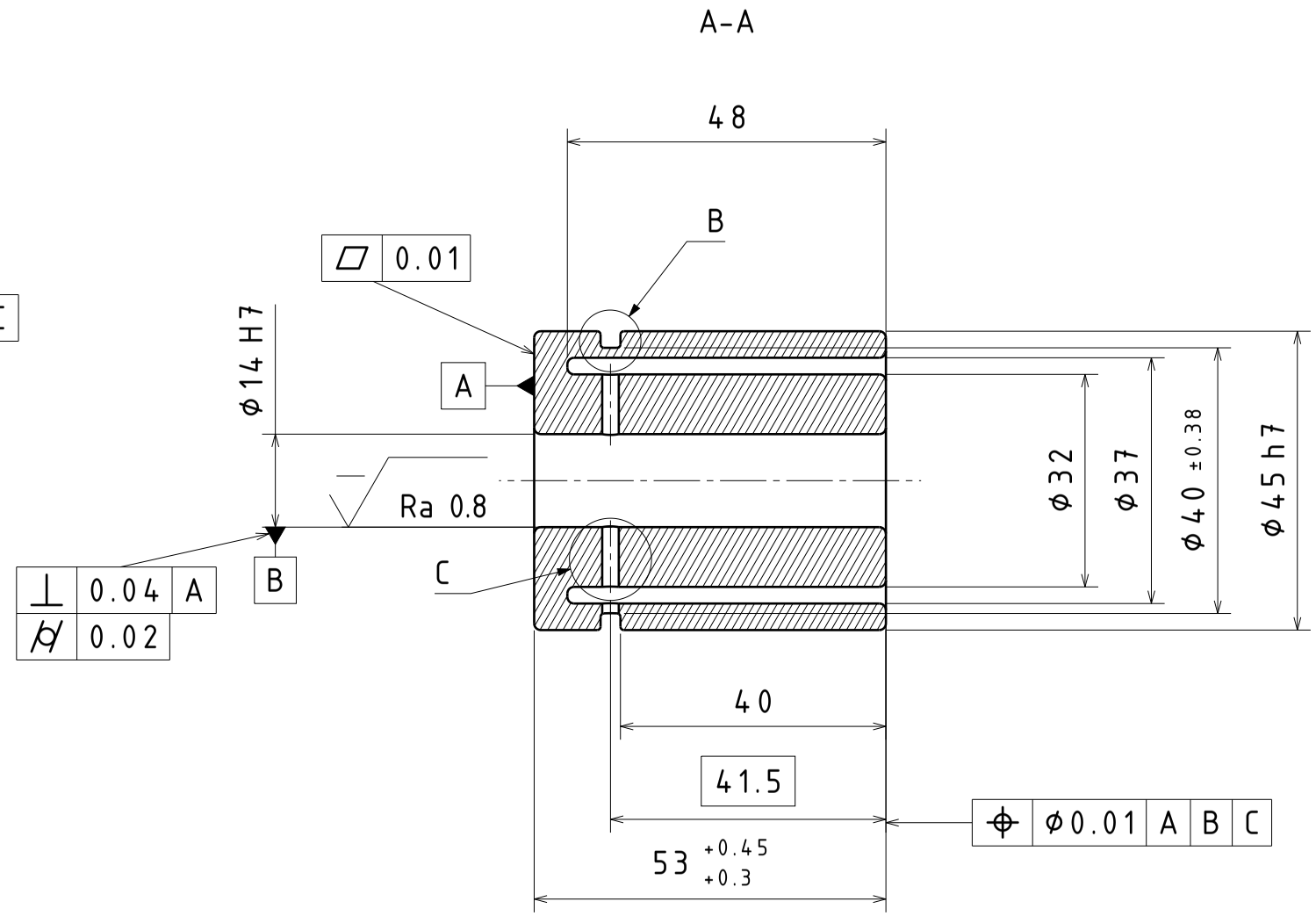
The first sub-assembly to be presented is the so-called "Piston and Rod Assembly". Obviously, it is composed by the main piston that allows the MR fluid to flow so the damping function, and the rod. More, all the shims related to both bump and rebound as well as the O-ring for a sealing function, nuts and Seager to fix everything together are here presented. The magnetic coil is represented too to give a practical idea about sizing and overall dimensions of coils and wires.

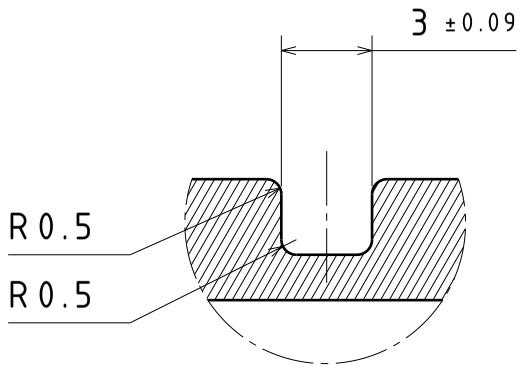


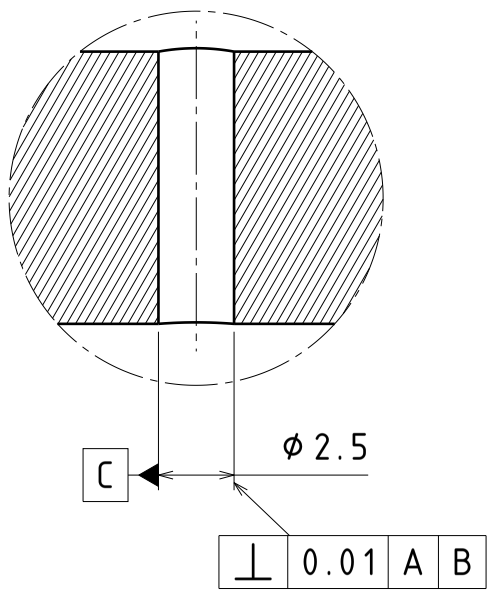
Assembly Drawing 1 – Piston and Rod Assembly

In the following pages, the related drafts are added:
Draft 1 – Main Piston
Draft 2 – Rod
Draft 3 – Bump Shims
Draft 4 – Rebound Shims









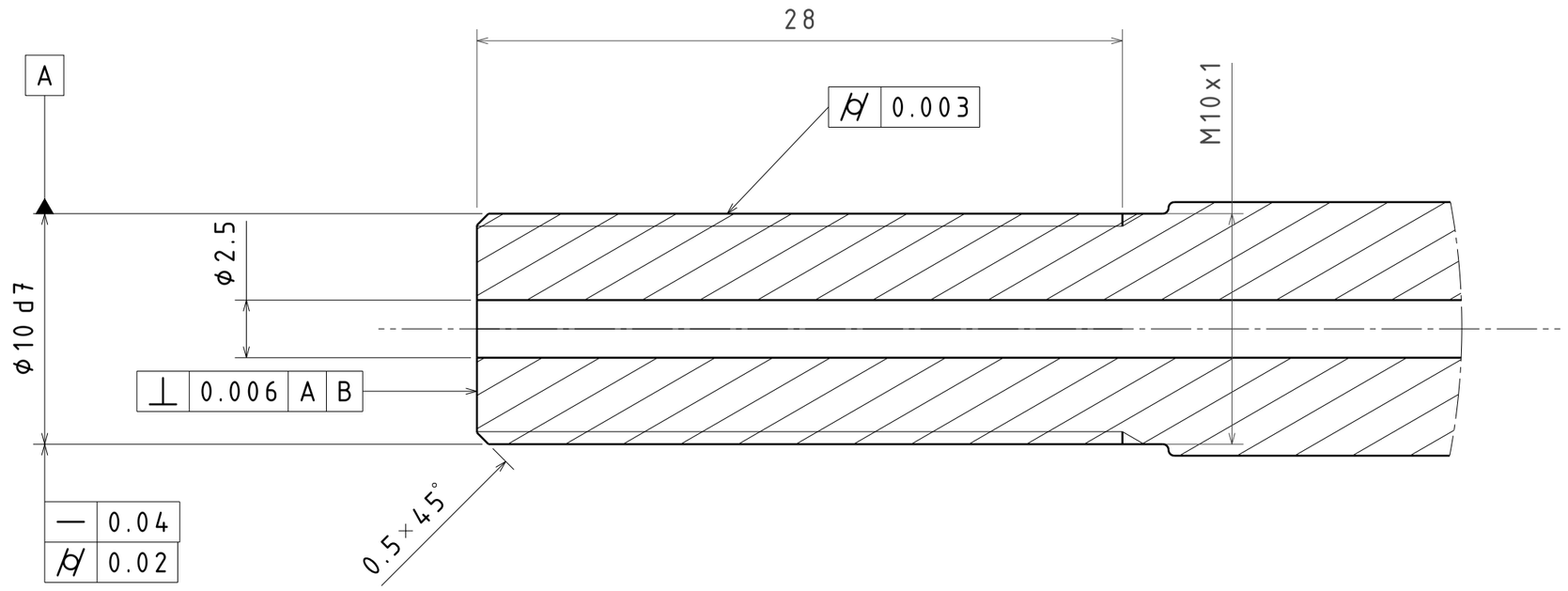
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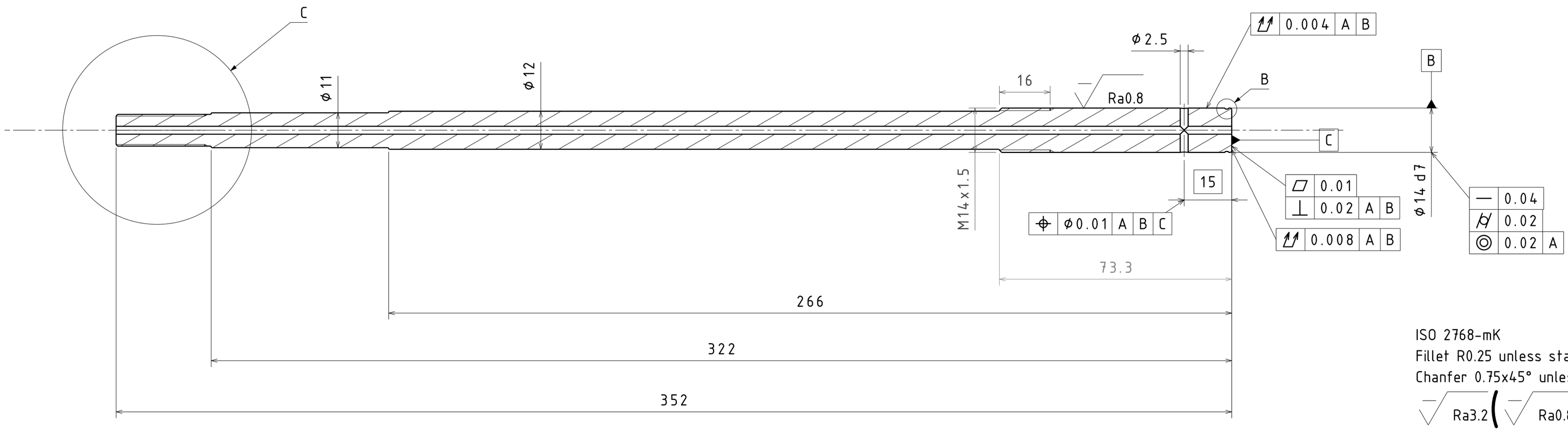
DESIGNED BY: Group 3 DATE: 02/02/2025		Main Piston	M _ L _ K _ J _ I _ H _ G _ F _ E _ D _ C _ B _ A _
CHECKED BY: Group 3 DATE: 11/02/2025			
SIZE: A3			
SCALE: 1:1	WEIGHT (KG): XXX	MATERIAL: Al 6082	PROCESS: Machining

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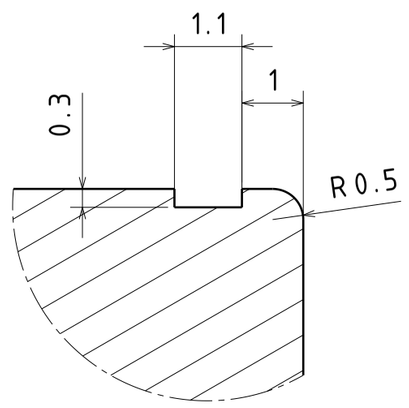


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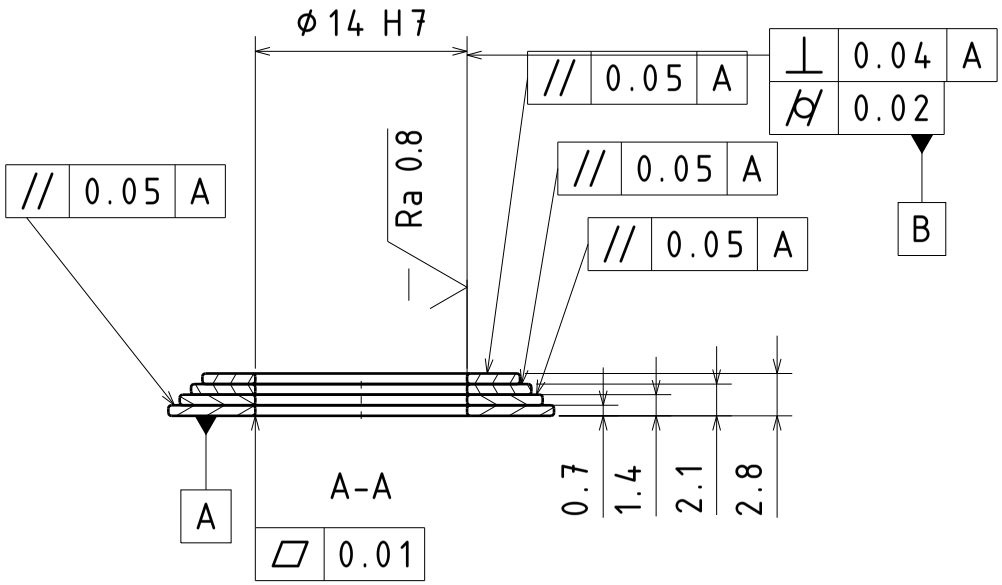
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Chanfer 0.75x45° unless stated
Ra3.2 (Ra0.8)

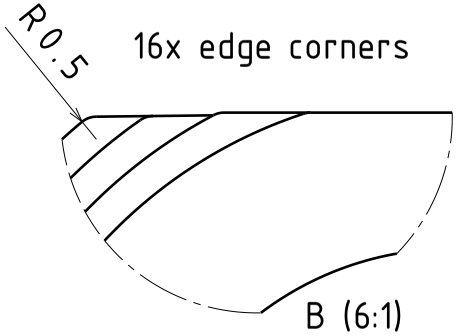
B (10:1)

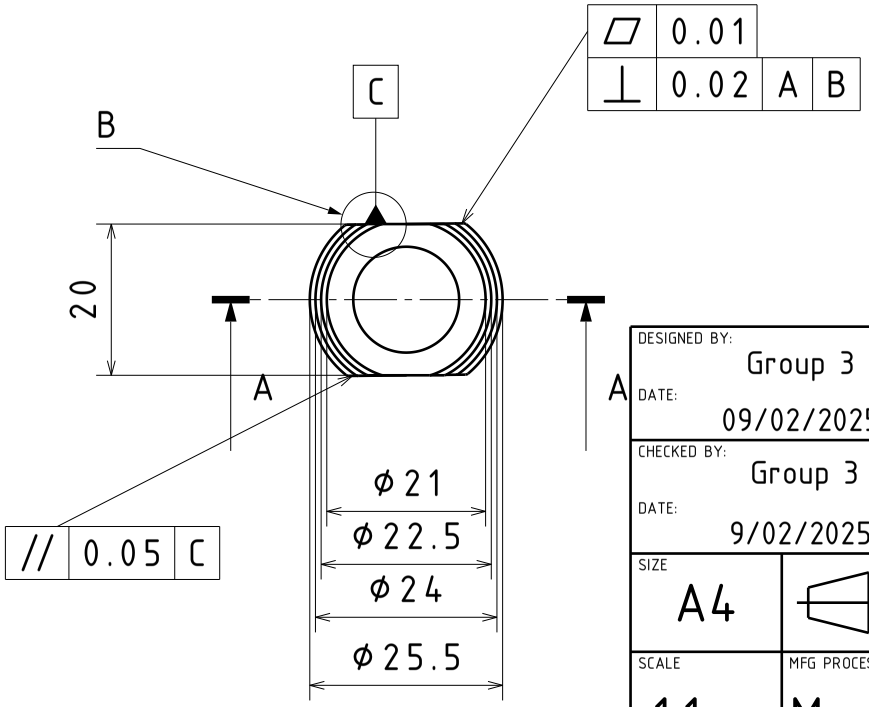


DESIGNED BY: Group 3		Rod	
DATE: 05/02/2025			
CHECKED BY: Group 3			
DATE: 11/02/2025			
SIZE: A2			
SCALE: 1:1	WEIGHT (KG): XXX	MATERIAL: SS (AISi 304)	PROCESS: Machining

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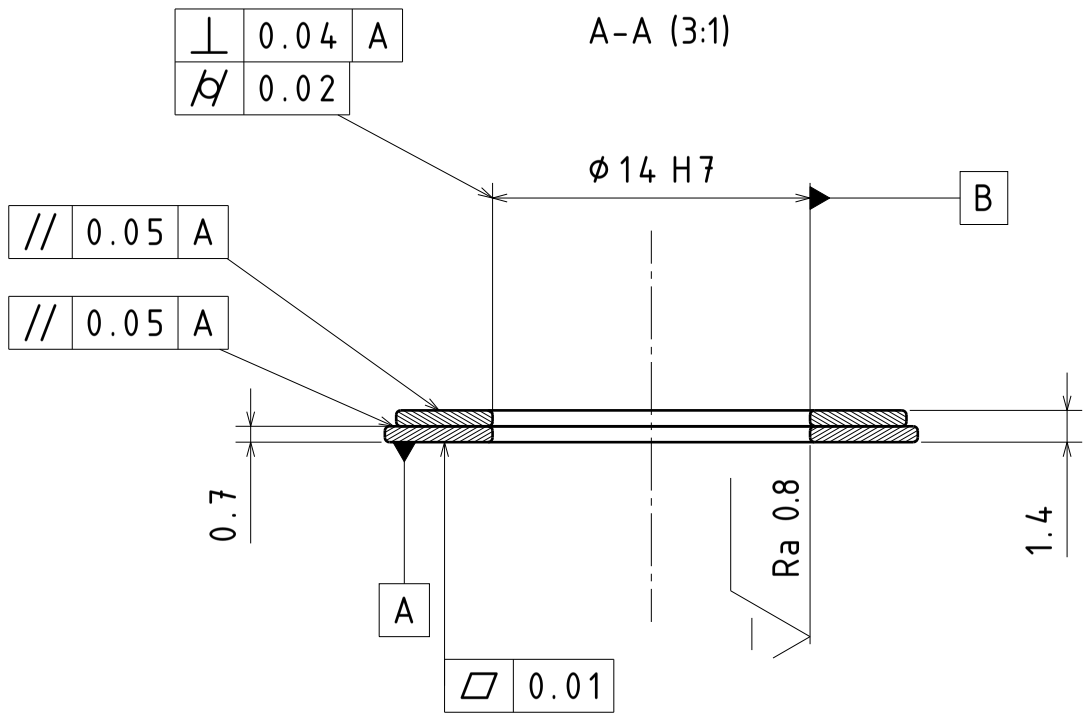
ISO 2768-cK
Fillets R0.2 unless stated

Ra 3.2 Ra 0.8

DESIGNED BY: Group 3	Bump shims pack
DATE: 09/02/2025	
CHECKED BY: Group 3	
DATE: 9/02/2025	
SIZE A4	DASSAULT SYSTEMES
SCALE 1:1 / MFG PROCESS Machining / MATERIAL Steel / SHEET 1/1	

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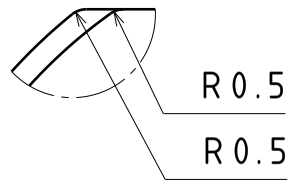
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A	_

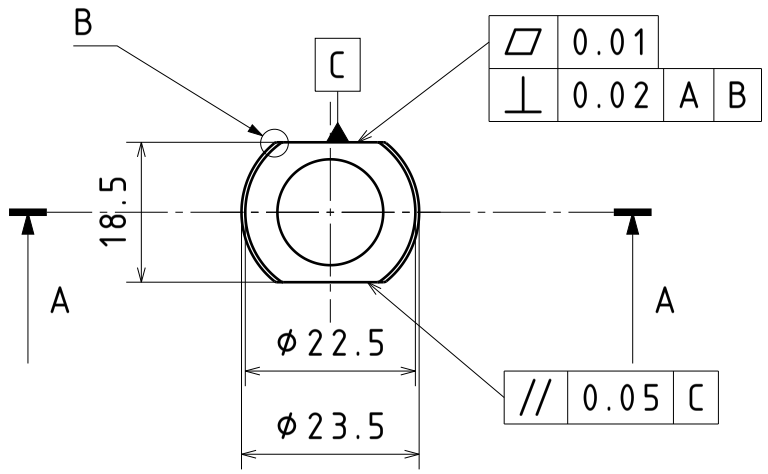


A-A (3:1)

B (6:1)

4x edges (corners)





ISO 2768-cK

Fillets R0.2 unless stated

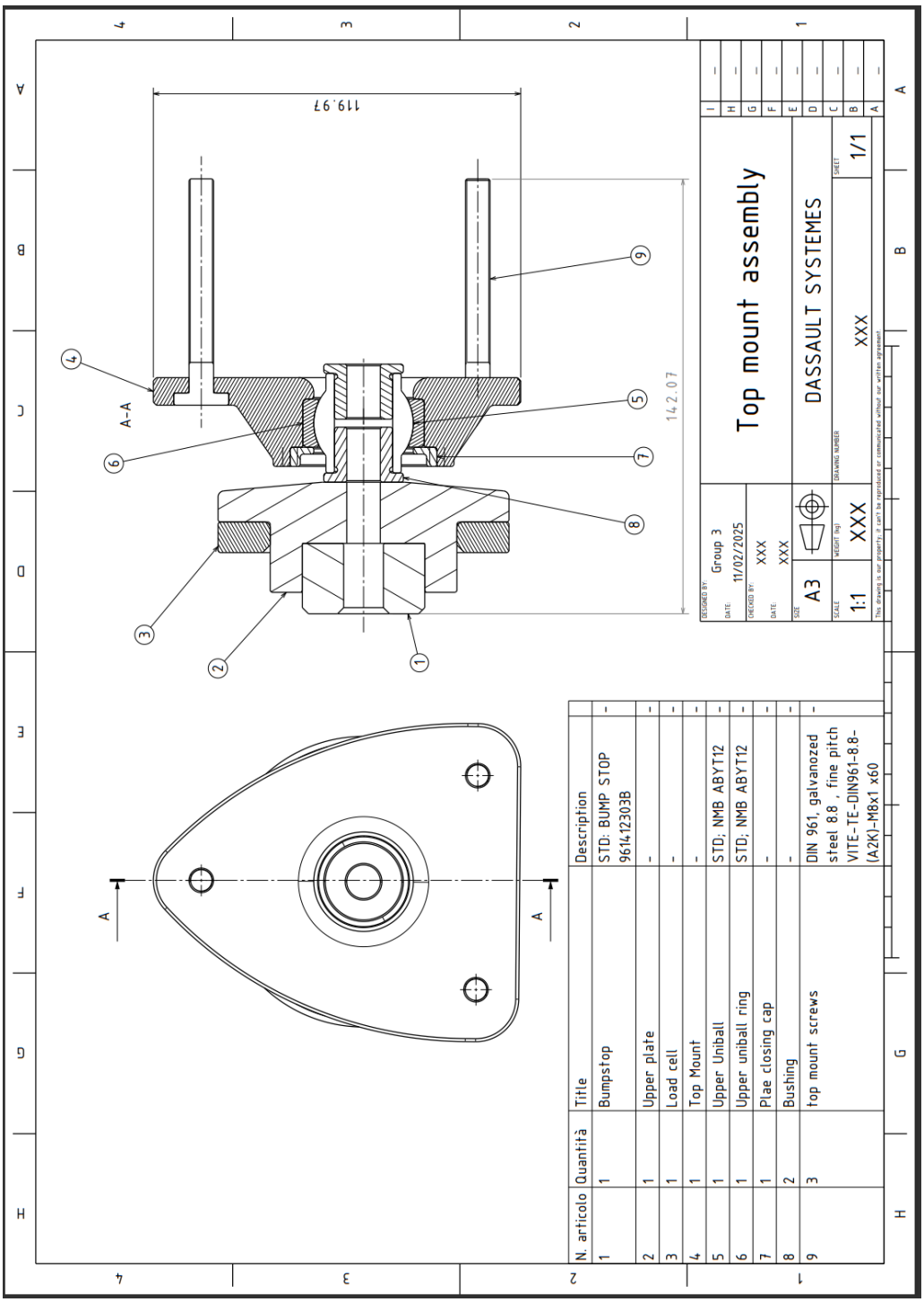
$\sqrt{Ra\ 3.2}$ ($\sqrt{Ra\ 0.8}$)

DESIGNED BY: Group 3 DATE: 02/02/2025		Shims Rebound		M —
CHECKED BY: Group 3 DATE: 11/02/2025				L —
				K —
				J —
				I —
				H —
				G —
				F —
SIZE: A3				E —
				D —
SCALE: 1:1	WEIGHT (KG): XXX	MATERIAL: Steel	PROCESS: Machining	C —
				B —
This drawing is our property; It can't be reproduced or communicated without our written agreement.				A —

A3

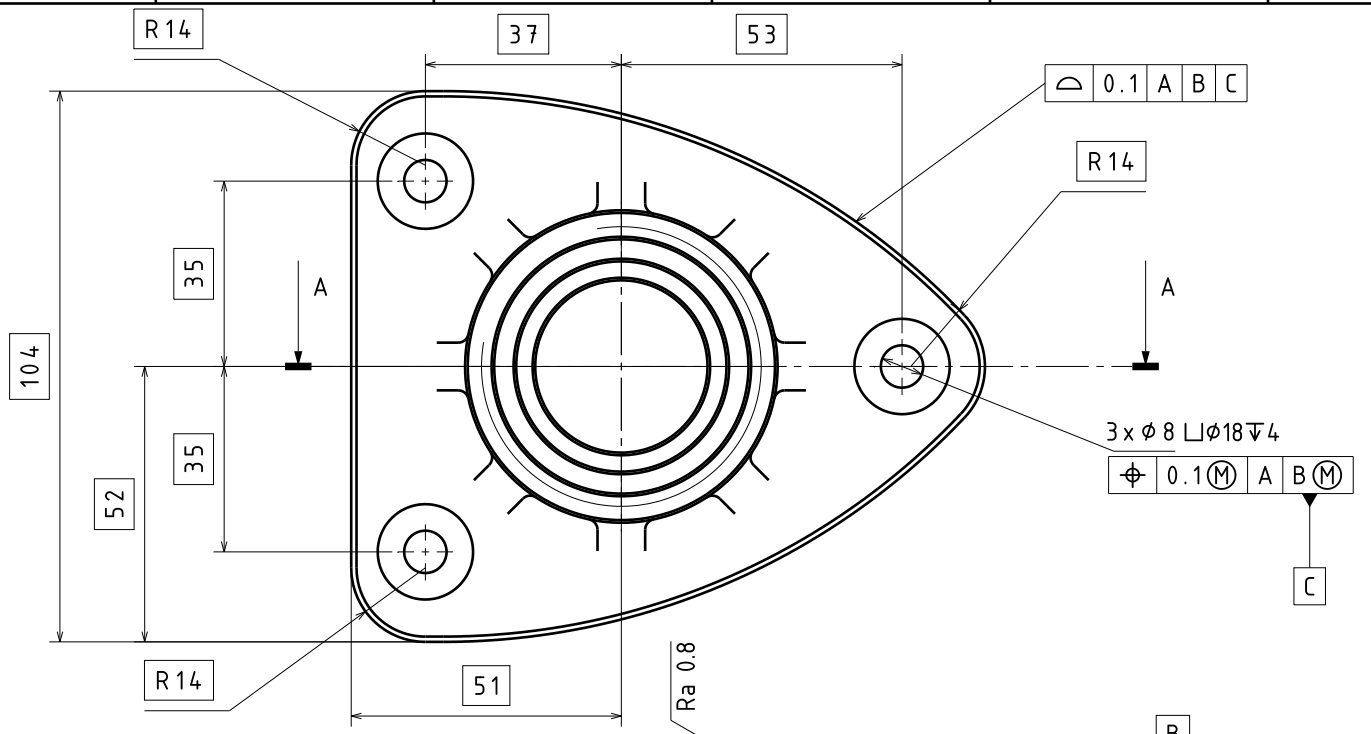
9.2. Top Mount and Upper Plate Assembly

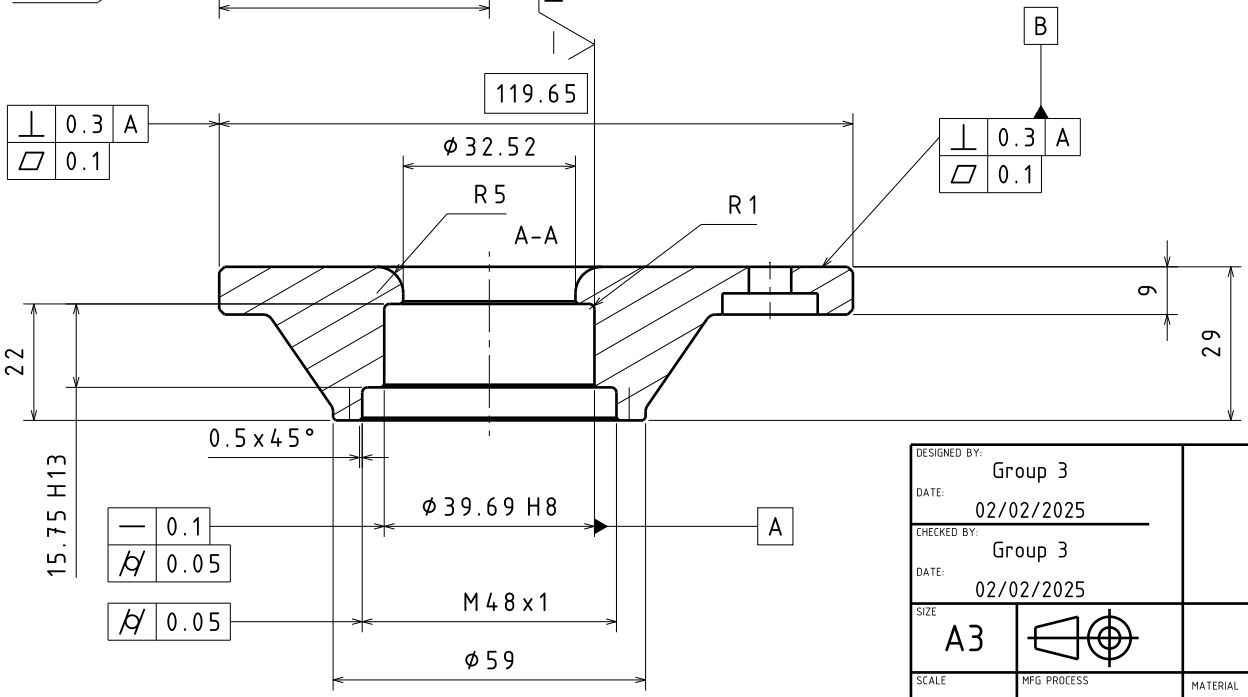
Going on, the "Top Mount and Upper Plate Assembly" is a fundamental sub-assembly to connect the designed damper top the chassis of the car. In addition to the top mount and the upper plate themselves, it consists of upper unibal, closing cap, bumpstop avoiding excessive compression of the damper and load cell providing the feedback control. Screws, rings, nuts and plates are included too.



Assembly Drawing 2 – Top Mount and Upper Plate Assembly

In the following pages, the related drafts are added:
Draft 5 – Top Mount
Draft 6 – Upper Plate
Draft 7 – Upper Plate Bushing
Draft 8 – Plate Closing Cap





iso 2768 mH

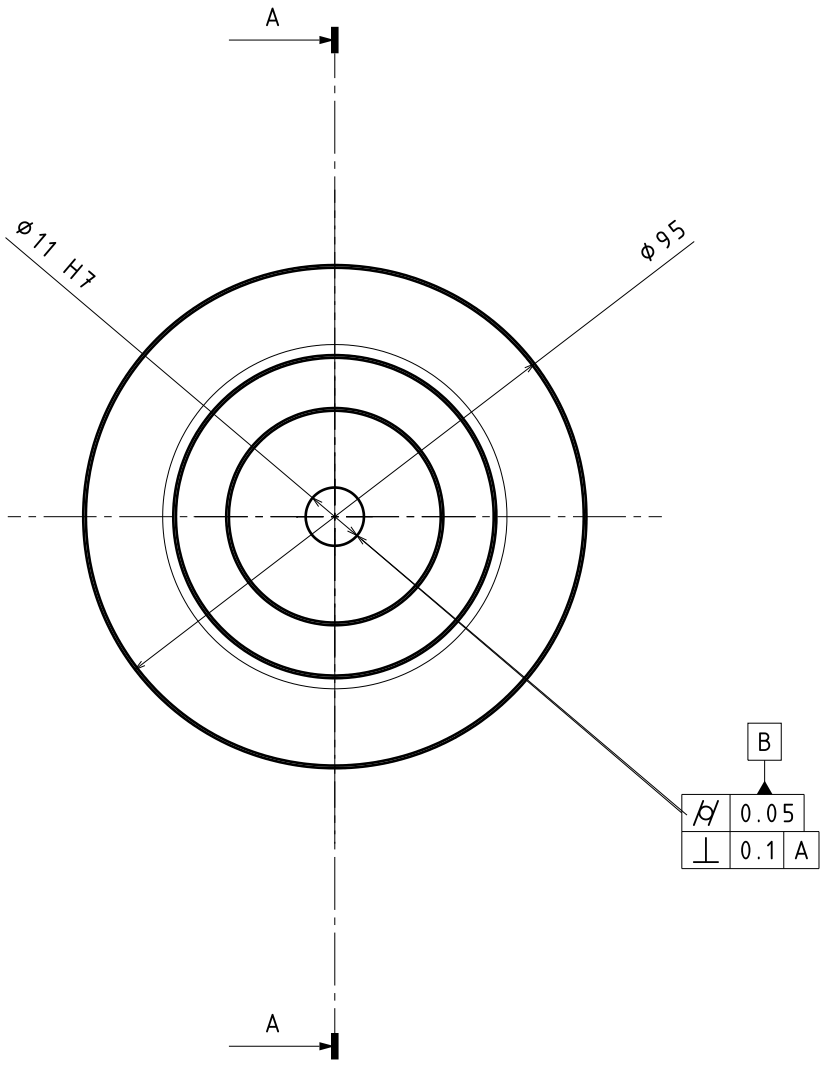
fillet radius R2 unless stated

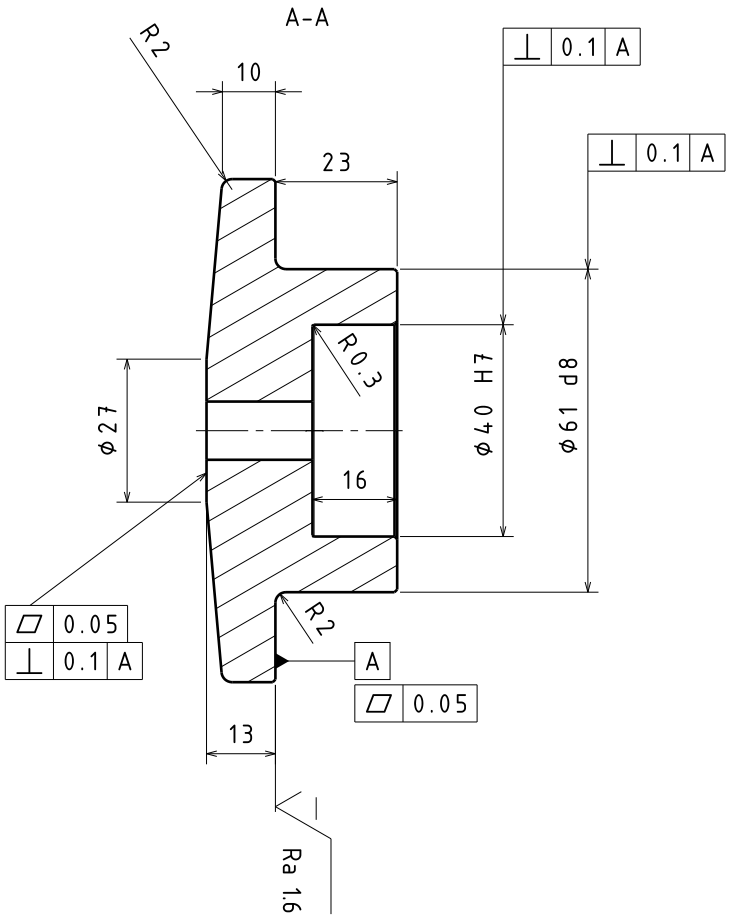
chamfer 1x45° unless stated

Roughness $\sqrt{}$ Ra 3.2 ($\sqrt{}$ Ra 0.8)

DESIGNED BY: Group 3	Top Mount
DATE: 02/02/2025	
CHECKED BY: Group 3	
DATE: 02/02/2025	
SIZE A3	DASSAULT SYSTEMES
SCALE 1:1 / MFG PROCESS Machining / MATERIAL Al 7075 Hard anodized / SHEET 1/1	

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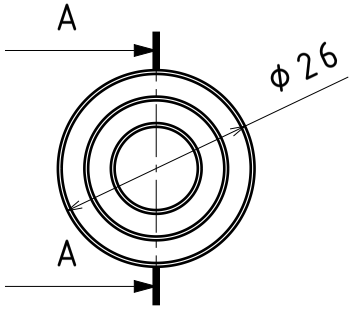


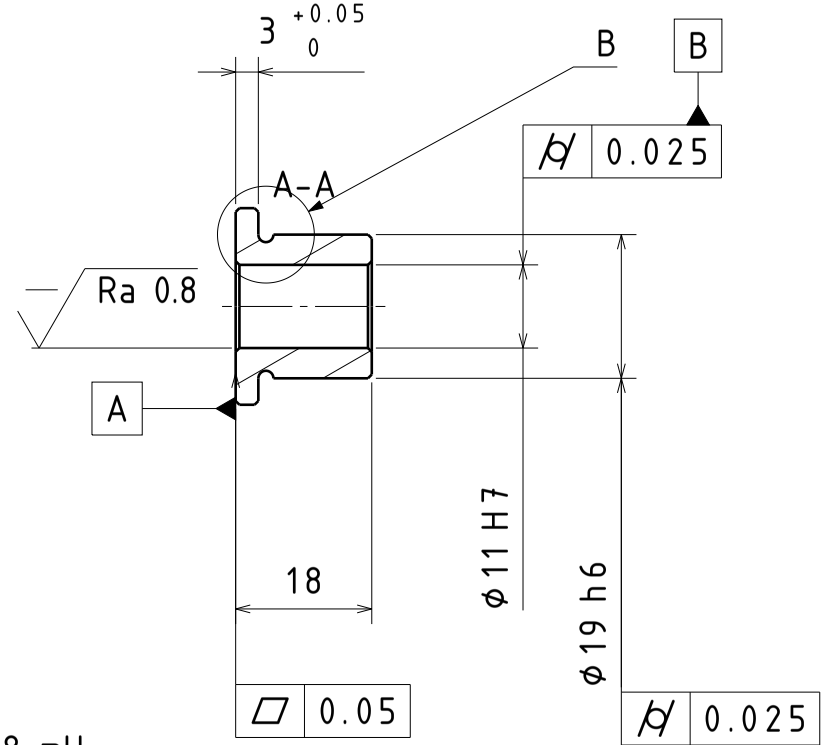
ISO 2768-mK
Chamfers 0.5x45° unless stated

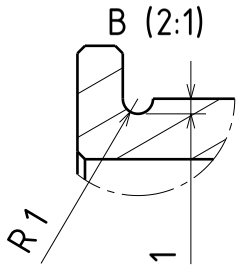
Ra 3.2 (Ra 1.6)

DESIGNED BY: Group 3 DATE: 28/01/2025	Upper Plate		
CHECKED BY: Group 3 DATE: 04/02/2025			
SIZE A3	DASSAULT SYSTEMES		
SCALE 1:1	Material Al 7075	MFG Process Machining	SHEET 1/1

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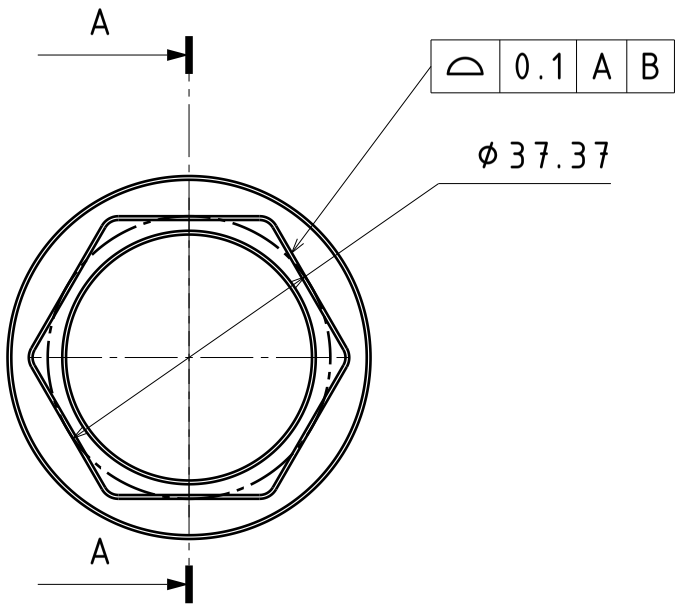


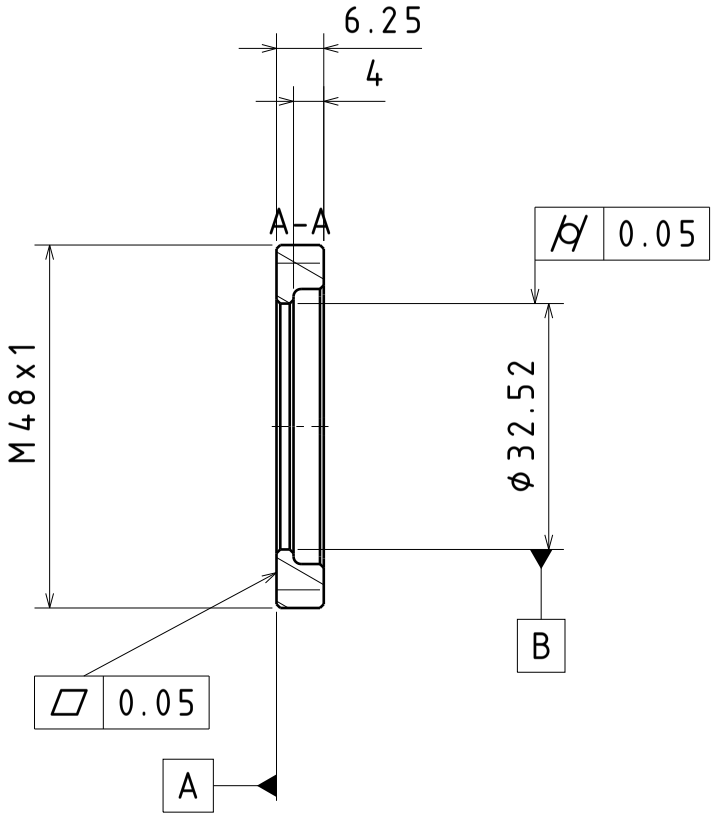
ISO 2778-mH
Chamfer unless stated: 0.5x45°
Roughness: Ra 3.2 (Ra 0.8)

DESIGNED BY: Group 3 DATE: 29/01/2025		Upper Plate Bushing	
CHECKED BY: Group 3 DATE: 29/01/2025			
SIZE A4		DASSAULT SYSTEMES	
SCALE 1:1	MFG PROCESS Machining	DRAWING NUMBER Steel	SHEET 1/1

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ISO 2768 mH
Chamfer unless stated: 0.5x45°
Fillet unless stated: R1
Roughness: Ra 3.2

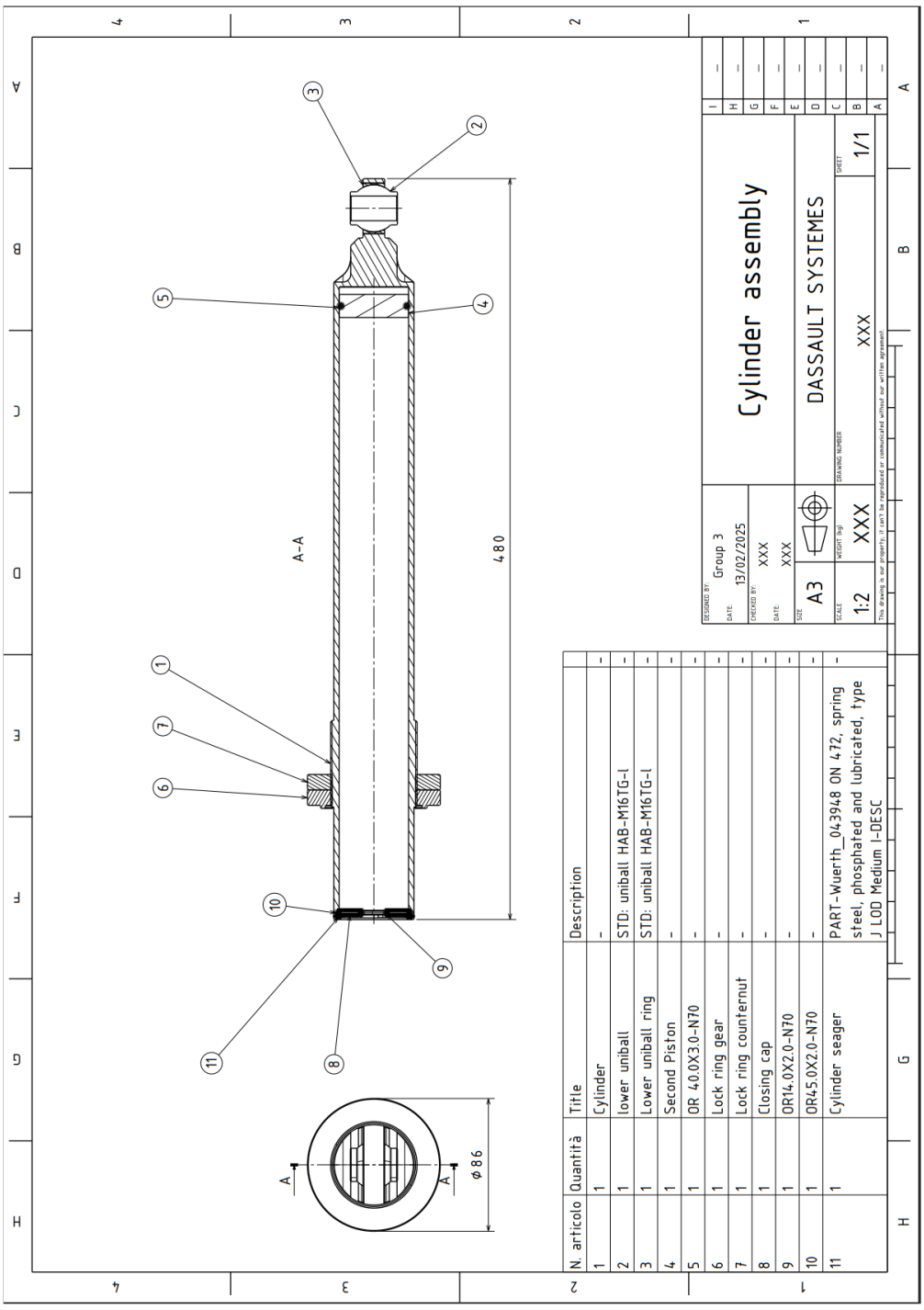
DESIGNED BY: Group 3 DATE: 02/02/2025 CHECKED BY: Group 3 DATE: 2/02/2025	Upper Plate Closing Cap		
SIZE A4	DASSAULT SYSTEMES		
SCALE 1:1	MFG PROCESS Machining	MATERIAL Al 7075	SHEET 1/1

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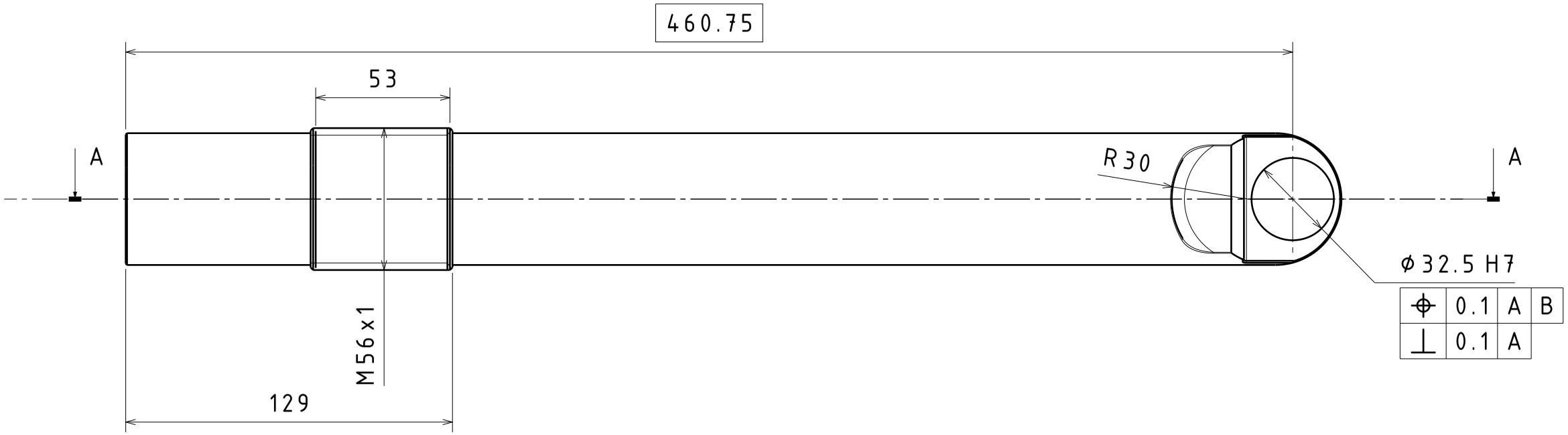
9.3. Cylinder Assembly

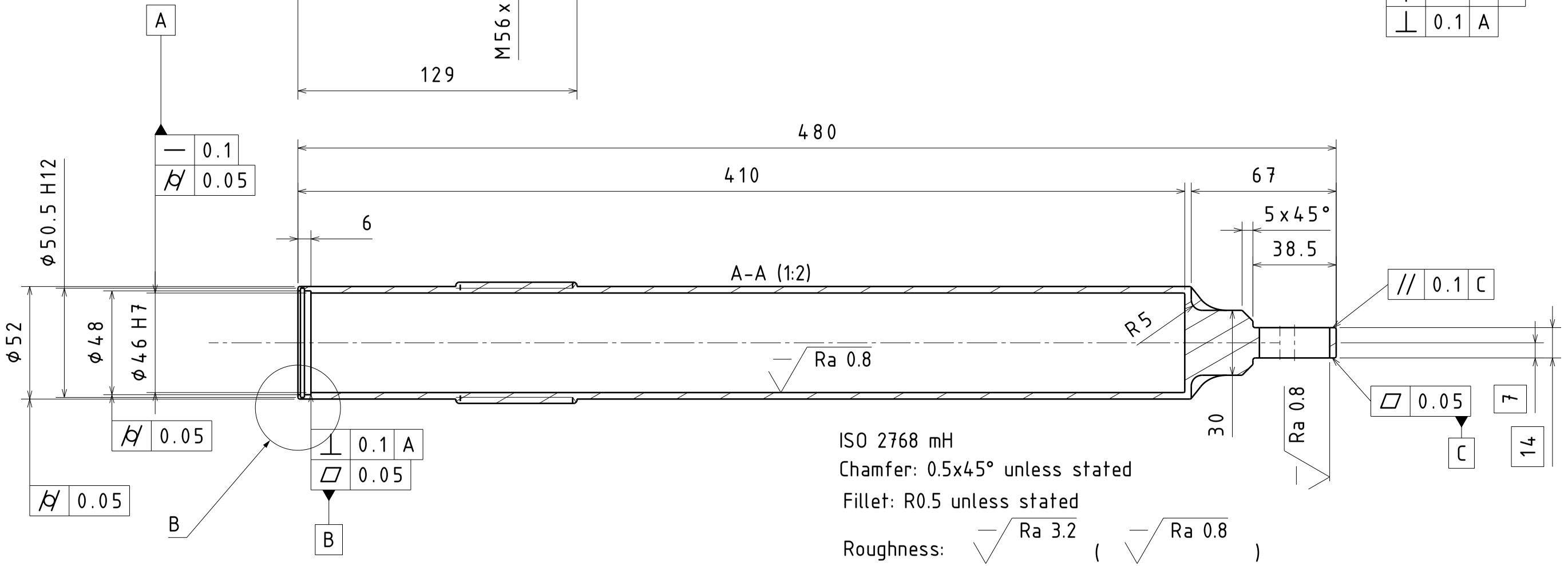
Then, it is possible to go through the "Cylinder Assembly" including the case that envelops the piston sub-assembly. It includes also the second piston dealing with the volume compensation function and its O-ring, the lower unibal, the closing cap together with its O-ring and Seager to seal and fix everything as discussed in Paragraph 9.1. More, the locking gear is here presented too.



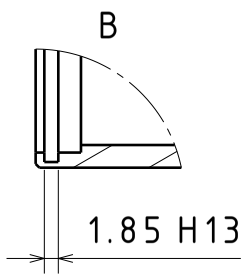
Assembly Drawing 3 – Cylinder Assembly

In the following pages, the related drafts are added:
Draft 9 – Case
Draft 10 – Second Piston
Draft 11 – Cylinder Closing Cap
Draft 12 – Lock Ring Counternut
Draft 13 – Lock Ring





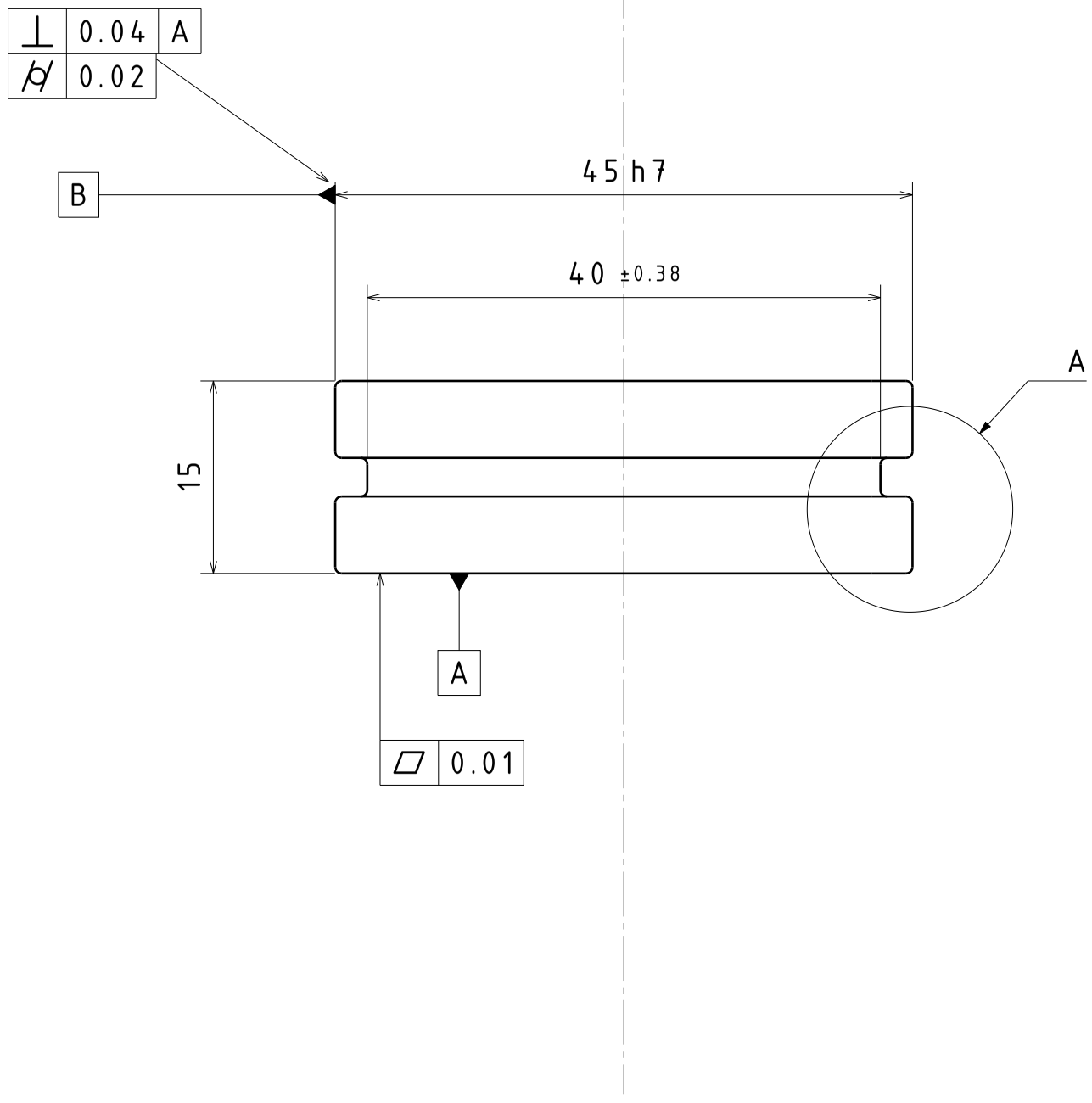
ISO 2768 mH
Chamfer: 0.5x45° unless stated
Fillet: R0.5 unless stated
Roughness: Ra 3.2 (Ra 0.8)



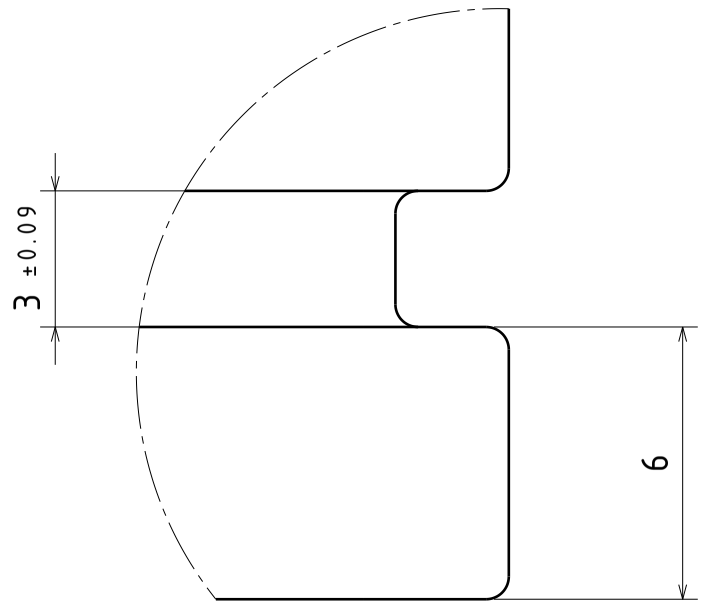
DESIGNED BY: Group 3	Case	
DATE: 02/02/2025		
CHECKED BY: Group 3		
DATE: 2/02/2025		
SIZE A3	DASSAULT SYSTEMES	
SCALE 1:2	MFG PROCESS Machining	MANUFACTURING PROCESS Al 7075 / SHEET 1/1

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A	–



A (6:1)



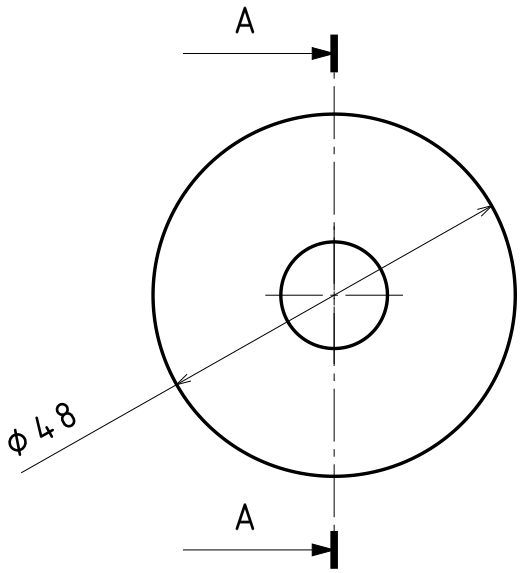
ISO 2768-mK
Fillets R0.5 unless stated

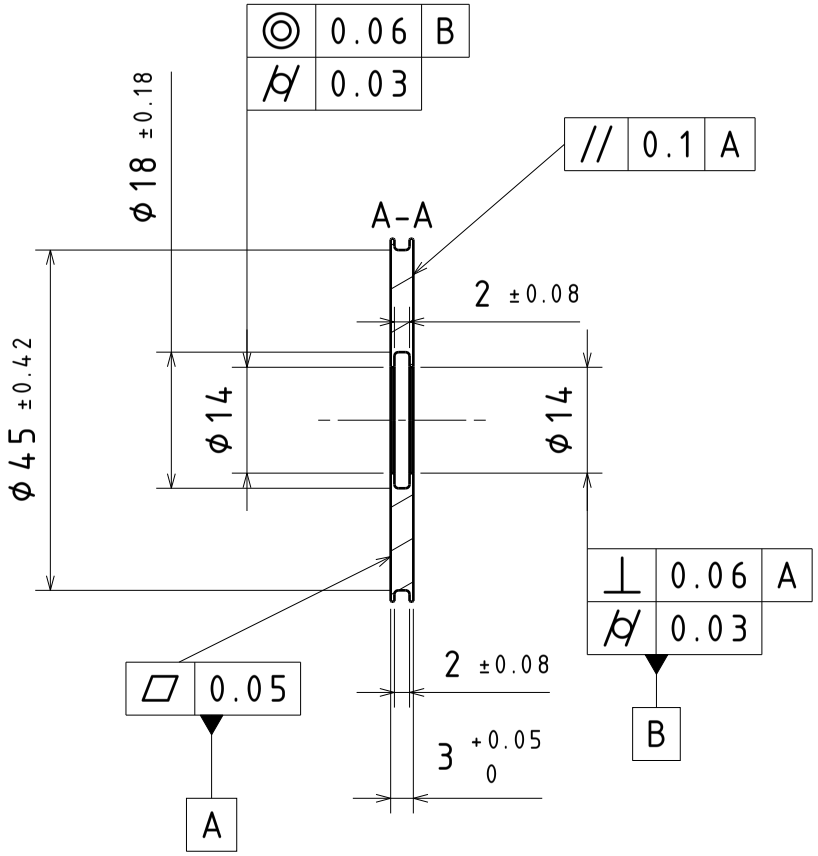
Ra 3.2 (Ra 0.8)

DESIGNED BY: Group 3		Second Piston	
DATE: 02/02/2025			
CHECKED BY: Group 3			
DATE: 11/02/2025			
SIZE: A3			
SCALE: 1:1	WEIGHT (KG): XXX	MATERIAL: Ergal (Al7075)	PROCESS: Machining

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M	
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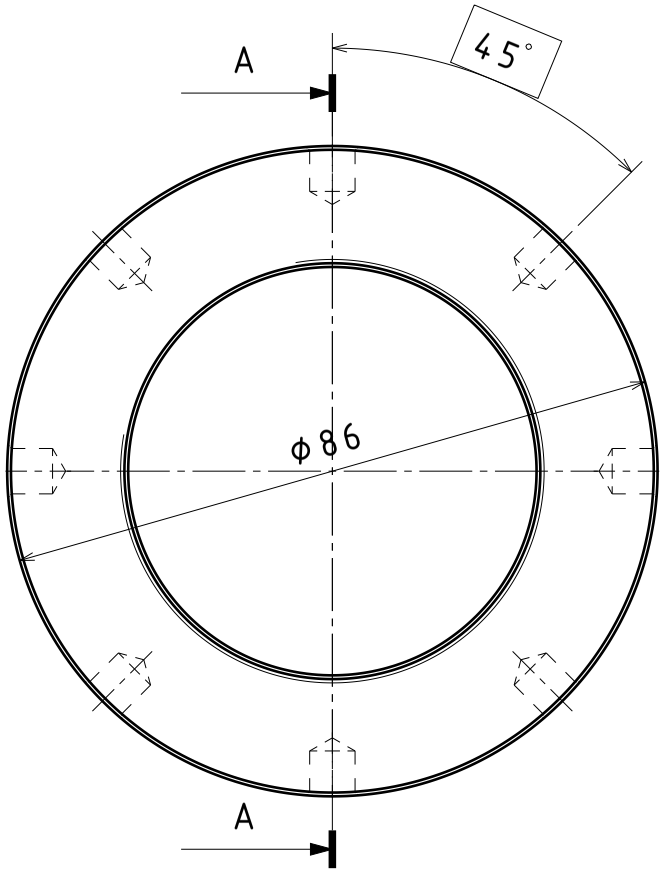


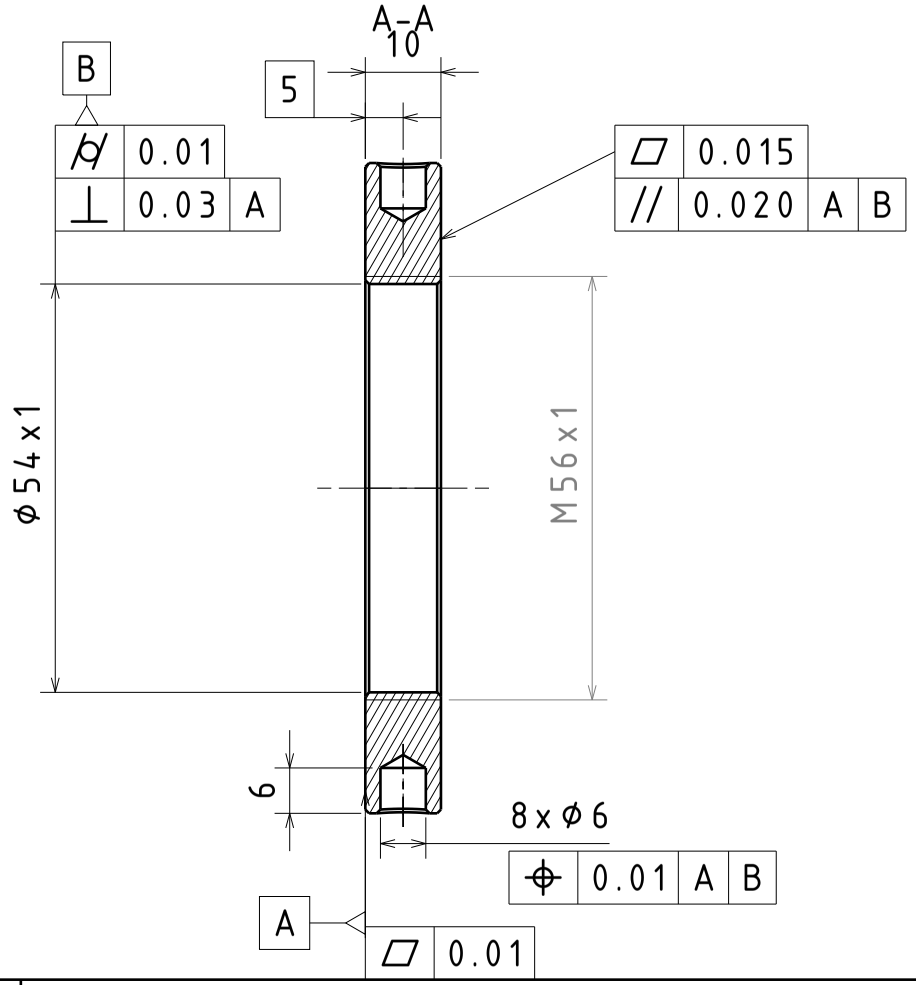
ISO 2768 mH
Camfers 0.1x45°
Fillets R0.5
Roughness Ra 3.2

DESIGNED BY: Group 3 DATE: 09/02/2025		Cylinder closing cap	
CHECKED BY: Group 3 DATE: 9/02/2025			
SIZE A4		DASSAULT SYSTEMES	
SCALE 1:1	MFG PROCESS Machining	MATERIAL Steel	SHEET 1/1

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Fillets unless stated: R 0.5

Chamfers unless stated: 0.5x45

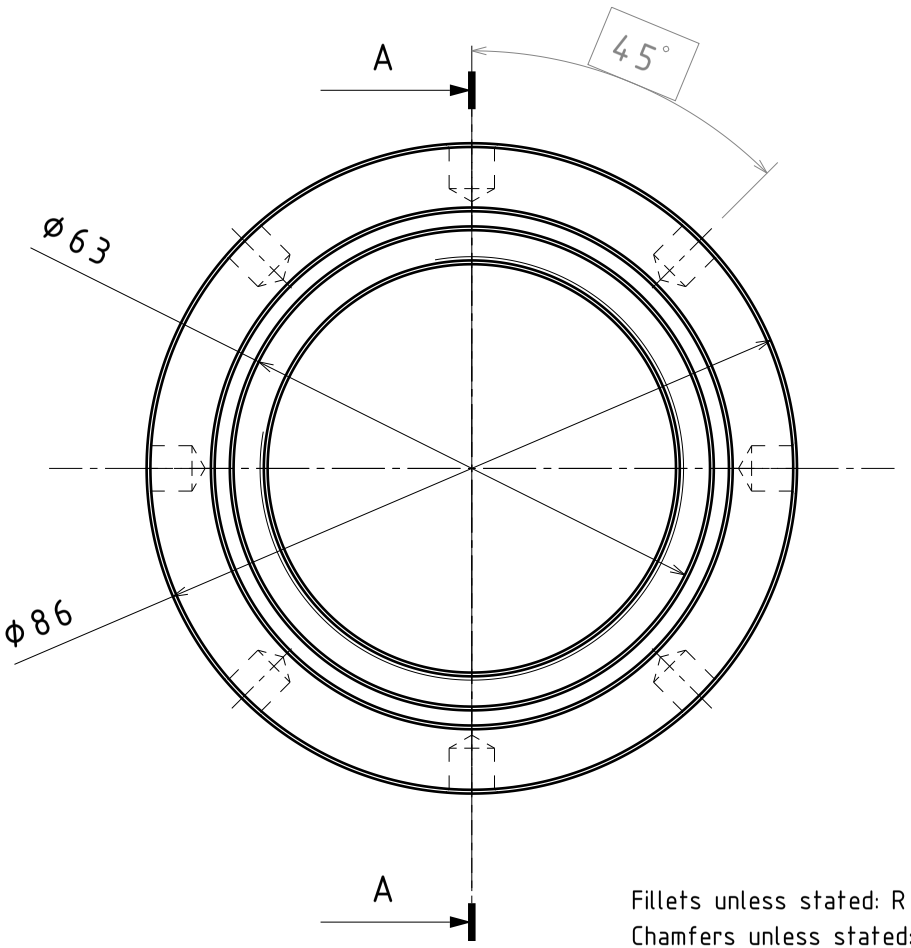
General tolerances: ISO 2768-mK

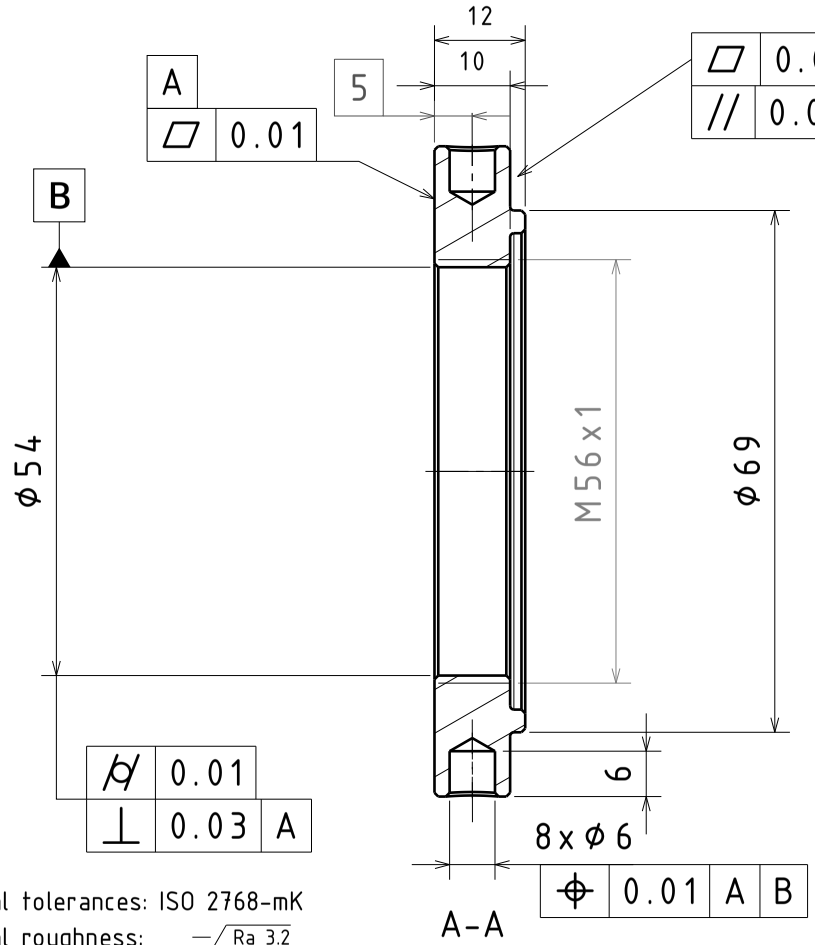
General roughness: Ra 3.2

DESIGNED BY: GROUP 3		Lock ring counternut	
DATE: 04/02/2025			
CHECKED BY: XXX			
DATE: XXX			
SIZE A4		MATERIAL AL 7075 hard anodized	DRAWING NUMBER
SCALE 1:1	WEIGHT (kg) XXX	PROCESS Machining	SHEET 1/1

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Fillets unless stated: R 0.5
Chamfers unless stated: 0.5x45

General tolerances: ISO 2768-mK
General roughness: Ra 3.2

Lock ring gear

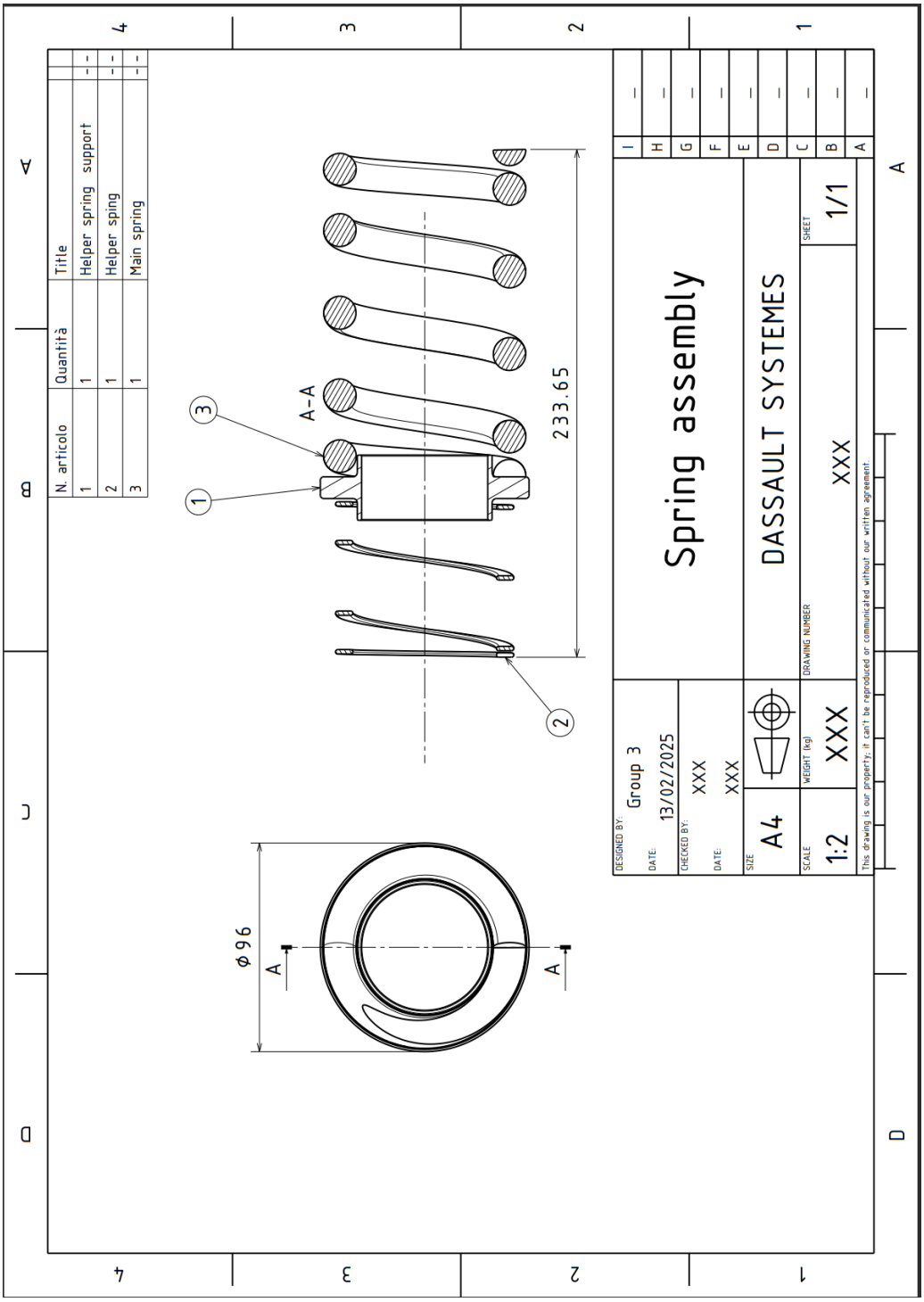
DESIGNED BY: GROUP 3	
DATE: 03/02/2025	
CHECKED BY: GROUP 3	
DATE: XXX	
SIZE: A4	MAERIAL: Al 7075 hard anodized
DRAWING NUMBER:	
SCALE: 1:1	WEIGHT (kg): XXX
PROCESS: MACHINING	SHEET: 1/1

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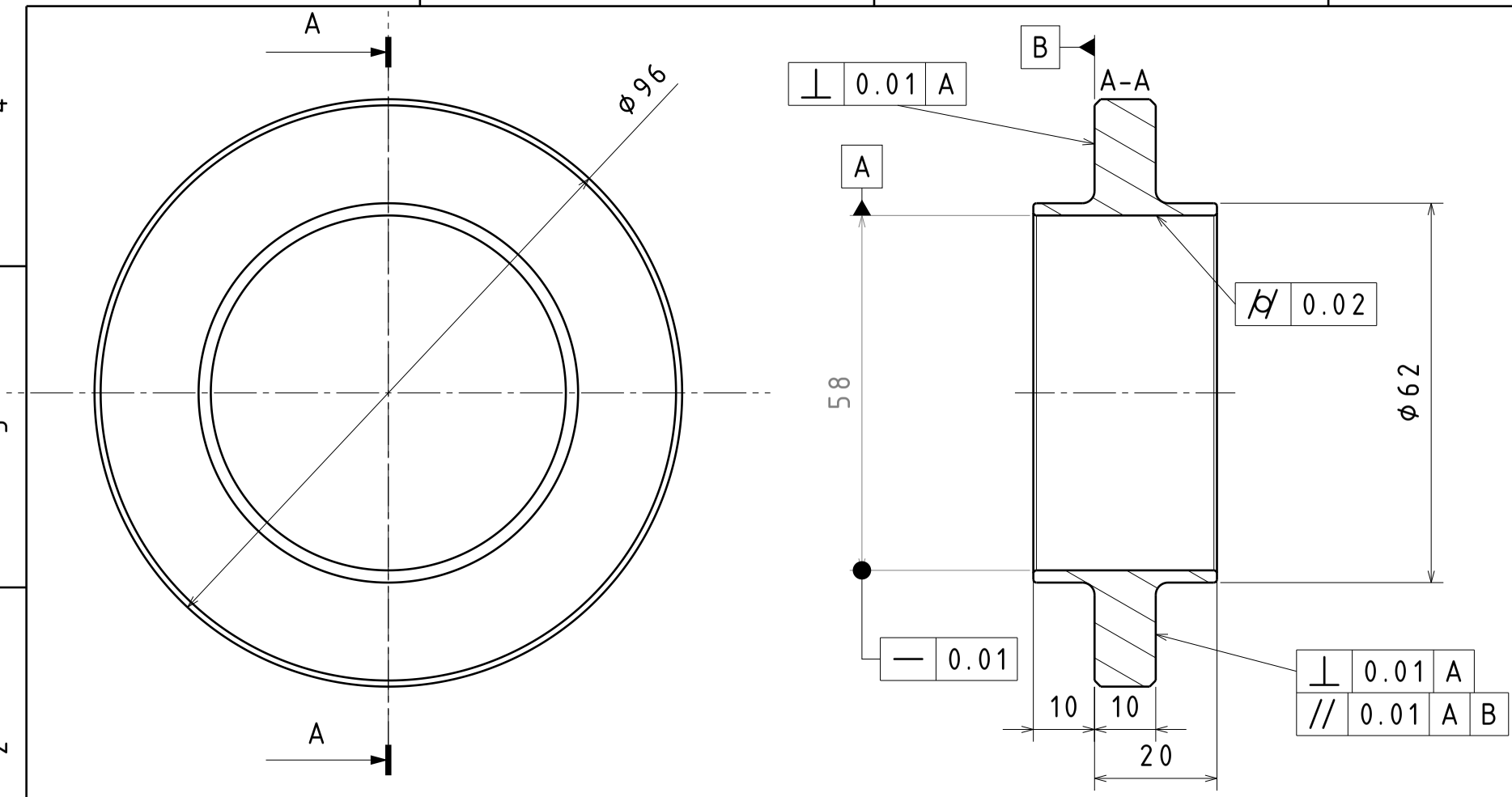
9.4. Spring Assembly

Finally, in the "Spring Assembly" all the springs are considered so both the main and the helper with its support. Springs are not directly related to the MR damper technology since their mechanism and integration are entirely mechanical, but it is good to represent them to have then a clear, immediate and practical idea about the entire system, its dimensions and its possible functioning.



Assembly Drawing 4 – Spring Assembly

In the following page, the related draft Is added:
Draft 14 – Helper Spring Support



Fillets unless stated: R 0.5
Chamfers unless stated: 0.5x45

General tolerances: ISO 2768-mK
General roughness: Ra 3.2

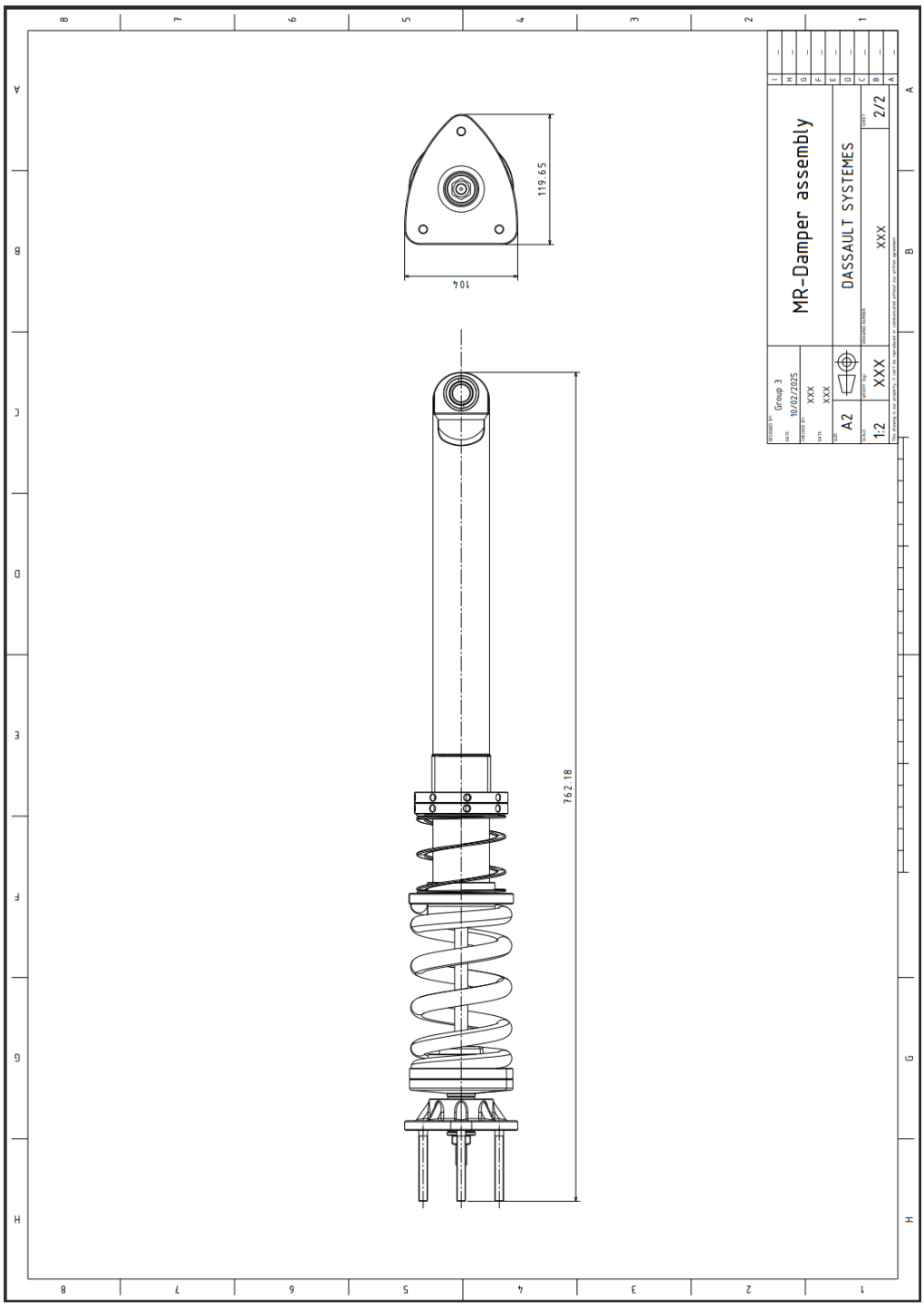
DESIGNED BY: GROUP3		# Helper Spring Support	
DATE: 04/02/2025			
CHECKED BY: XXX			
DATE: XXX			
SIZE A4		MATERIAL Aluminum	DRAWING NUMBER
SCALE 1:1	WEIGHT (kg) XXX	PROCESS MACHINING	SHEET 1/1

This drawing is our property; it can't be reproduced or communicated without our written agreement.

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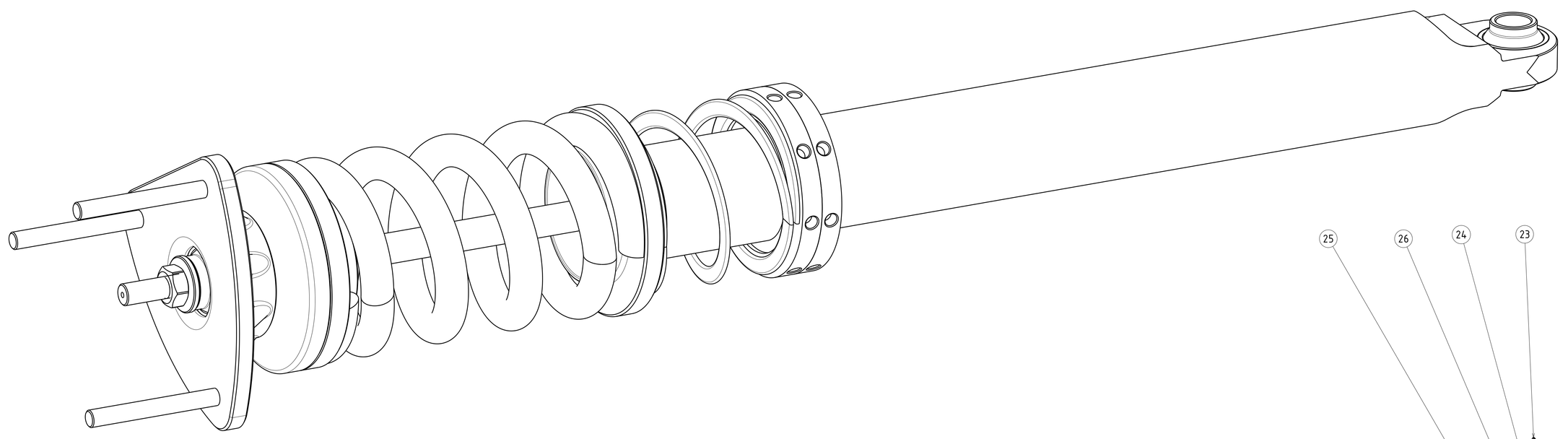
9.5. MR Suspension Assembly

Here below, the final exploded view of every sub-system is shown as well as the final aspect of the entire MR assembly. An extra table is presented just to underline the actual overall dimensions of the entire system.

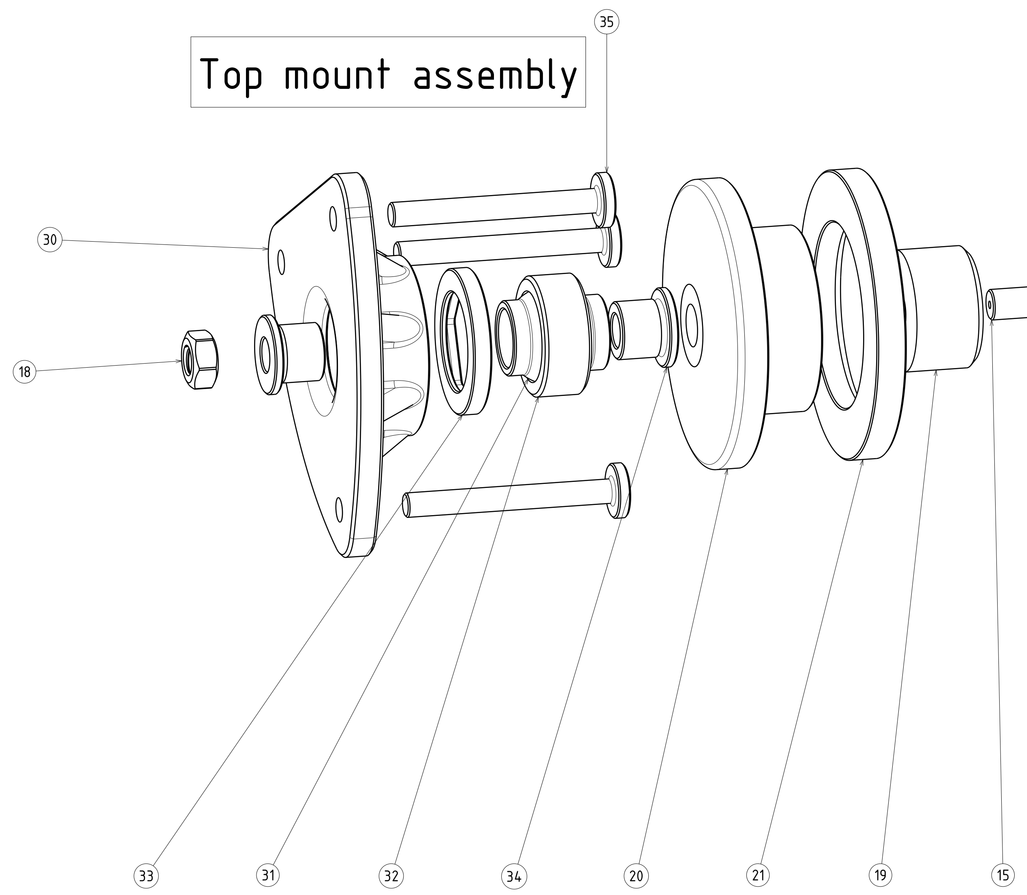


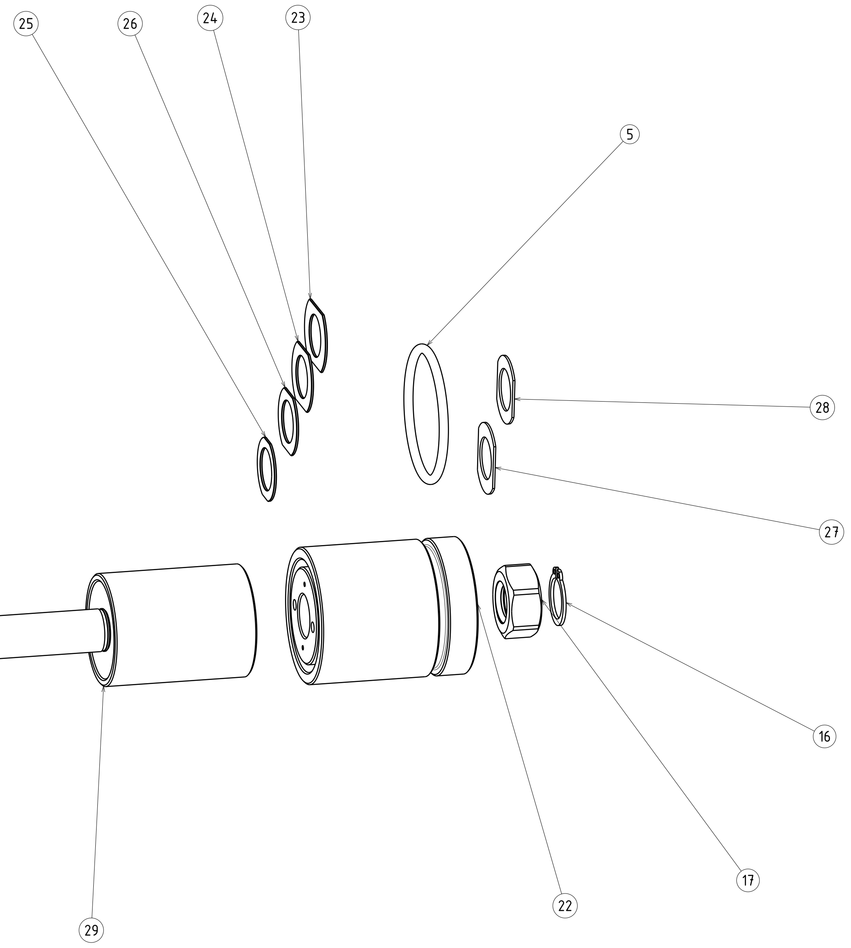
Assembly Drawing 5 – MR Assembly Overall Dimensions

In the following page, the final exploded view of the entire system and of each sub-assembly is added:
Exploded View 1 - MR Assembly

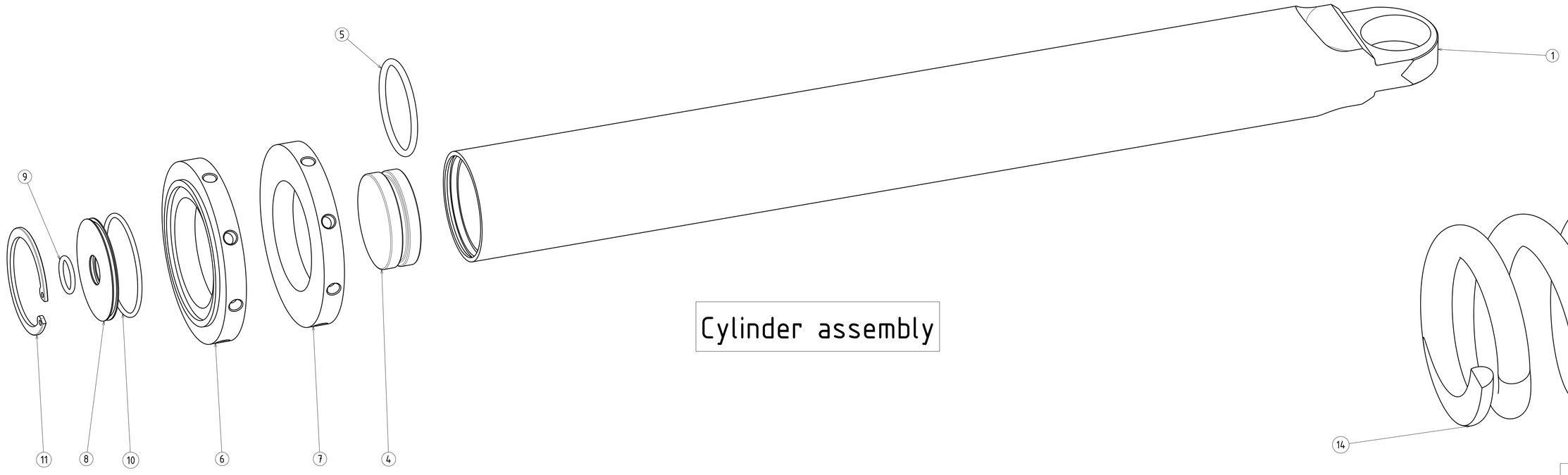


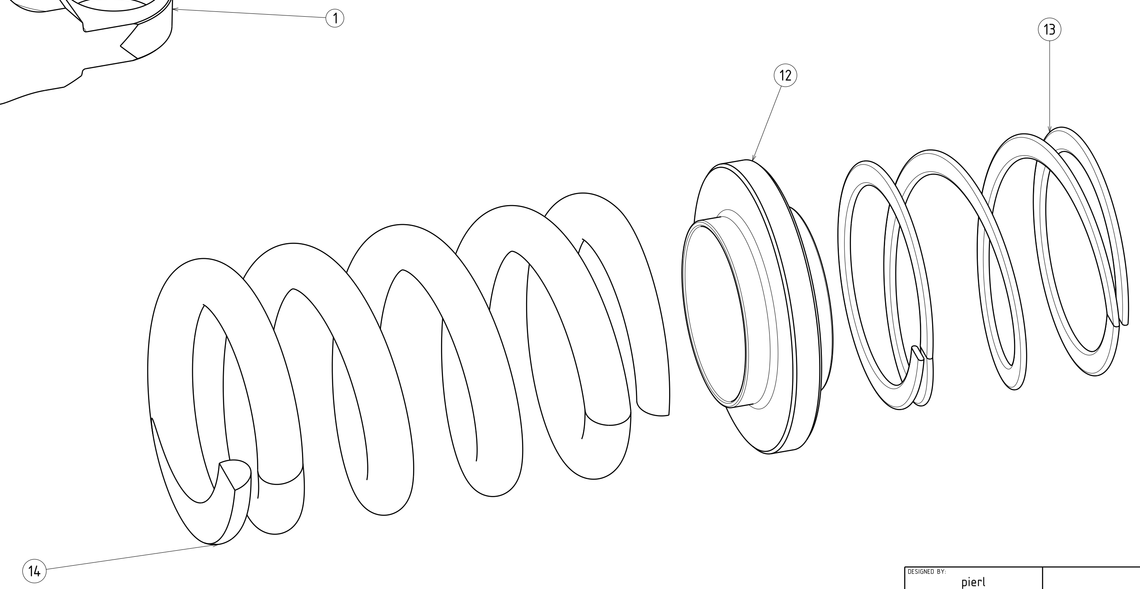
N. articolo	Quantità	Title	Description	
1	1	Piston cylinder	Al7075 T6 hard anodized	-
2	1	lower unibal	STD: uniball HAB-M16TG-1	-
3	1	lower uniball ring	STD: uniball HAB-M16TG-1	-
4	1	Second piston	Al6782	-
5	2	pistons O-rings	STD: OR 40.0X3.0-N70	-
6	1	locking gear	CNC Al7075 anodized	-
7	1	locking gear counternut	CNC Al7075 anodized	-
8	1	closing cap	-	-
9	1	cap inner O-ring	STD: OR 14.0X2.0-N70	-
10	1	cap outer O-ring	STD: OR 45.0X2.0-N70	-
11	1	cylinder seager	PART-Wuerth_043948_DIN 472, spring steel, phosphated and lubricated, type J_LOD_Medium_1-DESC	-
12	1	Helper spring support	CNC Al7075	-
13	1	helper spring	STD: OEM coil spring	-
14	1	Main spring	STD: OEM coil spring	-
15	1	Piston rod	AISI 304	-
16	1	piston seager	PART-Wuerth 043814 DIN 471 spring steel, phosphated standard vers. LOD Medium I-DESC	-
17	1	piston nut	M14x15 DIN 934, galvanized steel 181, fine pitch LOD High I-DESC	-
18	1	upper uniball plate nut	M10x1 DIN 934, galvanized steel 181, fine pitch LOD High 1 DESC	-
19	1	Bumpstop	STD: BUMP STOP 9F1412303B	-
20	1	Upper plate	Al 7075 hard anodized	-
21	1	Load cell	-	-
22	1	Main piston	Al	-
23	1	Bump shim 1	L=25,5 mm	-
24	1	Bump shim 2	L=24 mm	-
25	1	Bump shim 3	L=22,5 mm	-
26	1	Bump shim 4	L=21 mm	-
27	1	Rebound shim 1	L=23,5 mm	-
28	1	Rebound shim 2	L=22,5 mm	-
29	1	Magnetic coil	copper wire Φ1,024 x 50 turns	-
30	1	Top mount		-
31	1	Upper uniball	STD: NMB ABYT12	-
32	1	Upper uniball ring	STD: NMB ABYT12	-
33	1	Plate closing cap	-	-
34	2	Bushing	steel	-
35	3	Topmount screws	STD: M8 x 60	-
36	1	Rebound stop	-	-











DESIGNED BY: pierl	MR-Damper assembly		
DATE: 09/02/2025			
CHECKED BY: XXX			
DATE: XXX			
SIZE: A0	DASSAULT SYSTEMES		
SCALE: 1:1	WEIGHT: XXX kg	DRAWING NUMBER: XXX	SHEET: 1/2

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9.6 Tolerance stack-ups

Tolerance stack-ups in the assembly can be found at both rod's ends. The first one to be analysed is the one placed in proximity of the upper plate and top mount. The performance metric G has to be in between 2 and 3.5 mm. In this way it is ensured that the assembly composed by the elements A, B, C, and D is always well tightened by the nut. Therefore, the assembly should cover at least 2 fillets, being the thread pitch of 1 mm. The upper value of G limits the number of unused fillets to 3.5.

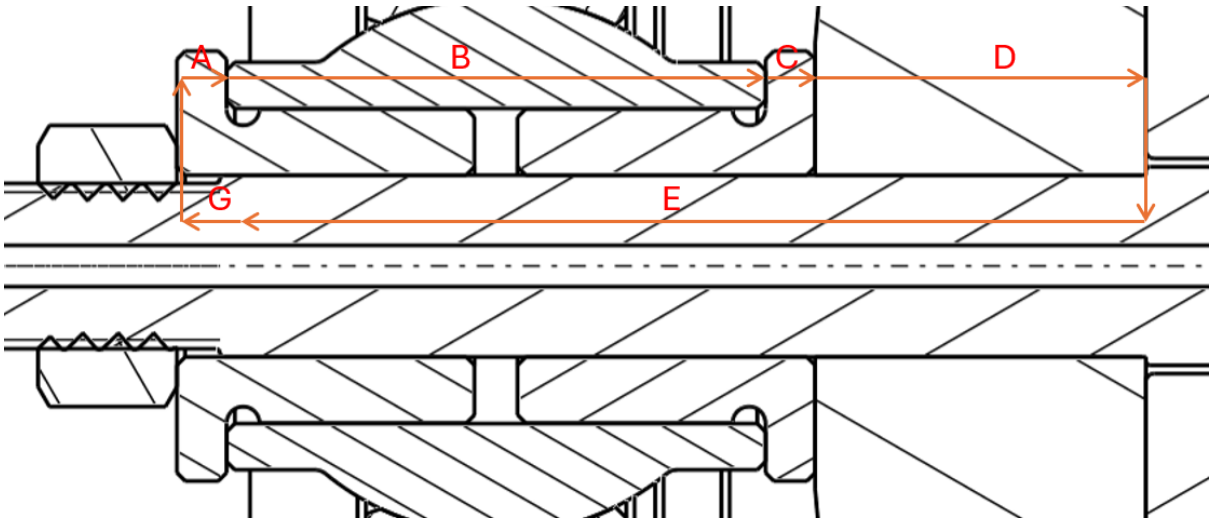


Figure 46 - Tolerance Stack-Up (at upper plate and top mount)

NAME	+ / -	+MAX / -MIN	+ / -	+MIN / -MAX	TOLL.
A	+	3.05 + 0.195	+	3	0.05
B	+	32.51	+	32.499	0.011
C	+	3.05 + 0.195	+	3	0.05
D	+	20.2	+	19.8	0.4
E	-	55.7	-	56.3	0.6
TOT		3.11		1.999	
G		3.5		2	1.5
Diff		0.39		0.001	

Table 37 - Tolerance Stack-Up for defined components

The tolerances in the table above come from:

- Specific dimensional tolerance on the two bushings
- Tolerance on the Unibal coming from the SFK Table for outer rings of outer diameter 30÷50
- General Tolerance for the Upper plate ISO 2768-mH
- General tolerance for the Rod ISO 2768-mH

The first calculation from the stack-up gives a difference of 0.39 mm between the total value and G in the upper boundary of tolerance.

A good way to solve this issue is to evenly distribute this difference between the two bushings by relaxing their dimensional tolerance. In this way a refined table is obtained, and the final differences are null or very close to zero in the case of the lower boundary.

Outer ring

D over	D incl.	Δ_{Dmp} high	Δ_{Dmp} low	V_{Dp}[1] Diameter Series 7, 8, 9 max	0, 1 max	2, 3, 4 max	Sealed bearings[2] max	V_{Dmp}[1] max	$\Delta_{Cs}, \Delta_{C1s}, V_{Cs}$	K_{ea} max
mm		µm		µm				µm		µm
2,5	18	0	–8	10	8	6	10	6	Values are	15
18	30	0	–9	12	9	7	12	7	identical to	15
30	50	0	–11	14	11	8	16	8	those for inner ring	20
50	80	0	–13	16	13	10	20	10	of same	25
80	120	0	–15	19	19	11	26	11	bearing	35
120	150	0	–18	23	23	14	30	14		40

Figure 48 - Outer Ring Tolerance Detection

Linear dimensions:

TOLERANCE CLASS		LIMIT SWITCHES FOR NOMINAL DIMENSIONS (1)							
Designation	Name	Over 0,5 to 3	Over 3 to 6	Over 6 to 30	Over 30 to 120	Over 120 to 400	Over 400 to 1000	Over 1000 to 2000	Over 2000 to 4000
f	Fine	±0,05	±0,05	±0,1	±0,15	±0,2	±0,3	±0,5	-
m	Media	±0,1	±0,1	±0,2	±0,3	±0,5	±0,8	±1,2	±2,0
c	Coarse	±0,2	±0,3	±0,5	±0,8	±1,2	±2,0	±3,0	±4,0
v	Very coarse	-	±0,5	±1,0	±1,5	±2,5	±4,0	±6,0	±8,0

Figure 47 - General Tolerances Detection

NAME	+ -	+MAX -MIN	+ -	+MIN -MAX	TOLL.
A	+	3.245	+	3	0.12
B	+	32.51	+	32.499	0.011
C	+	3.245	+	3	0.12
D	+	20.2	+	19.8	0.4
E	-	55.7	-	56.3	0.6
TOT		3.5		1.999	
G		3.5		2	1.5
Diff		0		0.001	

Table 38 - Refined Tolerance Stack-Up for defined components

As already said, a similar situation can be found close to piston, where it's required to leave a small gap in between the start of the thread and the relative nut in order to guarantee that the assembly composed by the piston the two sets of shims (element L1, P and L2) is always tightened by the nut. On this purpose, since the bump shims pack is nominally thick 2.8 mm and the thread is characterized by a pitch of 1.5 mm, to guarantee the correct tightening the shims pack should cover at least one fillet and a half, thus requiring the performance metric G to be in between 2.5 mm and 3 mm.

$$G = B - A = (L1 + P + L2) - A$$

Equation 22 - Tolerance Stack-Up formula

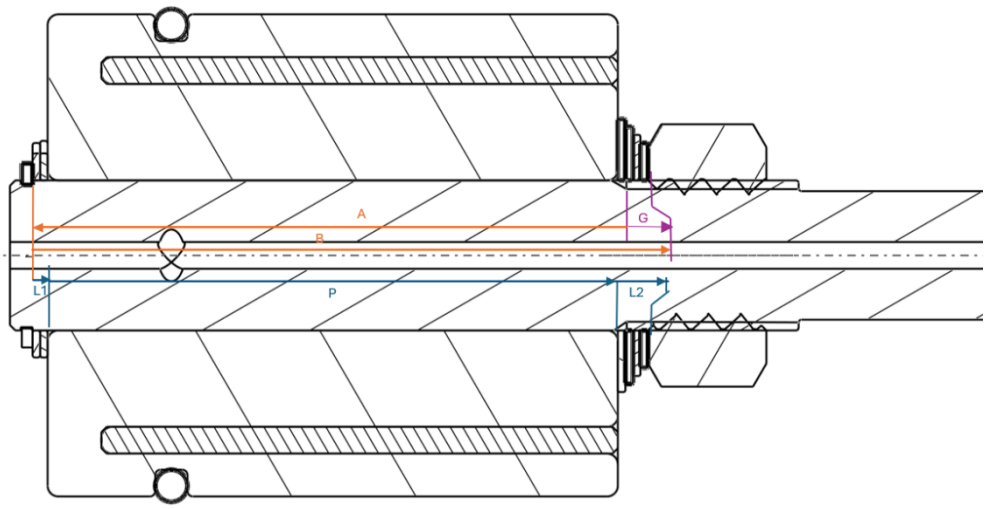


Figure 49 - Tolerance Stack-Up (close to piston)

NAME	+ -	+MAX -MIN	+ -	+MIN -MAX	TOLL.
A	-	54.9	-	55.5	0.6
L1	+	1.5	+	1.3	0.2
P	+	53.3	+	52.7	0.6
L2	+	2.9	+	2.7	0.2
TOT	+	2.8	-	1.2	1.6
G		3.00		2.5	1.5
Diff		0.20		1.3	

Table 39 - Tolerance Stack-Up for defined components

The tolerances in the table comes again from the general tolerances of Figure47.
The first iteration of the stack-up gives a difference of 0.20 mm between the total value and the G upper boundary of tolerance, while 1.3 mm between G and the lower one.
In order to solve this issue is to evenly distribute the difference of the upper boundary between the two stacks of shims by relaxing their general dimensional tolerance from class "m" to "c". However, in this way also the lower limit of the respective components' changes, thus increasing consequently the relative G difference. Despite of the increase, the difference is again evenly distributed between the P and A distance, thus obtaining final differences that are null, as shown in the refined table below.

NAME	+ -	+MAX -MIN	+ -	+MIN -MAX	TOLL.
A	-	54.9	-	55.5 – 0.75	-0.15
L1	+	1.5 + 0.1	+	1.3 – 0.1	0.4
P	+	53.3	+	52.7 + 0.75	-0.15
L2	+	2.9 + 0.1	+	2.7 – 0.1	0.4
TOT	+	3	-	2.5	0.5
G		3,00		2.5	0.5
Diff		0		0	

Table 40 - Refined Tolerance Stack-Up for defined components

10. Design Review

The design review process is very useful to improve the final design of the components. Some solutions need to be modified to work properly, even if they appeared to work during the embodiment phase. It could be the case of interference issues between parts, a simpler way of realization or a design error that can be spotted when all the components are merged in the final assembly.

10.1. Circlip at Piston Rod End

The original design involves two nuts at both piston ends to fix it to the rod, altogether with the shims. The issue with two nuts is that the rod has to be threaded in two sections, three if the other rod end is taken into consideration. A simpler solution is to lock the axial movement of the piston at the rod end with a circlip. In this way only one nut is needed while all the required functions are still fulfilled.



Figure 50 - Circlip instead of Nut

10.2. Rebound Stop

In the fully extended configuration of the damper there is a gap between the cylinder closing cap and the piston nut of at least 3mm. In this position both springs are at their fully extended length. Clash between nut and cap will occur if, for some reason, the damper extends even more. To avoid damage in the components, a rubber element called rebound stop is adopted. Its working principle is the same as the bump stop, just its activation happens in the opposite direction of motion.
The selected rebound stop has an external diameter of 22mm and a height of 3mm.

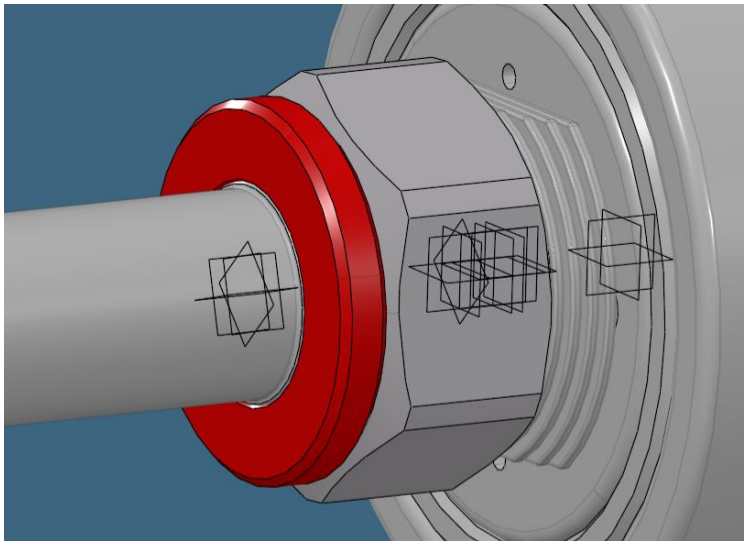


Figure 51 - Rebound Stop Usage

10.3. Rod and Case Thread Diameter

Both rod and case need to be threaded in their midparts. To achieve a correct coupling between the threaded part and the nut, the diameter of the former should be raised with respect to its surroundings. Therefore, the rod goes from ø 12 to ø 14 in the threaded part, and the case from ø 52 to ø 54.

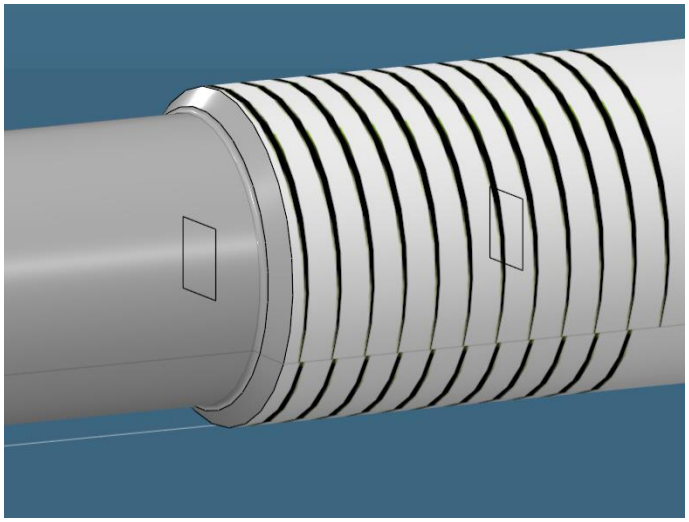


Figure 52 - Thread Diameter Update

11. Conclusions

This paper has covered the full development process of an MR damper for racing applications, particularly suited for a very demanding course as the Pikes Peak. The best solution has been achieved thanks to the tools provided by the Engineering Design. Each phase has been responsible for the definition of the final product, in particular:

- **Competitor and market analysis** set the initial benchmarks for performance, quality, reliability and safety.
- **Requirement list** to find the main function carriers with their properties.
- **Quality Function Development** to assess the interrelations within technical and customer requirements for the final benchmarks.
- Cost for the **cost estimation**, considering processes and materials.
- **Conceptual design**: from the main solution principle to function structures and working principles, which have been then combined to get the solution variants. The winning concept has been selected through **VDI2225 method**.
- **Embodiment** to get the final layout by defining dimensions, materials and processes

Once all possible errors and faults have been cleared out, the very final stage has been the **detail design** that has provided the technical drawings, useful for the manufacturing of the product.